COUNTRY REVIEWS

OECD Economic Surveys

Norway

ECONOMICS

OECD

February 2000

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OECD ECONOMIC SURVEYS

1999-2000

Norway

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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BASIC STATISTICS OF NORWAY (1998)

THE LAND

Area (1 000 km^2):	
Total	385.4
Mainland	323.8
Agricultural	10.5
Productive forests	73.9

Major cities (thousand inhabitants, 1.1.99):	
Oslo	502.9
Bergen	227.3
Trondheim	147.2

THE PEOPLE

Population (thousands, 1.1.99)	4 445.3
Number of inhabitants per km^2	11.5
Net natural increase (thousands, 1997)	15.2
Net migration (thousands, 1997)	10.7

Total labour force (thousands)	2 308.0
Civilian employment (thousands)	2 231.0
Civilian employment (% of total):	
Agriculture, forestry and fishing	4.4
Industry and construction	22.9
Services	72.5

PRODUCTION

Gross domestic product:	
NOK billion	1 107.1
Per head (US$)	33 007

Gross fixed capital investment:	
% of GDP	25.9
Per head (US$)	8 541

THE GOVERNMENT

Public consumption (% of GDP)	21.5
General government (% of GDP):	
Current and capital expenditure	46.4
Current revenue	50.3

Last general elections: 15.9.97
Next general elections: 2001

Composition of Parliament (number of seats):	
Labour	65
Progress	25
Christian Democrats	25
Conservatives	23
Centre	11
Socialist Left	9
Other	7
Total	165

FOREIGN TRADE

Exports of goods and services (% of GDP)	37.4
of which: Oil and gas	10.8
Main commodity exports (% of total):	
Fish and fish products	8.7
Base metals and products	12.5
Machinery and transport equipment (excluding ships)	11.7
Mineral fuels	43.3

Imports of goods and services (% of GDP)	37.2
Main commodity imports (% of total):	
Ships	4.3
Raw materials (including fuel and chemicals)	9.2
Base metals and products	9.9
Machinery and transport equipment (excluding ships)	35.4

THE CURRENCY

Monetary unit: Krone

December 1999, average of daily rates:	
NOK per US$	8.01
NOK per euro	8.10

Note: An international comparison of certain basic statistics is given in an Annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Norway by the Economic and Development Review Committee on 10 January 2000.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 2 February 2000.

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The previous Survey of Norway was issued in February 1999.

Assessment and recommendations

After overheating in 1998, a significant policy tightening contributed to a slowdown in economic activity in 1999

The Norwegian economy cooled off in 1999 after a period of exceptional strength that had led to overheated conditions. Growth of mainland GDP (*i.e.* excluding shipping, oil and gas activities) slowed significantly, following growth of 3¼ per cent in the previous year. Investment declined substantially reflecting lower investment by the offshore oil sector, the completion of major public infrastructure projects, a profit squeeze and tight monetary policy. The decline was cushioned by continuing, though more subdued, private consumption growth. However, labour and housing markets have remained tight. The unemployment rate, currently at 2.9 per cent, has so far increased only marginally. Despite moderate wage settlements in 1999, unit labour costs rose from the year before twice as fast as in the euro area, reflecting strong carry-over and weak productivity growth. Tight monetary policy, helped by the oil price recovery, has contributed to strengthen the krone and to reduce cost and price inflation. As the risks of currency depreciation and accelerating inflation receded, prospects for weak output development and lower inflation have led to cuts in the key central bank rates by 2.5 percentage points since the peak in late 1998 and early 1999. However, with the deposit rate at 5.5 per cent monetary policy has remained cautious.

Output should recover in 2001 with the main risk being wage rises that are too strong

Mainland GDP is projected to remain flat in 2000, with a further sharp drop in offshore and mainland investment partly offset by rising private and public consumption. Stronger world demand should lead to a pick-up in manufactured exports but the deterioration in cost competitiveness may cause continued market share losses. The unemployment rate is projected to increase and, as a consequence, wage rises should decelerate. This should

allow a reduction in the inflation differential with the euro area. This, together with support from foreign demand, should underpin a recovery in 2001. However, wage developments are a major risk to the outlook. Parts of the labour market are still very tight, and wage rises could be higher than projected. That would boost private consumption in the short term but would prevent a reduction in inflation. Thus, the forthcoming wage negotiations for 2000-01 will be crucial for alleviating the still strong labour cost pressures and for bringing cost developments down to those in the main trading partners.

A cautious monetary policy stance is therefore needed

Given the risk of strong labour cost pressures, a cautious monetary policy stance is needed. The present operational procedures should facilitate this. More than in the past, Norges Bank has emphasised the need to bring price and cost inflation down over time to the ECB's inflation objective in order to underpin exchange rate stability *vis-à-vis* the euro. From the turmoil in August 1998 and previous periods of currency turbulence, the Bank concluded rightly that it has become impossible to fine-tune movements of the krone. The emphasis now placed on low inflation in the medium-term as the fundamental precondition for exchange rate stability is welcome. It should prevent a pro-cyclical monetary policy stance, which on occasions occurred in the past.

Tight fiscal policy would underpin confidence in the policy framework

The government decision to pursue a neutral fiscal stance in 2000, reflecting the economic slowdown, is questionable, the more so given the record of fiscal slippage in recent years and the limited possibilities to impose restraint later in the budget year. Furthermore, it is regrettable that the fairly rapid rise in outlays in the approved 2000 Budget has even necessitated a tax increase to maintain a neutral fiscal stance. This tax increase could give rise to an inflationary impulse. Even if macroeconomic and incomes policy are successful in bringing about convergence of cost and price inflation to the rates in the euro area in 2000, new macroeconomic policy challenges could emerge quickly. In the early phase of the recovery that started in 1993, macroeconomic policy was facilitated by substantial spare capacity. After the current growth pause, however, slack in the

economy will be very limited. This could imply that labour market bottlenecks could reappear quickly and that an early policy tightening may be required.

The policy framework regained credibility in 1999…

Since 1992, policy has been based on a consensus, the so-called *Solidarity Alternative*, that wage moderation and exchange rate stability should safeguard the competitiveness of mainland industry, while fiscal policy should be geared towards smoothing domestic demand. Budget surpluses associated with the government's oil revenues are reinvested in financial assets abroad. This macroeconomic policy framework came under heavy strain in 1998 as wage rises were excessive and fiscal policy did not prevent an overheating of the economy. Indeed, in these conditions it was perhaps fortuitous that with falling oil prices the krone fell sharply, thereby opening up scope for monetary policy tightening; this was needed to cool off the economy. The framework has regained credibility in 1999 due to the agreement on moderate wage settlements and a fiscal policy tightening, although the fiscal policy stance was less restrictive than originally intended. Moreover, the emphasis on the fundamental preconditions for exchange rate stability in the implementation of monetary policy should serve to strengthen the framework for the future. But the evidence remains mixed on the viability of the Solidarity Alternative as regards both wage-setting and the assignment of fiscal policy to smooth demand.

… but the current highly centralised incomes policy may be increasingly difficult to implement and…

The highly centralised wage negotiations with substantial government involvement worked well in the early stages of the economic recovery; and while they did not prevent a wage explosion in 1998, they did succeed in restraining cost pressures in 1999. However, the ongoing shift from the manufacturing to the service sector, where union density is much lower, shortages of skilled labour in parts of the public sector and rapid international integration will pose increasing challenges to the negotiating framework. The current broad review of incomes policy to strengthen its role in preventing strong cyclical movements is therefore welcome. Furthermore, while policy concessions to arrive at a reasonable central wage agreement can have short-term benefits these could easily be outweighed by their long-term costs.

This is, for instance, the case with the two-year reduction in the early-retirement age agreed in 1997.

… fiscal policy may again be insufficiently tight in the future

Fiscal policy was insufficiently tight in 1997 and 1998 to prevent the overheating of the economy. Moreover, the tightening in 1999 was smaller than envisaged in the draft Budget. Fiscal policy could fail again to smooth domestic demand in the future as pressures to increase expenditures are substantial – especially with petroleum revenues soaring – and debt is not a constraint. Moreover, fiscal policy is not a very flexible instrument as possibilities to adjust the budget during the budget year to offset stronger or weaker than expected demand are limited.

Better expenditure restraint is needed to cope with population ageing and avoid tax increases

Government outlays have outstripped budget projections in the past four years. Such slippage needs to be avoided from here on to assure credibility and to avoid an excessive stimulative fiscal stance. Norway should enhance its medium-term framework, for instance by setting more explicit medium-term expenditure targets – while retaining flexibility for short-term counter-cyclical action if that proves necessary. Within this framework, given the limited potential for further total employment growth, the development of public employment needs to be carefully monitored. Greater budgetary discipline is also needed to cope with the fiscal implications of the ageing of the population and to reduce the tax burden. The 2000 budget paper indicated that tighter expenditure restraint would be needed to avoid a considerable rise in the tax burden of future generations to cope with the ageing of the population. Several policy options concerning pensions and early retirement were outlined in the 2000 Budget but no decisions have been taken on these. Early action would not only alleviate future imbalances but also stimulate labour supply.

Urgent measures are needed to stop the rise in disability and early retirement…

Overall, the Norwegian labour market performance has been excellent in most respects in recent years. The employment rate is the third highest in the OECD, the unemployment rate is among the lowest and long-term unemployment hardly exists. The effective retirement age is still higher than in most other OECD countries. A weak spot,

however, reinforced by policy measures, is the trend increase in the number of disability pensioners and early retirees. As a consequence, the effective retirement age is falling rapidly, making the ageing problem in the coming decades more difficult to deal with. In the budget paper, the government underlined the need to reverse this trend. As noted in previous *Surveys*, a more rigorous medical screening is needed to stem the inflow into the disability scheme. Concerning early retirement, measures discussed in the budget – but not implemented – such as the elimination of pension accruals during the period of early retirement and the reduction of the favourable tax treatment of early-retirement benefits, would go in the right direction. Furthermore, the option to retire early in the public sector with a full pension should be reviewed. Over time, pension and early-retirement schemes need to be put on an actuarially-neutral footing.

... while labour market programmes should underpin restructuring

The government increased active labour market programmes substantially in the 2000 Budget in response to the problems of the shipyards and supply industries caused by the drastic drop in investment in the offshore oil sector. In designing these programmes, current over-capacity in these sectors should be taken into account. The labour market programmes should facilitate restructuring of the sector and shifts of the work force to other sectors and regions. Furthermore, complementing the proposed liberalisation of job placement services with deregulation of the current rigid rules on fixed-term contracts and working hours would improve the economy's adaptability to change.

With respect to product markets, structural reform should focus on reducing subsidies, especially to agriculture...

Liberalisation of the electricity market has been exemplary and there has been progress in deregulation in other areas. However, some product markets remain heavily regulated. Reflecting the predominance of regional policy goals, the agricultural sector is strongly subsidised, at a high cost to consumers and taxpayers. Putting more emphasis on financing through direct budget transfers or tax expenditures would allow convergence of food prices with those in neighbouring countries and reduce the direct costs for consumers, but would not reduce the total economic burden of the sector on the rest of the economy. Support to the

agricultural sector should become less linked to output and better targeted to reduce the overall costs. As in many OECD countries shipbuilding is another heavily subsidised sector, but subsidies will be phased out in 2001, as in the rest of the European Economic Area. Other sectoral and regional subsidies should be assessed and evaluated carefully. Opportunity costs such as the positive effects of lower taxes, should be included in these assessments.

... and on reducing government involvement in telecommunications...

Norway is one of the few OECD countries not to have started privatisation of its public telecommunications operator. A privatisation was planned after the merger between Swedish Telia and Norwegian Telenor. Even though the proposed merger has collapsed, Telenor should eventually be fully privatised. Continued public involvement in this sector would possibly have negative effects on efficiency and would lead to potential conflicts of interest in the regulation of the sector.

... and the oil sector

A comprehensive policy assessment of state ownership in the offshore oil sector is currently under way. Government policy concerning petroleum activities has been fairly successful up to now in extracting the resource rent and has achieved the industrial policy targets of establishing major oil companies and a substantial oil-supply sector. Moreover, involvement in the sector has helped to gain know-how in oil and gas policy matters. However, in current circumstances, partial privatisation of Statoil and more room for new entrants are likely to improve investment incentives for future resource developments, with positive consequences for oil production and the government's oil revenues.

At the same time, privatisation of banks should be speeded up

The considerable extent of state ownership in the two largest commercial banks constitutes a structural weakness of the Norwegian financial sector. An ownership structure more sensitive to market forces and to shareholder value would be important given the increasing institutionalisation of savings, the stronger incentives private ownership provides to raise efficiency and the greater room it gives to geographic diversification. Moreover, it would have the advantage of

avoiding the potential conflicts of interest inherent in state ownership. A majority in Parliament agreed in December 1999 that the government should join its interests as an owner in one entity and that the government should own at least one-third of this entity. Such a strong government presence could hamper further efficiency improvements in the financial sector. The international trend in the banking sector is towards greater size, but Norwegian banks are relatively small, even in a regional context. Most of the savings banks, in particular, are small although there has been some consolidation. Even so, the mutual character of most savings banks may make them less responsive to market forces. Competition will increase further in the financial sector, both internationally and in Norway. Already, most major Norwegian companies have one foreign bank as one of their core banks. The role of foreign banks could increase substantially if Norwegian banks cut costs insufficiently or cannot provide the international services firms require.

The introduction of a dual income tax in 1992 enhanced economic efficiency…

This *Survey* reviews tax policy initiatives and the scope for further reforms. Norway faces the challenge of containing expenditure to at least avoid future increases in the tax burden, which is already above the OECD average. The favourable tax regimes for some industries imply that the fully taxed sectors carry a comparatively higher burden, distorting resource allocation. Norway has made considerable progress in easing the distortions that are typically associated with a high tax burden. However, there has been some backsliding recently. Tax reform initiatives have been geared to considerably broadening the tax bases and lowering top marginal tax rates in the income tax system, with the introduction of a dual income tax system in 1992 being the centrepiece. This system taxes all capital income at a low flat rate and labour income at higher and progressive rates. Such a dual approach is favourable for economic efficiency insofar as the more mobile factor – capital – is taxed less heavily than less-mobile labour. It also facilitates the elimination of differences in tax treatment of different forms of capital income, such as dividends, retained profits, interest income and capital gains, thereby enhancing the neutrality of the system with regard to ways of doing business and funding investment. Moreover, the broadening of the

income tax base, in combination with a strong emphasis on indirect tax and a sizeable tax yield from oil and gas activities, allows Norway to maintain a smaller marginal tax wedge for lower-paid labour than most other OECD countries with a similar tax burden. Such a combination of high average and comparatively low marginal tax rates on labour income has encouraged labour market participation in the recent upswing, in particular of youth and part-time workers.

... but tax planning is inherent to the dual system and subsequent measures have diluted its thrust

The separation of labour and capital income of self-employed and small business owners proves to be the Achilles heel of the dual system. The wide difference in marginal tax rates between capital and labour income for these groups generates strong incentives for tax planning. In particular, the system prompts small business owners to avoid labour taxation by inflating the balance sheet of their company or by exploiting loopholes that allow them to qualify as a "passive" shareholder. It also induces liberal professions to incorporate and thus take advantage of these flaws. These incentives could be eased by a reduction of the difference between the tax rate on capital income and labour income, as recently recommended by an official committee. Unfortunately, the increase in the top rate for labour income by 6 percentage points in 2000, lifting it to well over 50 per cent, is going in the opposite direction. Apparently, the political consensus on tax policy is now favouring redistribution over efficiency considerations. This is also exemplified by the conversion in 2000 of the lowest tax bracket for labour income into a tax-free threshold. By both reducing the average and raising the marginal rate of taxation at the lower end of the pay scale, this measure risks discouraging the supply of temporary or part-time labour by secondary earners. This policy may prove costly both in view of the tight labour market conditions at the moment and the longer-run challenges associated with the ageing of the population.

Scope for reform remains regarding other strands of the tax system

In addition, the 1992 reform left some scope for further change. Most importantly, the differences in the valuation of assets for the wealth and property tax leads to distortions in investment and saving behaviour. Removing the current bias towards accumulation of wealth in dwellings could reduce these distortions and hence improve economic

efficiency substantially. Moreover, further steps to move towards more stable financing for local government would improve the conditions for local authorities' budget management. A shift from local income taxes to transfers from central government combined with, for instance, greater reliance on local property taxes should provide better incentives to contain expenditure at the local level. Furthermore, tax expenditures operating to redistribute income from urban to rural areas in the pursuit of regional policy objectives and to support specific industries should be reviewed. Meanwhile, since the tax take from indirect taxes is already high, the scope for a shift from income to indirect taxes, as recommended for many other OECD countries in order to better prepare for the financing requirements of population ageing, is limited in Norway. In fact, the need to bring some indirect taxes more in line with lower levels of indirect taxation in neighbouring countries to reduce cross-border shopping is a constraining factor. Nevertheless, widening of the indirect tax base by including service activities in the VAT base and by a further "greening" of taxes, for instance by removing exemptions for several industries, is feasible.

Summing up

Since the recession in the early 1990s, Norway's economic performance has been very good in most respects. Economic growth has been among the strongest in the OECD area and Norway is one of the few countries that has restored full employment. At the same time, a major share of the oil revenues has been saved for future generations and government outlays have been insulated from the swings in petroleum revenues through the Government Petroleum Fund. However, fiscal policy smoothed demand insufficiently, causing the economy to overheat in 1998 followed by low output growth in 1999 and 2000. The major challenge for the coming years will be to return to robust non-inflationary growth. The emphasis of monetary policy on low inflation as the fundamental precondition for exchange rate stability should reduce inflation expectations while it will be crucial that fiscal policy offsets any demand pressure. However, if moderate wage agreements for 2000-01 are not reached, a renewed tightening of macroeconomic policy will be inevitable. Additionnal labour and product market reforms are needed to

boost productivity over the medium term and help prepare the economy for the post-oil era. Measures are urgently required to reduce the inflow into the disability pension and early-retirement schemes. Further deregulation and privatisations are essential to increase productivity. Concerning the tax system, the 1992 reform has had favourable effects but subsequent measures have diluted its thrust. Better expenditure restraint will be needed to at least avoid future increases in taxation. Advancing in these directions would help to boost living standards over the medium term.

I. Macroeconomic performance

In 1999, the Norwegian economy has cooled off after a period of exceptional strength that led to overheating. Tight macroeconomic policy, the squeeze on corporate profitability from sharp wage rises in previous years and the completion of major public and offshore investment projects induced an investment-led slowdown. Growth of mainland GDP was low while the strong rise in employment came to an end. Unemployment, however, has risen little so far and has remained among the lowest in the OECD. Driven by the oil price recovery and tight monetary policy, the Norwegian krone has appreciated substantially since its trough in December 1998 and has recovered roughly to the range of 8.1 to 8.4 kroner to the euro. This, in combination with more moderate wage rises and the economic slowdown, should lead to a decline in price inflation, although the positive inflation differential with the euro area is likely to remain sizeable in the short term.

As the risks of currency depreciation and accelerating inflation receded, prospects for weak output development and lower inflation induced cuts in the key central bank rates by 2.5 percentage points since the peak in early 1999. In the light of the continuing slowdown, the government aims at a neutral fiscal stance for 2000. Reflecting the easier macroeconomic policy stance, and with households, enterprises and banks in a much healthier financial position, the downturn is likely to be limited to a growth pause and will not lead, as at the end of the 1980s, to a pronounced recession. However, the recovery is not expected to gather momentum before 2001, as mainland investment is likely to continue to drop while, after years of strong growth, investment in the offshore oil sector is projected to contract dramatically.

A major challenge will be to secure a wage agreement for 2000-01 that is in line with developments in the main trading partners. This would be a crucial step towards reducing cost pressures and price inflation to the euro area average. New policy challenges may arise when the economy picks up in 2001 as the slack in the economy will be limited and bottlenecks could reappear quickly.

The following sections discuss recent macroeconomic developments and policy responses in more detail. The chapter concludes with a review of the prospects for 2000 and 2001, together with an assessment of both the risks surrounding these projections and the main challenges for macroeconomic policy in the years ahead.

Macroeconomic developments: the economy is cooling off

Investment contracts sharply

The current slowdown is investment-led and has hit the manufacturing and construction sectors most (Table 1). After expanding at an average annual rate

Table 1. **Demand and output**

Percentage change from previous period, volume

	1996 Current prices NOK billion	1997	1998	1999[1,2]	Annual rate[2] 1998 2nd half	Annual rate[2] 1999 1st half
Private consumption	490.3	3.7	3.1	1.8	1.6	2.1
Government consumption	206.8	2.8	3.7	1.8	2.3	1.6
Gross fixed capital formation	216.2	15.1	8.1	–5.9	13.5	–22.2
Public sector	31.6	18.1	3.4	–4.0	–7.4	–5.4
Petroleum production and pipeline transport	51.3	15.6	25.7	–5.6	56.3	–36.4
Residential	27.2	8.2	–0.6	–4.1	–10.8	–0.7
Other private[3]	106.0	15.7	3.1	–7.0	6.0	–21.7
Stockbuilding[4,5]	15.9	0.3	0.9	0.2	–0.7	0.3
Total domestic demand	929.2	6.4	5.4	0.0	3.9	–4.5
Exports of goods and services	414.4	5.7	0.5	–1.3	–7.8	2.1
Traditional goods	155.8	8.0	3.4	0.8	0.6	–0.8
Crude oil and natural gas	156.7	2.1	–3.8	–2.2	–14.6	3.3
Ships and platforms	9.1	11.5	5.0	–20.3	–68.3	170.9
Services	86.2	7.2	2.1	–1.4	–0.6	–4.4
Imports of goods and services	327.1	12.0	9.1	–3.5	2.6	–10.5
Foreign balance[4]	87.4	–1.6	–2.9	0.8	–4.2	4.9
GDP	1 016.4	4.3	2.1	0.8	–0.5	0.5
of which: Mainland	833.0	4.4	3.3	1.1	1.4	0.3

1. First three quarters compared with corresponding period one year earlier.
2. Seasonally adjusted data.
3. Including shipping.
4. Contribution to GDP growth.
5. Includes statistical discrepancy.

Source: Statistics Norway and OECD Secretariat.

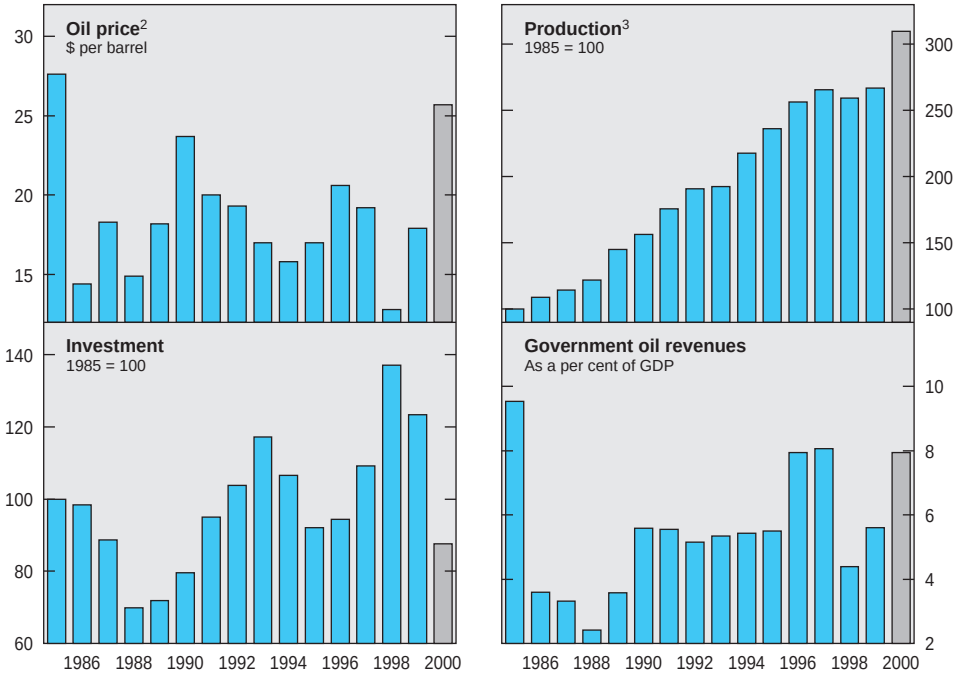
Box 1. The influence of the offshore oil sector on the mainland economy

The government has been relatively successful in insulating government outlays from swings in petroleum revenues through the Petroleum Fund, which aims to preserve an equitable share of the current oil and gas revenues for future generations. Nevertheless, the mainland economy can be strongly influenced by the offshore oil sector; in 1998, investment by this sector, which has a direct effect on mainland GDP, accounted for a third of total business investment. With such a high weight, the volatile oil investment flows have a major influence on mainland production, despite the high import content of oil investment of around a third. In 2000, the decline in oil and gas investment is estimated to yield a direct gross negative contribution of almost 2 percentage points to mainland domestic demand and a direct net (import-corrected) negative contribution of almost 1½ percentage points to mainland GDP. A smaller, but not negligible, direct effect of the oil sector on the mainland economy is through the offshore supply sector (maintenance of platforms and vessels, logistics, catering, operation of processing and landing facilities). In 1998, more than 20 000 persons were employed in this sector. The cost-cutting drive by oil companies initiated by the low oil prices in 1998, in combination with technological changes, will have a substantial negative effect on production and employment of the supply industry in 2000. Furthermore, the oil sector influences confidence in the state of the economy. For instance, higher oil prices could lead to more optimism among consumers, leading to less saving and more consumption.

Apart from these short-term effects, the mainland economy is influenced by the greater leeway to pursue anti-cyclical fiscal policy provided by the oil revenues. The room for manoeuvre is larger than in other OECD countries due to the strong asset position of the government associated with the oil wealth. Furthermore, the oil revenues and their expected drop in the coming decades has given a stronger long-term flavour to the setting of economic policy. As a consequence, Norway was one of the first countries to present inter-generational accounts in the budget papers. Finally, the oil sector has had a significant influence on the structure of the mainland economy: it has led to a larger sheltered sector, while the share of "traditional" export-oriented and import-competing activities has declined gradually.

of 11 per cent over the three previous years, total gross capital formation dropped by an estimated 9 per cent in 1999. All categories declined. The completion of major projects such as the National Hospital contributed to a large extent to the decline in public sector investment. Mainland business fixed investment contracted by an estimated 11 per cent in 1999, largely reflecting the slowdown in domestic demand, the completion of the main Oslo airport at Gardermoen, weak corporate profitability due to the sharp rise in labour costs, low non-oil commodity prices (especially aluminium and paper) and high interest rates. Manufacturing investment is estimated to have dropped even more strongly, by around 25 per cent. Investment in offshore oil and gas activities – which accounts for one-fourth of total fixed investment – declined by an estimated 10 per cent in 1999, offsetting only partly the strong rise in 1998 (Box 1). The fall in gross capital

Figure 1. **The oil sector**[1]



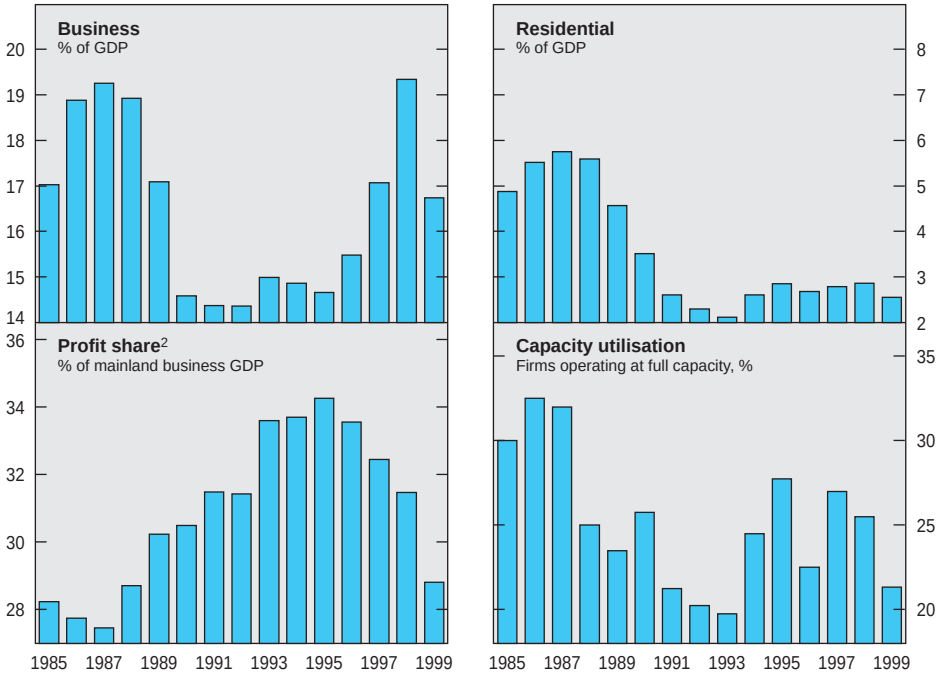
1. Data for 2000 are OECD Secretariat estimates.
2. Brent Blend. The estimate for 2000 shows the price of the latest available month.
3. Petroleum activities and ocean transport.

Source: Statistics Norway and OECD Secretariat.

formation on the Norwegian continental shelf – which is expected to continue and even deepen in the year 2000 – reflects an adjustment to more normal levels of investment activity after a period of very strong expansion (Figure 1).[1]

Residential investment, which already contracted slightly in 1998, is estimated to have declined by 9 per cent in 1999 (Figure 2). Housing starts began to fall well before the Autumn of 1998, when interest rates were hiked by 3 percentage points and before the slowdown in disposable income growth, indicating that supply-side factors played a considerable role. Investment was hampered by insufficient supply of skilled labour, lack of suitable building sites and administrative delays in processing construction permits. Improving orders indicate, however, that residential investment could pick up already in 2000. House prices, after rising vigorously for five years, fell in the second half of 1998 following the

Figure 2. **Business and residential investment**[1]



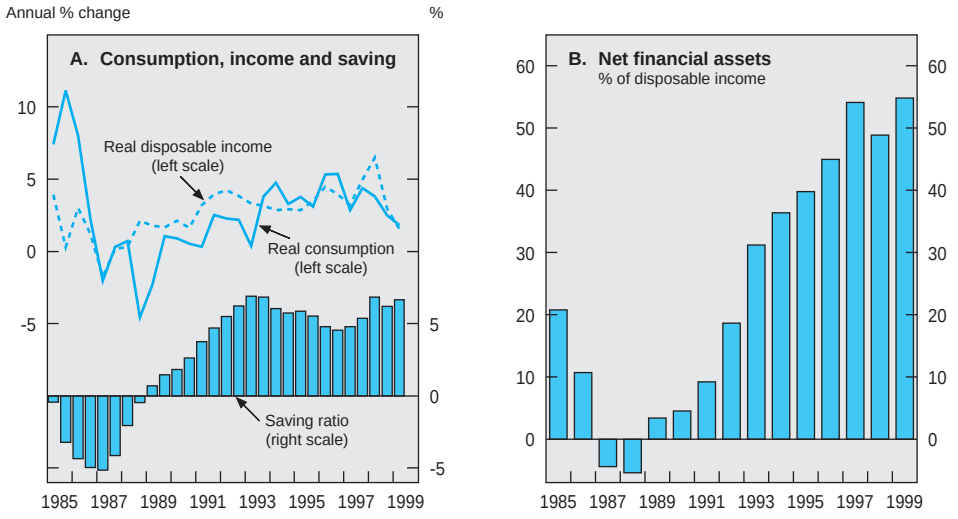
1. Data for 1999 are estimates.
2. Mainland business GDP less compensation and net indirect taxes.
Source: Statistics Norway; OECD, *Main Economic Indicators* and OECD Secretariat.

pronounced increase in mortgage rates, but have recovered strongly since then, increasing by an estimated 10 per cent in 1999. Real house prices have reached a record high, but the ratio between house prices and disposable income has stayed below the peak of the 1980s. The rapid recovery may have been induced by the expectation that the surge in interest rates was only temporary. The robust rise in house prices over recent years largely reflects strong employment growth and substantial increases in real wages, in combination with a limited supply of new dwellings. Migration to the main urban regions has increased the demand for housing in large cities, explaining, in conjunction with the lack of building sites, the above average rise in these areas: since the fall in house prices bottomed out in 1993, prices have risen by 160 per cent in the Oslo-region, by 90 per cent in the other major cities (Bergen, Trondheim and Stavanger), but by only about 70 per cent in the rest of the country.

Private consumption moderates the cyclical downturn

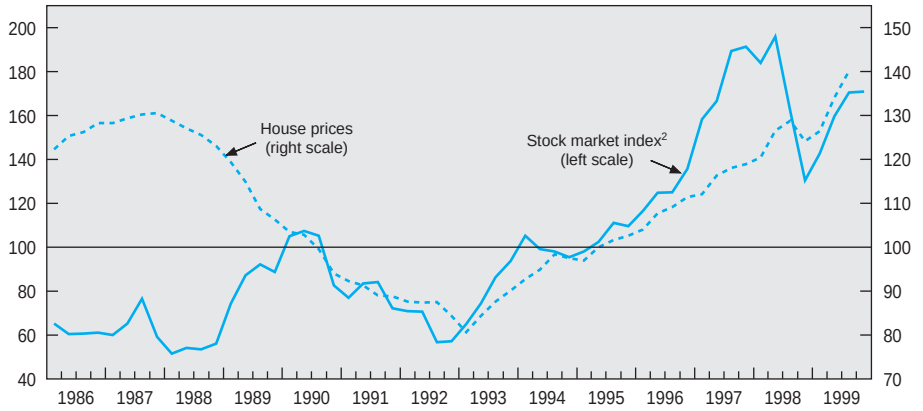
Private consumption softened in 1999 but still mitigated the downturn (Table 1 and Figure 3). Growth subsided to an estimated 1¾ per cent, from slightly above 3 per cent in 1998, due to stalling employment, a lower real wage rise and higher interest payments on mortgages and other consumer debt. The deceleration was most pronounced for durable goods, especially cars,[2] due to higher interest rates on consumer credit and substantial car replacement in previous years. After accelerating for several years, households' real disposable income growth receded from 4¾ per cent in 1998 to an estimated 2 per cent in 1999. The saving ratio is estimated to have remained broadly constant despite strong consumer confidence – the drop in confidence at the end of 1998 was only temporary – and the improvement in households' balance sheets due to the brisk house price rise and the rebound of the Norwegian stock market. Reflecting the relatively high saving ratio and steeply rising asset prices (Figure 4), households' net financial assets have increased to an estimated 55 per cent of disposable income, thus exceeding considerably the previous peak of 21 per cent in 1985 and contrasting sharply with the 1988 trough of negative net financial wealth of 5 per cent.

Figure 3. **Household consumption**



Source: Statistics Norway; Ministry of Finance and OECD, *Main Economic Indicators*.

Figure 4. **Asset prices**
Real prices,[1] 1990 = 100



1. Deflated by the consumer price index.
2. Oslo all share index, monthly average.
Source: Statistics Norway and Ministry of Finance.

Exports weaken while imports drop sharply

The Norwegian export structure is unusual, as oil, gas and fish products account for almost 40 per cent of total exports. In volume terms, oil and gas sold abroad – representing one-third of total exports – did not recover in 1999 from a drop in the previous year caused by technical problems in existing fields and delayed production starts at some new fields (Table 2). The government's decision to reduce oil production by 200 000 barrels per day from 1 April 1999 onwards (a cap of nearly 7 per cent compared to the production prognosis on an annual basis) prevented a recovery.[3] In December 1999, the government decided to delay the lifting of the production cap until April 2000. The broadly unchanged oil and gas exports in 1997-99 contrast with the total increase of almost 40 per cent in the three previous years. The exports of fish products – the second largest export category after oil and gas – has almost doubled over the 1990s. This reflects a strong increase in fish catches as well as soaring sales of farmed salmon to Europe, Asia and the United States.[4] In 1999, however, the total exports of fish products showed a weaker development, because fish catches, after booming in 1998, declined due to smaller quotas and fish stocks. On the other hand, exports

Table 2. **Exports by commodity**[1]

Percentage volume changes

	1998 % of total exports[2]	1993	1994	1995	1996	1997	1998	1999[3]
Total exports	100.0	3.2	8.7	4.3	9.3	5.7	0.5	–1.4
Goods	72.7	3.4	11.0	6.3	10.5	5.2	0.1	–1.3
Crude oil and natural gas	30.1	5.9	11.9	9.2	13.7	2.1	–3.8	–2.1
Traditional products[4]	38.9	3.2	12.5	4.5	10.0	8.0	3.4	0.7
of which: Manufacturing	36.1	3.2	13.2	3.6	10.4	8.5	3.3	0.2
of which:								
Food products, beverages and tobacco	3.7	10.0	18.3	2.7	11.6	6.5	2.5	1.7
Basic metals	8.6	2.6	9.6	–4.7	13.2	9.8	3.4	0.7
Machinery and other equipment	8.8	–1.4	8.5	15.2	13.1	8.0	11.1	1.5
Services	27.3	2.4	2.7	–1.3	5.8	7.2	2.1	–1.5
Gross receipts, shipping	15.5	–2.7	4.5	3.9	0.8	2.4	3.3	–0.9

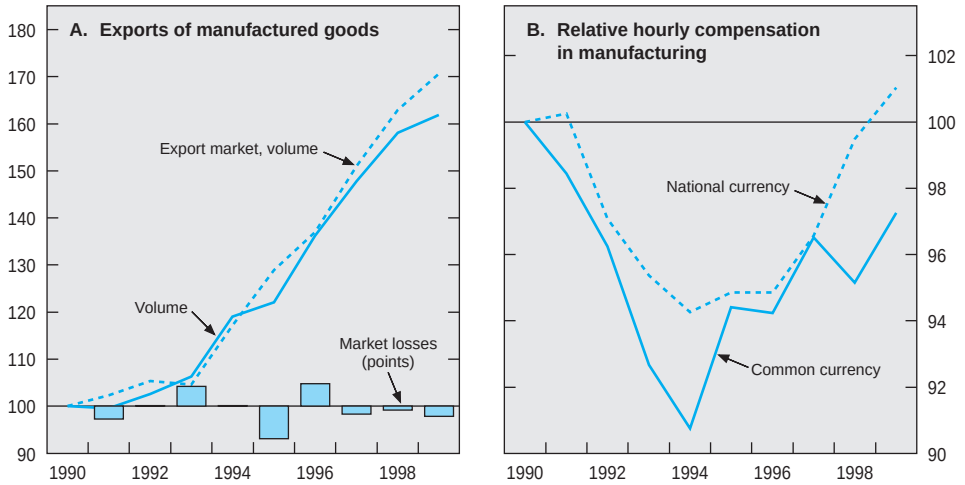
1. On a national accounts basis.
2. Current prices.
3. Average of annualised growth rate for first three quarters.
4. Excluding petroleum, oil rigs and ships.

Source: Statistics Norway.

of farmed fish continued to expand rapidly, although salmon exports to the European Union had to be reduced as prices fell below the minimum prices agreed with the European Union (EU).

Weaker global demand slowed the expansion of Norwegian manufactured exports from 3.3 per cent in 1998 to 0.2 per cent in the first three quarters of 1999, after average growth of nearly 9 per cent during the years 1994-97 (Figure 5, panel A and Table 2). The deterioration in labour cost competitiveness caused a further loss in market share in 1999. On average, exports lagged behind market growth by 1 percentage point per year between 1993 and 1999. This market share loss is relatively small compared with the deterioration in cost competitiveness of 2¾ per cent per year over the same period.[5] This partly reflects the strong improvement in competitiveness in previous years (Figure 5, panel B). Moreover, strong competition has meant that manufactured export prices could not be raised in line with cost developments, limiting the impact on the market share, but reducing profitability. In 1999, the deterioration in the competitive position was mainly caused by the appreciation of the krone *vis-à-vis* the euro, after an improvement in 1998 due to the depreciation of the currency, while in previous years, stronger wage increases were the main reason for weakening competitiveness. Notwithstanding the market loss, manufactured exports increased by 50 per

Figure 5. **Export performance and cost competitiveness**
1990 = 100



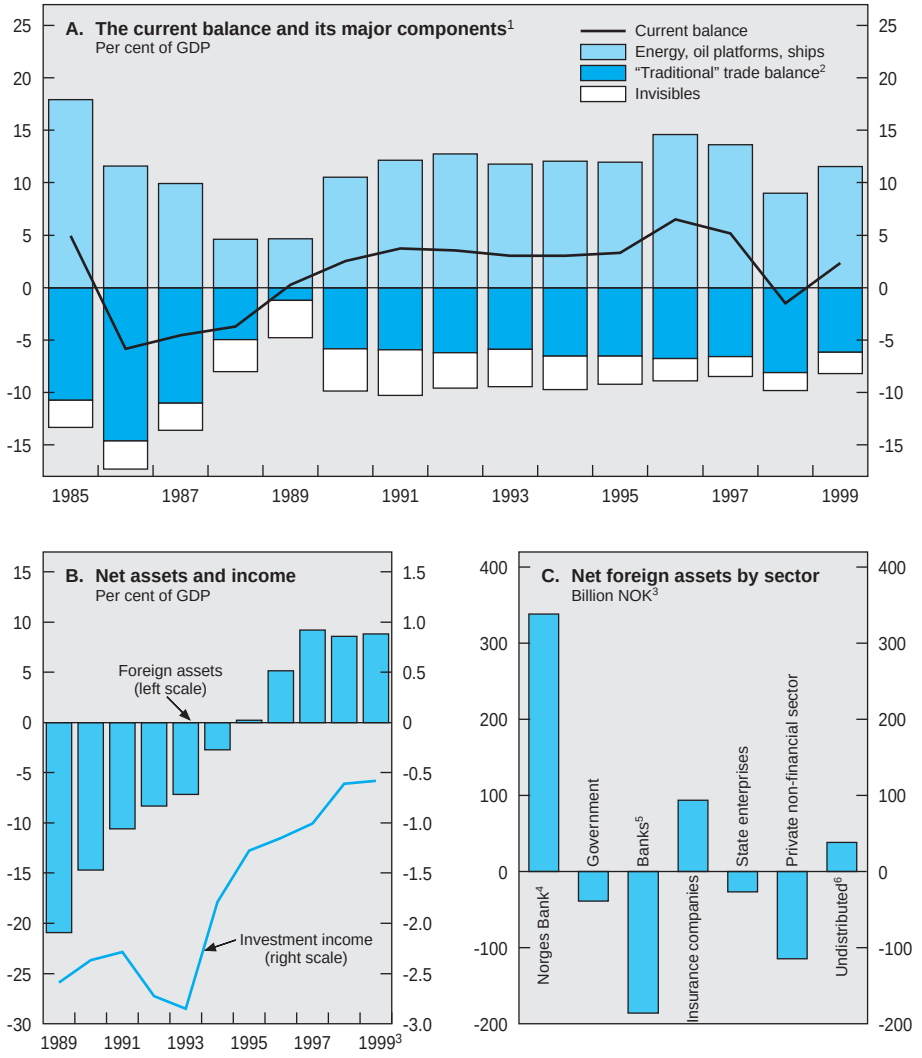
Source: Statistics Norway, Ministry of Finance and OECD Secretariat.

cent between 1993 and 1999. Growth was most pronounced for machinery and for chemical products, while it was below average for basic chemicals, refined petroleum products and basic metals.

Imports of goods and services decreased in 1999 by an estimated 2¼ per cent, the first drop since 1989. This was mainly due to the fall in capital goods imports, while weak growth in imports of consumption goods reflected the decline in car purchases. As a result of the drop in imports, the contribution of net exports to GDP was positive in 1999, after sizeable negative effects in the two preceding years.

Current account developments have largely mirrored the sharp oil price swings in 1998 and 1999. In 1998, the combination of the lower oil price and strong imports led to the first deficit since 1988 (Figure 6). The oil price recovery in 1999, together with the much lower imports of investment goods, moved the current account back into surplus, reaching an estimated 4 per cent of GDP. The service balance was in deficit for the second year running, after surpluses had been recorded since 1990. The swing from a comfortable surplus of nearly 1 per cent of GDP in 1996 to a deficit of the same magnitude in 1999 has mainly been due to a

Figure 6. **External position**



1. Break in series in 1990. Figures for 1999 are estimates based on data up to October.
2. Total exports and imports excluding petroleum, oil rigs and ships.
3. At 30 September 1999.
4. Including the Government Petroleum Fund.
5. Commercial and savings banks, mortgage and financial companies.
6. Including errors and omissions.

Source: Statistics Norway, Norges Bank and OECD Secretariat.

weaker travel balance and stronger imports of financial and business services. Net investment income revenues remained negative despite substantial net foreign assets and lower dividend payments by Norwegian companies due to the economic slowdown. However, net investment income paid has rapidly declined from the 2 per cent of GDP peak in 1993 to almost zero in 1999. Net foreign assets are expected to have increased in 1999 as the impact of the current account surplus was only partly offset by revaluation losses due to the appreciation of the krone. In September 1999, net foreign assets amounted to NOK 104 billion, 9 per cent of GDP.

Employment is flat but unemployment edges up only marginally

With mainland production growth low, employment practically stopped increasing in 1999, after surging by 2¼ per cent per year on average since 1993 (Table 3 and Figure 7). Sectoral developments diverged considerably reflecting differences in production developments. The number of jobs and vacancies declined in the manufacturing sector while temporary lay-offs increased strongly. In building and construction, employment was fairly flat and the number of vacancies declined somewhat. In contrast, labour demand continued to be buoyant in the service industries and in the public sector, especially in education and health care. Strong labour demand led to a further shift from temporary to permanent jobs.[6]

Labour supply continued to react strongly to the employment situation. In previous years, labour supply rose steeply in reaction to rapid job creation and the participation rate reached a record high. In 1999, the lack of job creation was accompanied by a stable labour force and a slight drop in the participation rate. Policy measures such as the early retirement scheme and the cash benefits scheme for parents with young children contributed to restrain labour force growth.[7] Moreover, the completion of major construction projects and the restructuring in the offshore oil sector led to the departure of foreign workers. As a consequence of the elastic labour supply, flat employment only resulted in a marginal increase in registered unemployment from May onwards.[8] In the third quarter, the unemployment rate stood at 2.9 per cent (on the basis of the labour force survey), far below the average rate in the rest of Europe and among the lowest in the OECD.

Although the annual unemployment rate has remained unchanged, labour market tensions have eased somewhat. In December 1999, unfilled vacancies were down by 17 per cent from the peak reached in August 1998, while the labour supply has become less of a production bottleneck – only 4 per cent of manufacturing enterprises reported it as a bottleneck in the third quarter of 1999 compared with 11 per cent in the first quarter of 1998. Nevertheless, the labour market situation can still be characterised as tight. One-third of all companies

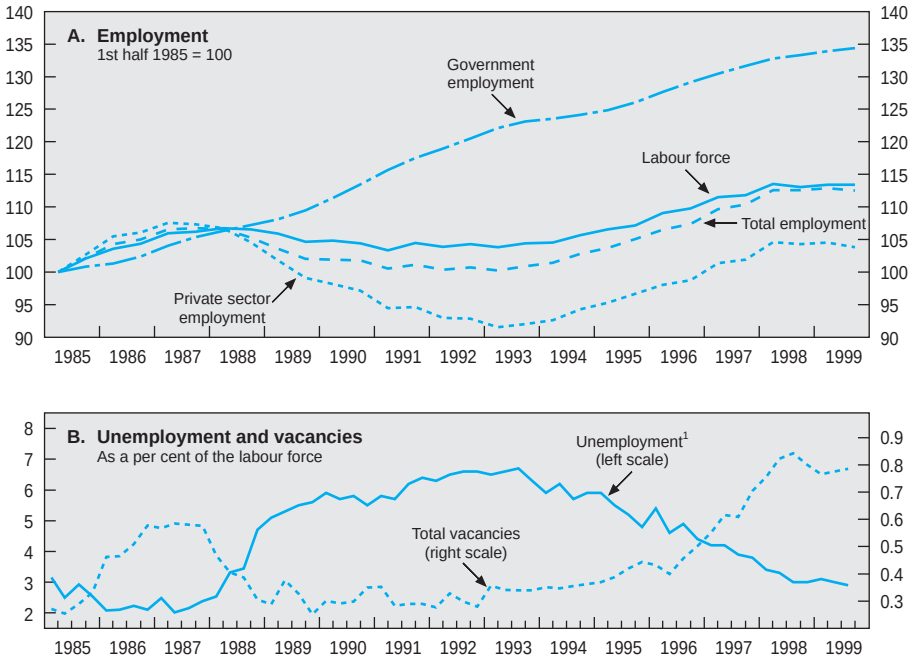
Table 3. **Labour market developments**

	1993 Thousand persons	1994	1995	1996[1]	1997	1998	1999[2]
		Percentage change					
Labour force	2 131	1.0	1.7	2.4	2.0	1.5	0.1
Employment	2 004	1.5	2.2	2.5	2.9	2.4	0.1
By sector[3]							
Private sector	1 405	1.2	1.6	2.6	2.0	1.5	1.5
of which: Construction of oil platforms and ships	33	–2.7	3.7	–2.7	3.7	3.8	–0.7
Government	632	1.2	1.6	2.6	2.0	1.5	1.5
Central	154	–2.1	–0.7	1.5	0.5	–0.5	0.5
Local	478	2.3	2.3	2.9	2.5	2.0	1.8
Labour productivity	..	4.2	1.7	2.8	1.4	–0.2	0.5
		Per cent of labour force in same category					
Unemployment	127	5.4	4.9	4.8	4.1	3.2	3.2
By sex							
Male	77	6.0	5.1	4.8	3.9	3.2	3.3
Female	50	4.7	4.6	4.9	4.2	3.3	3.1
By age group							
16-24	42	12.6	11.8	12.4	10.9	9.5	9.7
25-54	78	4.5	4.1	3.8	3.2	2.3	2.4
55-74	7	2.5	2.6	2.4	1.7	1.9	1.2
By duration							
Persons unemployed for more than 6 months (as a per cent of total)	49	45	40	31	24	15	12
		Per cent of working age population[4]					
Active labour market programmes							
Participation	..	1.7	1.3	1.1	0.7	0.4	0.2
Including special measures[5]	..	3.4	2.9	2.5	2.0	1.7	1.6

1. Break in series for Labour Force Survey data.
2. Estimate based on average of first three quarters of data except for total labour force, total employment, productivity and total unemployment which are OECD Secretariat estimates.
3. National accounts data. For 1999, growth is based on the first three quarters over the same period of the previous year.
4. Population aged 16-74.
5. Figures before 1996 include occupationally disabled in waiting phase or undergoing assessment.

Source: Statistics Norway, Ministry of Labour and Government Administration, and OECD Secretariat.

Figure 7. **Labour market**



1. Break in series in January 1996 due to a broadening of the definition of unemployment in the Labour Force Survey which is estimated to have raised the unemployment rate by 0.5 percentage point.

Source: Statistics Norway and OECD Secretariat.

reported recruitment problems in the second quarter of 1999 and labour shortages even increased somewhat in the health sector. Moreover, 25 per cent of full-time employed were working overtime in the second quarter of 1999, only 2 percentage points less than a year earlier. In addition, the unemployment rate has remained below 4 per cent, the Secretariat's estimate of the NAWRU (Non-Accelerating Wage Rate of Unemployment).[9]

Wage inflation slows down

After the sharp wage rise in 1998 – wages in December were 8 per cent higher than a year earlier – contractual wages changed little in 1999. The wage settlements implied an average pay rise of 0.4 percentage point in 1999.[10]

However, given the substantial carry-over from 1998 of around 3¼ percentage points, the wage drift must, on average, be lower in 1999 than in 1998 to attain the estimated wage growth of 5 per cent in 1999 (Table 4 and Figure 8). This estimate is slightly above the 4½ per cent rate recommended by the *Arntsen Committee*.[11] As a result of the moderate pay settlement, annual wage inflation in 1999 slowed by 1¼ percentage point compared with 1998. The deceleration in unit labour costs was, however, smaller due to the weak productivity development induced by the slowdown. Unit labour cost increases in the manufacturing sector and the business sector as a whole, are estimated at close to 5 per cent, more than twice as high as in the euro area.

Price inflation remains above the euro area average

Consumer price inflation was stable in 1999 with the impact of the appreciation of the Norwegian krone offset by the still rapid rise in labour costs. As a consequence, inflation has remained well above the rate in the euro area (Figure 8, panel C and Table 4).[12] Price rises in services dominated by labour cost developments came down during 1999 but, with a rise of 5 per cent in December, remained high (Figure 8, panel A). The direct impact of the higher oil prices on consumer prices is limited to around 0.1 percentage point, as taxes account for around 80 per cent of the petrol price. Underlying inflation in 1999 is estimated at 2¼ per cent, unchanged from the previous year.[13]

Table 4. **Prices, wages and costs**

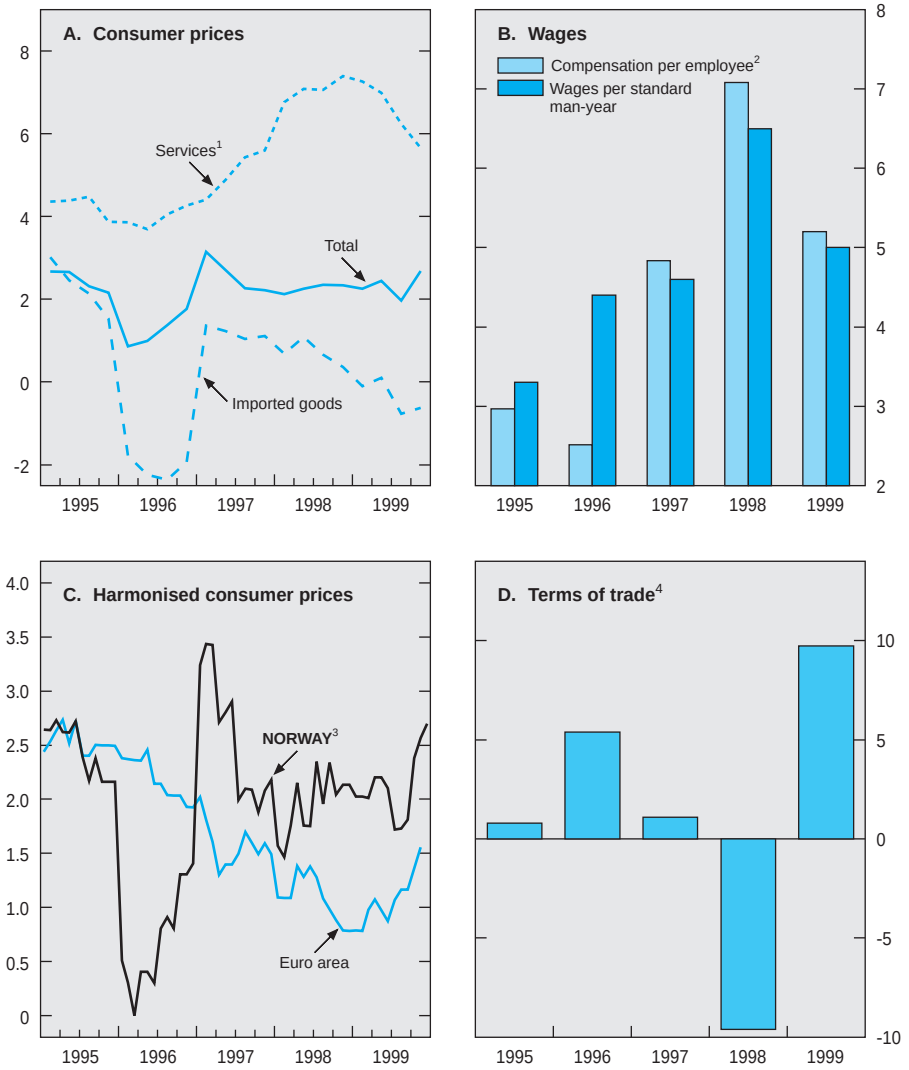
Percentage change

	1992	1993	1994	1995	1996	1997	1998	1999[1]
Consumer price index	2.3	2.3	1.4	2.5	1.3	2.6	2.3	2.3
Harmonised consumer price index	..	..	..	..	0.7	2.6	2.0	2.1
Hourly earnings per employee	2.7	3.5	3.0	4.6	4.7	5.0	6.2	..
of which:								
Mainland	2.7	3.3	3.0	4.6	4.7	4.9	6.2	..
Manufacturing	2.8	2.8	2.9	4.6	4.7	4.5	5.6	..
Personal services	2.9	3.0	2.8	4.7	4.4	4.5	5.5	..
Public sector	2.8	3.4	3.2	4.6	4.6	4.9	5.9	..
Relative compensation per hour[2]								
Local currency	–3.2	–1.7	–1.2	0.7	0.0	1.8	3.1	1½
Common currency	–2.2	–3.8	–2.0	4.0	–0.2	2.4	–1.4	2¼

1. Estimate.
2. Manufacturing.

Source: Norges Bank, Ministry of Finance, Statistics Norway and OECD Secretariat.

Figure 8. **Wage and price developments**
Annual percentage change



1. Services with wages as the dominating price factor.
2. Business sector.
3. Non-harmonised data for 1995.
4. The change in the ratio of prices of exports/prices of imports for goods and services.

Source: OECD, *Main Economic Indicators*, Statistics Norway and OECD Secretariat.

The prices of other domestic expenditure categories are estimated to have risen faster due to a smaller import and a higher labour content, with the strongest price rise for general government consumption (3.9 per cent in the first three quarters of 1999 compared with a year earlier). Due to easing wage inflation and narrowing profit margins caused by the slowdown, the rise in the mainland GDP deflator decelerated from 4¼ per cent in 1998 to around 2½ per cent in 1999. By contrast, the overall GDP deflator, having been negative in 1998, surged in 1999 by an estimated 6½ percentage points to 6 per cent, reflecting the sharp rise in oil prices.

Norway in the 1990s: strong performance ends in a downturn

With GDP per capita increasing by 30 per cent in real terms, the Norwegian economy performed well in the 1990s (Figure 9 and Table 5). Output and employment growth were considerably stronger than in the European Union, inflation was slightly lower on average – although somewhat higher in recent years – while the government budget and the current account were in surplus, largely reflecting strong oil revenues. In the early 1990s, the Norwegian economy recovered only slowly from the severe recession of the end of 1980s as consumers, whose wealth was severely hit by the drop in house prices, rebuilt wealth positions and were reluctant to spend. Due to low profitability and weak domestic demand, it took till 1993 before mainland business investment started rising. The impetus for the recovery came from improved competitiveness (reflecting an effective depreciation of the currency and moderate wage rises), rising oil production and an expansionary fiscal policy made possible by the considerable

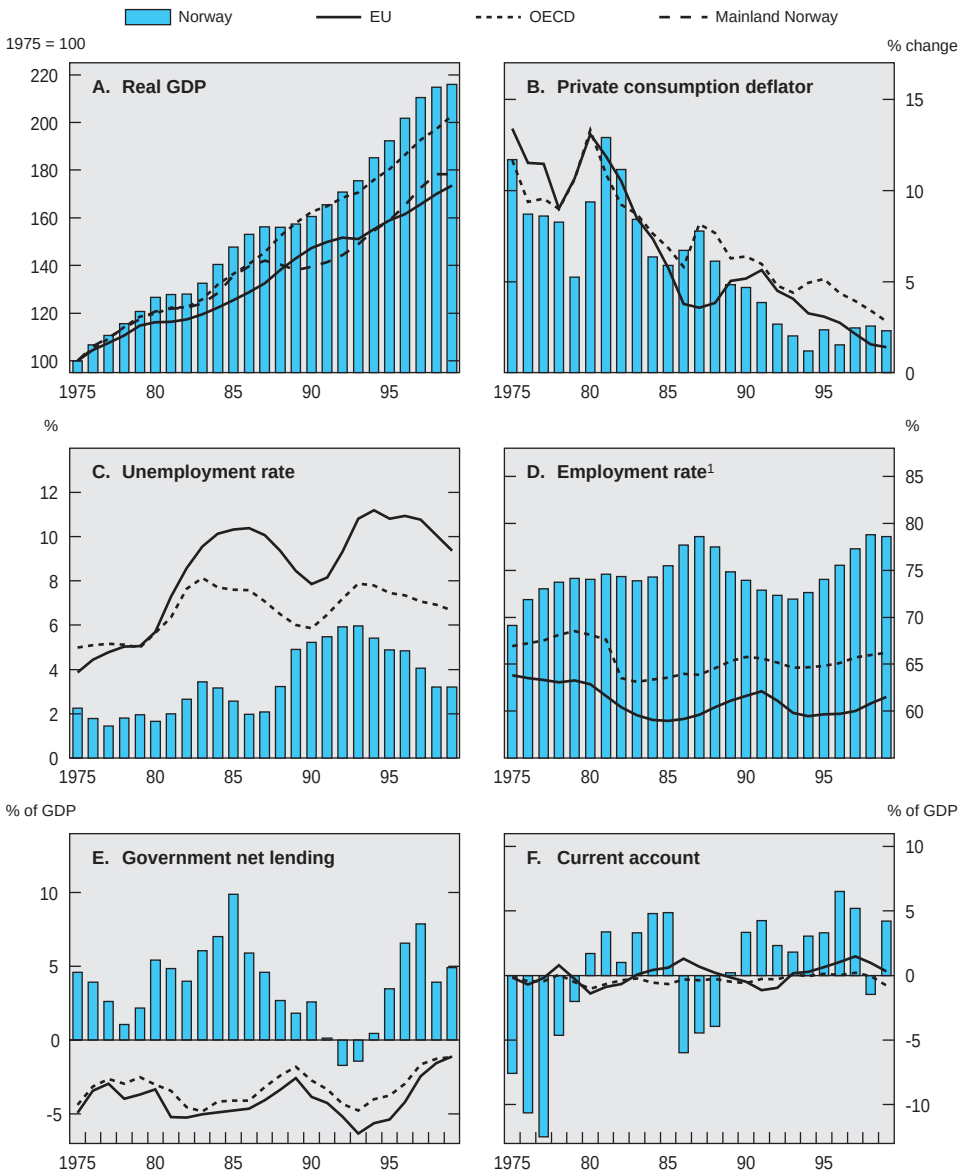
Table 5. **International comparison of key figures**

1990s average

	Units	Norway	Other Nordic countries[1]	Euro area	EU	OECD
Real GDP	% growth	3.2	1.7	2.0	2.0	2.5
Private consumption deflator	% growth	2.6	2.9	3.0	3.3	4.6
Employment	% growth	0.9	–0.5	0.3	0.3	0.9
Unemployment rate	% of labour force	4.8	8.5	10.5	9.9	7.1
Government net lending	% of GDP	2.7	–2.4	–3.9	–4.0	–3.0
Current balance	% of GDP	3.3	0.8	0.5	0.2	–0.2

1. Denmark, Finland and Sweden.
Source: OECD (1999), *Economic Outlook* No. 66.

Figure 9. **Key indicators in long-term and international perspective**



1. Employment as a per cent of working age population (aged 16-64).
Source: OECD Secretariat.

government oil revenues. In 1994, the upturn gathered momentum and the strong cyclical upswing that followed led to a drop in the unemployment rate from a peak of 6 per cent in 1993 to 3 per cent in 1999 and a rapid expansion of the labour force. The recovery was accompanied by a major fiscal tightening, also by international standards. The OECD Secretariat estimate of the cyclically-adjusted mainland deficit narrowed from 6.1 per cent of mainland potential GDP in 1993 to 0.2 per cent in 1999. On the other hand, monetary policy remained relaxed until mid 1998. The Norwegian economic cycle was out of phase with that of the euro area. After recovering from the 1988-92 slump, the Norwegian economy outperformed the euro area, except for 1999. After registering considerable slack in the early 1990s, the output gap of the Norwegian economy became positive already in 1996 and by 1998 excess demand had widened to 3¼ per cent. This is in stark contrast to the euro area where excess demand in 1990 had completely melted away by 1992 and the output gap remained negative for the rest of the decade.

The Norwegian upswing ended in the classic way: excess supply vanished due to strong economic growth, and overheating became a major problem. This was followed in 1999 by a slowdown due to lower profitability, a deterioration of competitiveness and a belated tightening of fiscal and monetary policy. The offshore sector exacerbated the end of the cycle, as oil investments provided a substantial demand impulse in 1997-98, but then impacted negatively in 1999 and 2000.

Macroeconomic policy: the framework regains credibility

The macroeconomic policy framework, the so-called *Solidarity Alternative*, has three pillars: *i)* co-operation between the social partners at the central level aimed at moderate cost inflation and full employment; *ii)* monetary policy oriented towards a stable krone exchange rate against the euro; *iii)* fiscal policy aimed at stabilising the economy, with budget surpluses associated with the government's oil revenues reinvested in financial assets abroad. It was laid down in a policy document in August 1992, at a time when there was broad consensus on the need to improve cost competitiveness, through a lower wage growth than in the main trading partners, in order to end the recession. This set-up reflects the long-standing Norwegian tradition of centralised wage bargaining with government often playing an active role in the negotiations. While the economic situation has changed substantially since 1992, government, unions and the employers' organisations have repeatedly confirmed their support for this framework. Most recently, the *Arntsen Committee* – with high level representatives from social partners and government – underlined the need to continue to work along these lines.[14]

The framework came under heavy strain in 1998 as wage rises were excessive, fiscal policy did not prevent an overheating of the economy and the exchange rate dropped substantially. By prompting the sharp rise in interest rates, which may have nipped a vicious wage-price spiral in the bud, the weakening of the krone may even appear fortuitous but it masks the failure of fiscal policy to cool off a booming economy in a timely fashion. Credibility of the framework was already hurt in 1997 by a too easy fiscal policy stance in the face of rapidly vanishing slack and by interest rate cuts in response to an appreciation of the krone that led to a pro-cyclical monetary policy stance. In 1999, the macroeconomic policy framework regained some credibility with the agreement on a "zero" wage round and fiscal policy tightening, which was, however, less than originally envisaged.

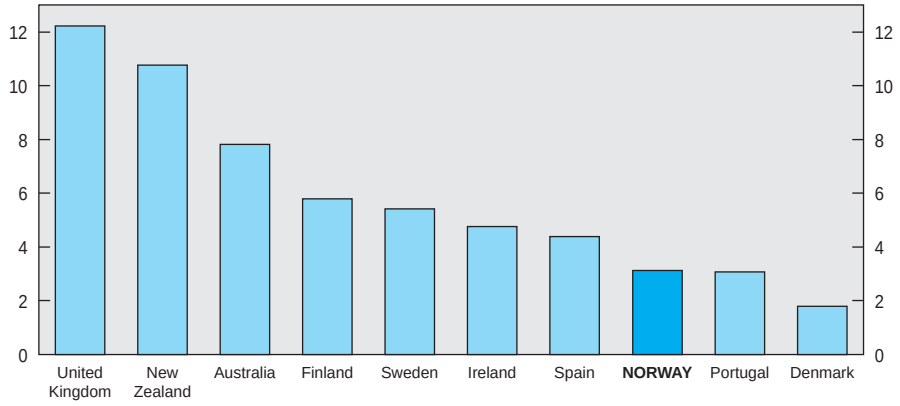
The policy framework will be tested again in early 2000, as negotiations on a new two-year wage agreement will start while substantial tensions still exist in parts of the labour market. The social partners and government representatives, via the *Arntsen Committee*, agreed early in 1999 to bring down wage growth to the level of the trading partners in 2000. However, on average in the past, wage rises in the first year of the two-year agreement were 1½ percentage points higher than in the second year. Moreover, as the Norwegian central bank pointed out in its Inflation Report, a low wage round is usually to some extent compensated for in subsequent years. On the other hand, the manufacturing sector is traditionally the wage-leader in Norway and is the sector most hit by the current slowdown.[15]

Despite the broad support for the current macroeconomic policy framework, another committee (the *Holden Committee*) has been created to review the *Solidarity Alternative*. Apart from the poor results in 1998, the main reason for the review is the trend shift from manufacturing to services which undermines the strength of the two main negotiating organisations, the LO and the NHO. The LO mainly represents blue-collar workers in the manufacturing sector and has a very weak membership base among white-collar workers in the service sector. As recommended by the *Arntsen Committee*, the diminishing representation of the LO and NHO led already in 1999 to the participation of other labour market organisations in the *Technical Reporting Committee on Income Settlements*, the committee preparing the central wage negotiations. The *Holden Committee* has been given a broad mandate, comparable with that of the *1992 Employment Committee* that formulated the Solidarity Alternative. In its report to be published in mid-2000, it will review whether changes in the policy framework could increase economic growth and whether changes, in income policy in particular, could reduce cyclical fluctuations. Another committee (the *Stabel Committee*) is working on a more technical review of the present wage negotiation system and may propose improvements in its institutional set-up at the end of 2000.

Monetary policy reduces demand pressures

Norges Bank is the executive and advisory body for monetary, credit and foreign exchange policy. The government set official policy guidelines for the bank in the Royal Decree of 6 May 1994, stipulating that "the monetary policy to be conducted by the Norges Bank shall be *aimed at* maintaining a stable krone exchange rate against European currencies, based on the range of the exchange rate maintained since the krone was floated on 10 December 1992. In the event of *significant* changes in the exchange rate, monetary policy instruments will be *oriented with a view* to returning the exchange rate *over time* into its initial range. No fluctuation margins are established, nor is there an appurtenant obligation on the Norges Bank to intervene in the foreign exchange market." (Italics added.) The monetary policy regime differs substantially from the fixed exchange rate regime that collapsed in December 1992 with its unlimited obligation to defend the established fluctuation margin for the krone exchange rate. Based on the 1994 guidelines, Norges Bank has aimed to stabilise the ECU/krone exchange rate, based on its initial range (8.2-8.4 kroner per euro). It has been relatively successful in doing so (Figure 10).[16] While exchange rate volatility has increased since the end of 1996, the normal trading range for the euro/krone rate has not changed significantly.

Figure 10. **Exchange rate volatility**
1993-99[1]



1. Standard deviation of the index level of the daily exchange rate (January 1998 = 100), over the period January 1993 to December 1998 for Finland, Ireland, Portugal and Spain; January 1993 to December 1999 for other countries. Per US dollar for Australia and New Zealand, per deutschemark for the other countries.

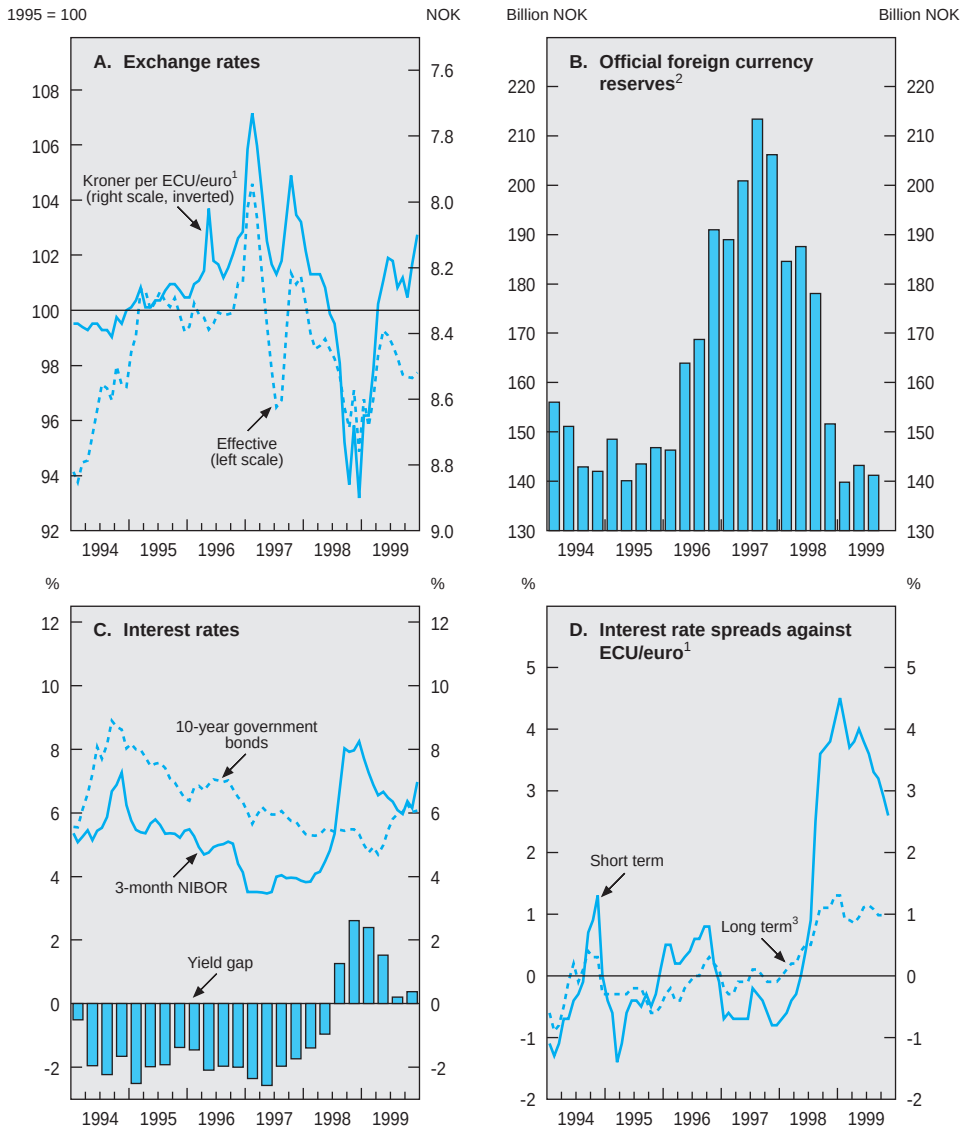
Source: OECD Secretariat.

More than in the past, Norges Bank has emphasised since early 1999 the need to bring price and cost inflation down over time to the corresponding objective for the euro area as a fundamental precondition for exchange rate stability.[17] The inflation projections and economic analyses in Norges Bank's Inflation Reports provide an important basis for the bank's monetary policy decisions.[18] The bank also emphasised that it cannot fine-tune movements in the krone exchange rate. The experience gathered during the various episodes of international financial turbulence in recent years has played a role in this. Exchange rate flexibility and focus on economic fundamentals is in line with the recommendations on monetary policy in the previous *Survey*. The room for discretion provided by the guidelines means that the distinction often made between exchange rate and inflation targeting will, under most circumstances, not be relevant in the Norwegian case and that the potential for policy dilemmas is limited, although it exists. If the situation arises whereby Norges Bank is incapable of returning the exchange rate into the initial range without the risk of triggering higher inflation (or a deflationary recession), the Bank will inform the government authorities that measures other than those available to the central bank are required (Gjedrem, 1999).[19]

Tight monetary policy, helped by the oil price recovery, has contributed to strengthen the krone (Figure 11).[20] Due to the low oil price, concerns about the fiscal policy stance, signs of overheating and turbulence in international financial markets, the currency had depreciated in 1998 from 8.1 kroner per ECU in January to 9.2 kroner late in the year.[21] The downward pressure was especially strong during the turbulence in international financial markets in August, which induced a depreciation of 9 per cent. Norges Bank reacted to the slide in the krone, and to accelerating inflation, by more than doubling its deposit rate between March and August, from 3.5 to 8 per cent.[22] The tightening was not sufficient for an immediate return of the krone into the desired range. Norges Bank did not increase rates further as this would not have been perceived as credible. The tight monetary policy played an important role in reducing demand pressures and preventing a further acceleration of inflation. According to Norges Bank, a 1 percentage point increase in interest rates leads, after two years, to the same reduction in demand as a fiscal tightening of 0.75 percentage point of GDP (Norges Bank, 1997).[23] In January 1999, based on prospects of reduced inflationary pressures, the central bank started to cut interest rates. In five steps of 0.5 percentage point, key rates were reduced by 2.5 percentage points between January and September. With the deposit rate at 5.5 per cent, monetary policy has remained cautious.

As a consequence of the central bank's key rate cuts, short-term market rates have declined, with the three-month interbank rate falling by 2 percentage points from 8 per cent at the beginning of the year to 5.9 per cent in December 1999.[24] The long-term interest rate, however, has moved up by 1 percentage

Figure 11. **Exchange and interest rate developments**



1. Theoretical ECU up to end 1998, euro as from January 1999.
2. Quarterly average of monthly data.
3. Differential between the effective yield on Norwegian government bonds and the rate on ECU/euro denominated government bonds, both with 10 years to maturity.

Source: Norges Bank and OECD, *Main Economic Indicators.*

point to 6.1 per cent (10-year government bond rate) by December 1999 shadowing the rise in rates in the euro area and the United States. The long-term interest rate differential *vis-à-vis* Germany has narrowed by 50 basis points to 90 basis points, while the forward rate differential against Germany ten years ahead is around 50 basis points. This is somewhat higher than for Denmark and Sweden (25 basis points) but indicates, nevertheless, a high level of confidence of financial markets in the conduct of Norwegian monetary policy.

Broad money supply growth (M2) accelerated, from around 5 per cent (year-over-year) at the end of 1998 to 9 per cent in November 1999 (Table 6 and Figure 12, panel A). Private sector net sales of foreign currency and central government budget transactions contributed most to this acceleration, which was only partly offset by weaker credit growth by banks. Domestic credit growth, which decelerated in 1998, slowed down only marginally in 1999, from 8.3 per cent in December 1998 to 8.1 per cent in November 1999. Due to stronger growth from foreign sources, total credit growth was more rapid (10 per cent in September), but has also decelerated.

Fiscal policy shifts from somewhat tight in 1999 to neutral in 2000

The relatively easy fiscal stance in 1998 contributed to the downward pressure on the Norwegian krone and the overheating of the product and labour

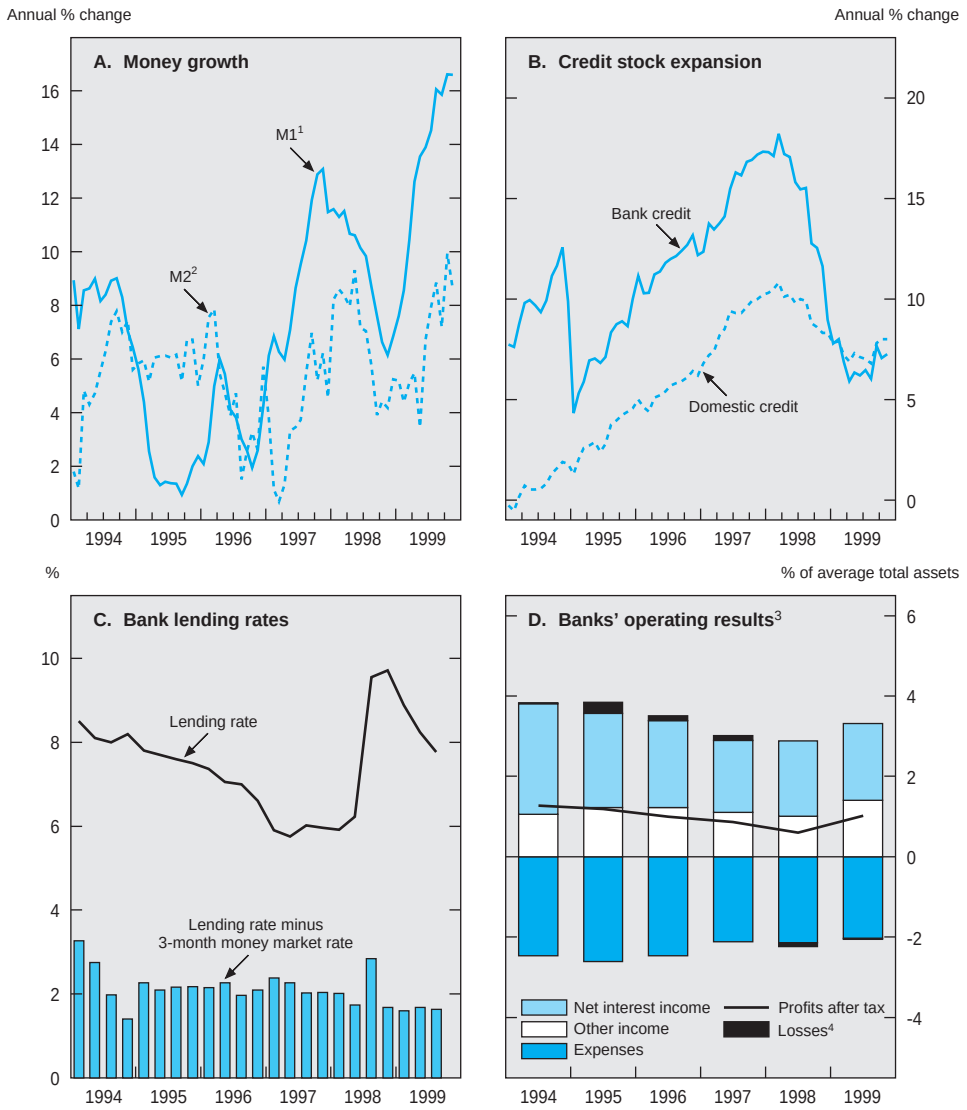
Table 6. **Money and credit**

Percentage change

	End-1993 NOK billion	1994	1995	1996	1997	1998	1999[1]
Total money stock (M2)	**546.0**	**6.4**	**5.0**	**5.7**	**4.6**	**5.2**	**8.6**
Percentage point contribution from:							
Central authorities	..	7.2	3.3	–6.9	–8.5	1.7	..
Commercial and savings banks	..	3.8	8.0	9.7	14.2	9.3	..
Other sectors	..	–4.6	–6.3	2.9	–1.1	–5.8	..
Total domestic credit (C2)	**877.7**	**2.3**	**4.9**	**6.2**	**10.2**	**8.3**	**8.0**
Commercial and savings banks	449.9	6.9	9.6	16.9	16.8	8.7	6.8
State lending insititutions	180.8	–2.2	–0.6	–1.9	0.2	4.4	4.7
Mortgage companies	73.1	–11.8	–1.0	–4.4	17.6	31.5	18.5
Life insurance companies and pension funds	74.8	–7.3	–16.2	–8.0	–28.7	–12.2	–5.6
Bond and certificate market	70.6	4.3	6.4	10.5	1.5	–2.3	12.6
Other credit	28.4	16.2	18.4	–3.7	17.6	18.8	22.7

1. November. Growth over same period of previous year.
Source: Norges Bank.

Figure 12. **Money and credit**



1. Cash, demand deposits and unused credit held by the public; each observation represents a 3-month right adjusted moving average.
2. M1 plus time deposits.
3. Commercial banks. Data for June in 1999.
4. Losses on loans and guarantees; write-downs and gains/losses on securities held long term.

Source: Norges Bank, Kredittilsynet and OECD, *Main Economic Indicators*.

Table 7. **Fiscal balance: projections and outcomes**

	Outcome			1999		2000
	1996	1997	1998	Proposed[1]	Estimate[2]	Approved[3]
Central government[4] (NOK billion)	**47.2**	**66.7**	**27.5**	**52.2**	**26.6**	**71.8**
Net proceeds from petroleum activities	69.9	86.8	45.0	57.7	44.6	85.1
Balance excluding petroleum activities	–22.7	–20.1	–17.5	–5.5	–18.0	–13.3
Petroleum Fund (NOK billion)	**44.3**	**67.3**	**34.2**	**58.3**	**33.4**	**79.2**
Transfer from central government	44.2	64.0	28.0	52.2	26.6	71.8
Return on financial assets	0.1	3.3	6.2	6.1	6.8	7.4
General government[4] (per cent of GDP)	**6.6**	**7.9**	**3.9**	**6.0**	**4.8**	**7.4**
Central government	6.6	7.8	4.9	6.3	5.0	7.7
Local government	0.0	0.1	–1.0[5]	–0.3	–0.2	–0.3
Memorandum items:						
Mainland GDP volume growth, per cent	4.1	4.4	3.3	1.3	0.5	0.7
Real "underlying" central government expenditure,[6] per cent change	1.0	2.2	1.9	1	2	2½
Change in adjusted central government surplus[7]	¾	¾	¼	1	½	0
Oil price (kroner per barrel)	134	136	96	110	125	125
Oil and gas production[8]	222	229	223	251	232	271

1. October of the preceding year.
2. December 1999 except for general government which is the October estimate.
3. December 1999. Same as proposed in October.
4. Net lending.
5. Roughly half the deficit reflects a timing change in accrued revenues.
6. Deflated by the mainland GDP price index.
7. Corrected for cyclical developments, net interest payments, petroleum revenues and technical adjustments; as a per cent of mainland GDP.
8. Million standard cubic metres of oil equivalents.

Source: Ministry of Finance.

markets (Table 7). In contrast, in 1999, a fiscal policy tightening supported the recovery of the krone and helped to dampen tensions in the economy (Box 2). As in previous years, there was some slippage in the central government budget and stronger than expected growth of local government outlays. On average during 1996-99, the underlying growth in central government spending has been 1¼ percentage points higher than projected in the draft Budget.[25] Despite the slippage in 1999, total government outlays excluding interest payments remained fairly stable as a percentage of GDP, at an estimated 44 per cent, which is slightly above the euro area average (Figure 13).[26] Petroleum revenues were unchanged after the strong drop in 1998, with the impact of the recovery in the oil price offset by that of the lower oil production. The general government budget surplus is estimated to have increased by 0.9 percentage point to 4.8 per cent of GDP in 1999. Norway's budgetary position is unique: the second-highest surplus in the

Box 2. Fiscal developments in 1999

The fiscal stance in 1999 has been less tight than projected in the draft 1999 Budget. The fiscal tightening for 1999 is estimated to amount to ½ per cent of mainland GDP (estimate of December 1999), as measured by the change in the central government cyclically-adjusted non-oil budget deficit net of interest payments, compared with the 1 per cent projected in the draft 1999 Budget.* The increase in the central government cyclically adjusted real underlying expenditure is estimated at 2 per cent, 1 percentage point up from the draft 1999 Budget. Total government consumption is projected in the draft 2000 Budget to increase by 2.0 per cent, 0.9 percentage point more than projected a year earlier.

The draft 1999 Budget presented to the Parliament in October 1998 projected the central government surplus to almost double to NOK 52 billion in 1999, reflecting oil revenues rising to NOK 58 billion and a considerable reduction in the non-oil budget deficit to NOK 5½ billion. Non-oil revenues were expected to rise by 7 per cent to NOK 427 billion, of which NOK 4 billion was supposed to result from increases in tax rates on income, electricity, tobacco, cars and heating oil and the remainder from the widening of the tax base and other proceeds. The NOK 4 billion tax increase was, however, not approved by Parliament. Non-oil expenditures were projected to increase by 4 per cent to NOK 432 billion. The bulk of the spending rises was due to growing social security appropriations and other government transfer programmes, including regional support, childcare cash benefits and pension schemes.

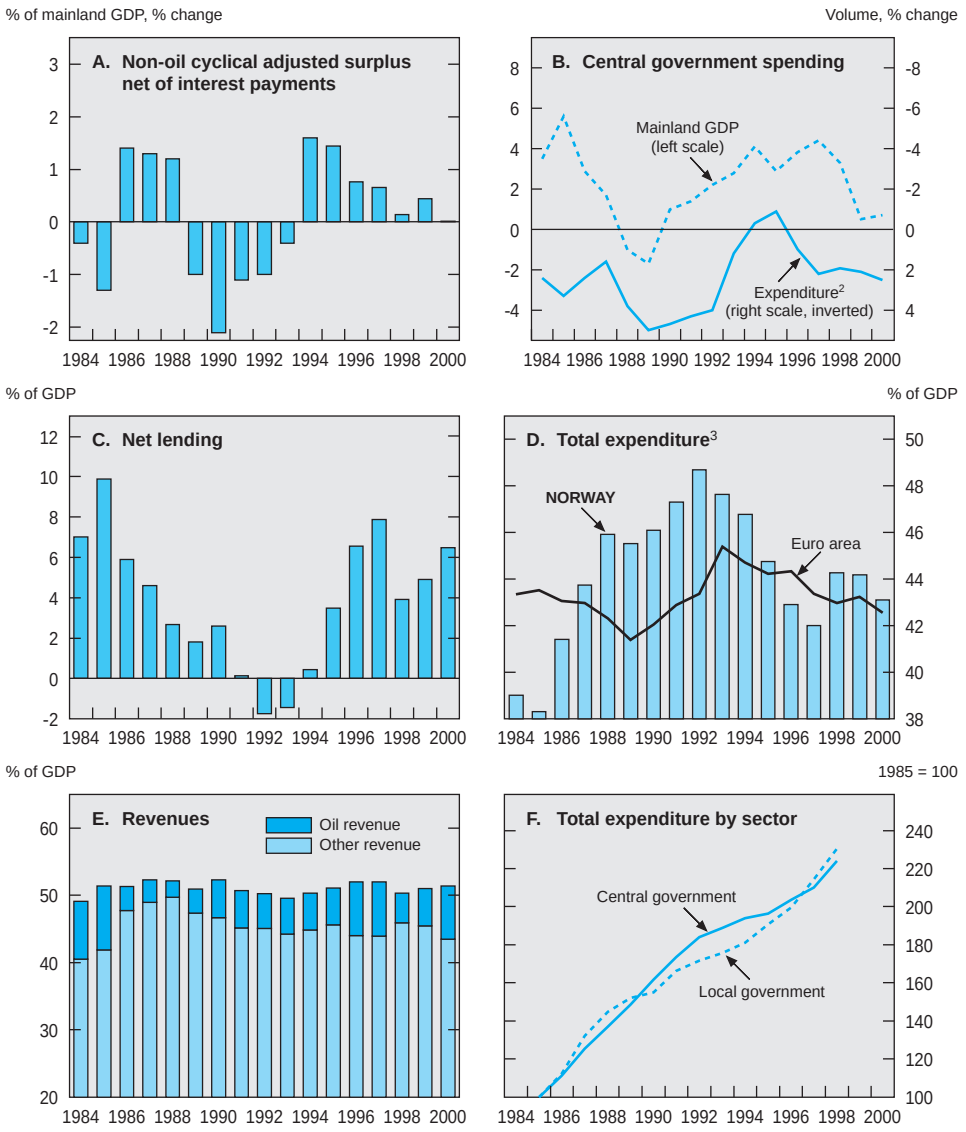
Since the draft 1999 Budget, estimated net revenues from petroleum activities have been lowered by NOK 13 billion (estimate of December 1999). This results from significantly lower oil and gas production than anticipated in the budget, offset partly by higher oil prices, as the price estimate has been increased by 14 per cent in local currency (from NOK 110 per barrel to NOK 125 per barrel) in recent projections. At the same time, non-oil revenues declined by NOK 10 billion, due mainly to lower tax receipts and interest revenues and postponement of the further privatisation of *DnB* (Den norske Bank). The non-oil deficit is estimated to increase by NOK ½ billion from 1998 – instead of declining by NOK 13 billion as envisaged in the draft 1999 Budget – to NOK 18 billion in 1999. As a result, the estimated central government fiscal surplus for 1999 is projected to decline by NOK 1 billion to NOK 26½ billion, or 2¼ per cent of GDP. The surplus will be allocated to the Government Petroleum Fund which will have accrued NOK 203 billion (17 per cent of GDP) by the end of 1999.

* 0.1 percentage point of the difference is due to technical changes in the methodology (Ministry of Finance, 1999). The indicator does not take into account local government spending.

OECD area in 1999 was recorded by Ireland (3.4 per cent of GDP) while the OECD countries still showed a deficit of 1.1 per cent of GDP on average.

In the light of the economic slowdown and the projected fading of excess demand, the government proposed a neutral fiscal stance in its draft 2000 Budget, presented to Parliament in October 1999 (Box 3). Despite the large budget surplus, the government found it necessary to propose some tax increases, of around NOK 2.9 billion (accrual basis), to finance higher spending on health care,

Figure 13. **General government finances**[1]



1. Data for 1999 and 2000 are projections or estimates.
2. Real underlying spending is defined as total spending excluding petroleum activities, unemployment benefits and interest payments.
3. Excluding gross interest payments.

Source: Ministry of Finance, Statistics Norway and OECD Secretariat.

Box 3. The 2000 Budget

The draft Budget presented to Parliament in October 1999 projected a doubling in the central government surplus to NOK 79 billion in 2000. The surge in petroleum revenues accounted for practically the whole improvement. Around a fifth of central government revenues will originate in 2000 directly from the petroleum sector. Non-oil expenditures are expected to increase by NOK 22¼ billion (or approximately 5 per cent). The bulk of the spending increases reflect growing appropriations for social security and other high priority areas such as regional policy, research and education. The draft Budget did not include any significant expenditure cuts in low priority areas. The non-oil revenues were projected to rise by NOK 24¼ billion (or around 5¾ per cent), of which NOK 2.9 billion (accrual basis) will result from tax increases, mainly of excise taxes on energy. Part of the tax increase is of an accounting nature as a transfer replaces an income tax deduction for children.

Parliament agreed on an additional tax increase in 2000 of NOK 0.6 billion (accrual basis) and on a substantial shift of the tax burden from low-wage incomes to capital income and high-wage income. The following major measures were agreed: conversion for wage earners of the lowest income tax bracket into a standard personal allowance (revenue loss: NOK 1.2 billion); higher threshold in the central government income surtax (loss of NOK 600 million); introduction of an additional bracket in the central government income surtax with a rate of 19.5 per cent (gain of NOK 600 million); reduction in depreciation rates on buildings (NOK 500 million); repeal of tax credit for share-market savings (gain of NOK 500 million); reduction in rate of return of capital and tightening of identification rules in "split model" (gain of NOK 200 million); and increased taxation on shipping companies (gain of NOK 200 million).

In the approved 2000 Budget, the stance of fiscal policy is neutral, with an increase by 2½ per cent of the real underlying expenditure of central government, up by 1 percentage point from 1999. The OECD's estimates for the general government also point to a neutral fiscal stance in 2000. Total government consumption is projected to increase less (1.5 per cent), mainly due to slower consumption growth of local government (1.2 per cent). The non-oil budget deficit is projected to narrow by NOK 5 billion to NOK 13 billion.

Net revenues from oil and gas activities are estimated to rise by almost 80 per cent to over NOK 85 billion in 2000. The increase is underpinned by a projected increase of almost 17 per cent in petroleum production, while the oil price is expected to remain at NOK 125 per barrel. The allocation to the Government Petroleum Fund is estimated to rise by almost NOK 40 billion to NOK 72 billion in 2000. General government net financial assets, including assets in the Petroleum Fund and investment in state-owned enterprises, are estimated at around NOK 580 billion – 47 per cent of GDP – at the end of 2000.

education, foreign aid, research and regional subsidies. Deliberations in Parliament have raised the tax increase by NOK 0.6 billion (accrual basis) as well as the tax burden for capital income and top wage incomes. The top rate for labour income will be increased by 6 percentage points in 2000, lifting it to well over

50 per cent, while at the lower end of the pay scale the measures will reduce the average but increase the marginal rate of taxation. The additional revenues will be spent on active labour market programmes and on transfers to local governments.[27] Due to the oil price recovery, the general government surplus is officially projected to rise by 2.6 percentage points to 7.4 per cent of GDP. The allocation to the Government Petroleum Fund will reach the record level of 5.8 per cent of GDP.[28]

Local government outlays increased more strongly in recent years than central government ones. Strong increases in tax revenues reflecting the booming economy made this possible. In 1999, local government consumption is estimated to have risen by 2.7 per cent in real terms, compared with a rise of 0.8 per cent for central government. The economic slowdown will restrain tax revenues in 2000, but revenue shortfalls will be partly offset by higher transfers from central government (see also Chapter III). Nevertheless, local government consumption growth is projected to decelerate to 1.2 per cent in real terms.

The Government Petroleum Fund, which is managed by the central bank, separately from the official reserves and ordinary central bank functions, invests exclusively in foreign bonds and since early 1998 also in foreign equities. The aim of the Fund is to accumulate financial reserves to preserve an equitable share of the current oil and gas revenues for future generations. The investment performance has been good up to now, with the total return exceeding that of the benchmark portfolio that follows from the government's guidelines. In 1999, the government proposed to Parliament the introduction of non-economic objectives in the investment strategy. Its proposal, to forbid investment in tobacco companies and to use the voting rights to promote human rights and environmental issues, was voted down but a separate portfolio based on environmental criteria of NOK 1 billion initially will be set up. There is a risk that returns of this portfolio will be lower than for the rest of the Fund.

Capital in the Petroleum Fund is projected to reach 23 per cent of GDP at the end of 2000.[29] General government net financial assets are even higher, an estimated 47 per cent of GDP, due to the stakes in state-owned enterprises and the considerable net foreign reserves held by the central bank.[30] This financial position is impressive compared with the other OECD countries. Nevertheless, as underlined in the draft 2000 Budget, there is no reason for complacency given the projected sharp rise in old-age and disability pensions in conjunction with falling oil revenues (see Chapter II for a more detailed discussion). Calculations presented in the draft Budget, based on inter-generational accounts, show that the budget surplus would need to be 0.4-1.6 per cent of GDP higher than foreseen in the draft Budget to avoid a heavier tax burden for future generations.[31] Other calculations presented in the draft Budget, based on a standard macroeconomic model, indicate the need for a somewhat bigger shift of 1½ to 2 per cent of GDP,

to keep taxes and transfers unchanged and achieve a moderate increase in public sector employment. Such calculations are of course very sensitive to the underlying assumptions, especially concerning the oil price, but indicate that there are substantial challenges for Norwegian fiscal policy which are long term in nature but would need to be addressed soon.

Prospects and challenges ahead

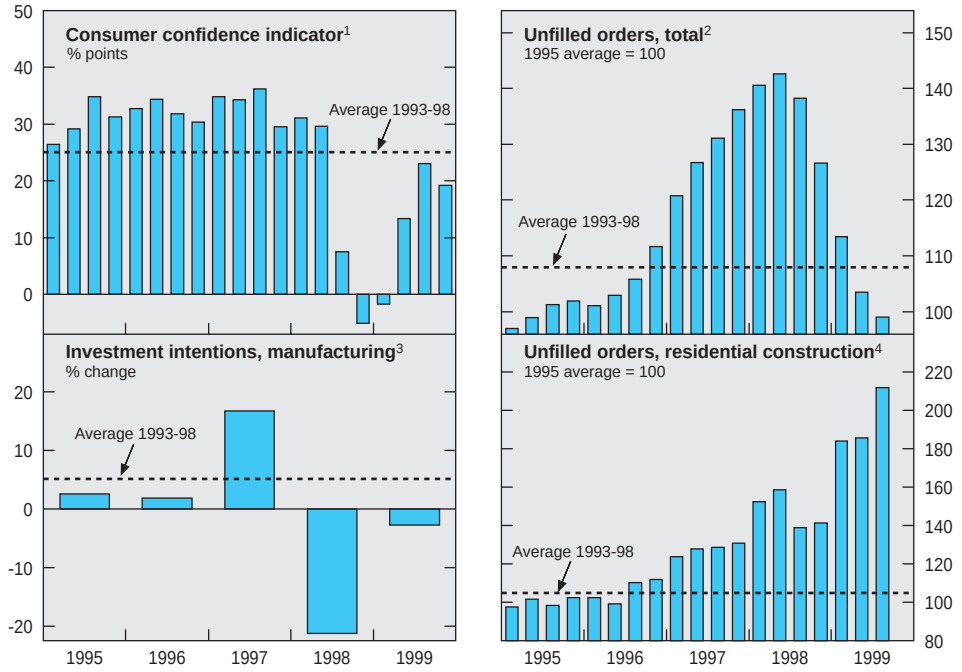
No recovery before 2001

The OECD Secretariat's projections for Norway, which are the same as those published in the OECD *Economic Outlook* No. 66 (December 1999), are based on a pick-up in world trade from 4.9 per cent in 1999 to 7.1 per cent in 2000 and 6.3 per cent in 2001. The oil price is assumed to remain stable, at slightly above US$22 per barrel, and it is expected that the Norwegian government will lift the production cap in 2000. The short-term interest rate in the euro area is predicted to increase from 3 per cent in the second half of 1999 to 4½ per cent in the second half of 2001. With excess supply falling, the euro area's wage growth is expected to accelerate gradually from 2 per cent in 1999 to 3 per cent in 2001. Currencies are assumed to remain at the level of their rates of 25 October 1999, implying an exchange rate of NOK 8.3 per euro. Based on the draft 2000 Budget, the fiscal stance is projected to be broadly neutral in 2000 and 2001, while monetary policy is assumed to be aimed at the fundamental preconditions for exchange rate stability, *i.e.* low inflation over the medium-term.

With inflation converging to the rate in the euro area, short-term interest rates are projected to drop close to the corresponding euro area rates, with the interest rate differential against the euro area rates expected to decline from 3½ percentage points in 1999 to ¾ percentage point in 2001. The short-term interest rate is projected to stand at around 5 per cent in 2001, 1¼ percentage points lower than in the second half of 1999. Although the long-term interest rate differential against the euro area is expected to narrow by a ¼ percentage point, long-term rates are predicted to rise by ¾ percentage point to 6¼ per cent, reflecting a projected rise both in the euro area and the United States.

Recent weak data for unfilled orders and investment intentions in the manufacturing sector, in combination with low business confidence, suggest that a turning point will not occur in the coming months (Figure 14). Mainland GDP growth will continue to be low in 2000, but is expected to regain momentum in 2001, in response to further monetary easing (Table 8). The growth rate for total GDP will, however, be substantially higher in 2000, a projected 3¼ per cent, due to the steep rise in oil production. Oil and gas production is projected to increase by almost a fifth as it is assumed that the production cap decided by the

Figure 14. **Leading indicators**



1. Quarterly average of monthly data.
2. Stock figures.
3. According to the November quarterly investment survey; investment intentions for the following calendar year.
4. Value.

Source: Statistics Norway and Ministry of Finance.

government will be lifted and new major fields will come on stream. After the Secretariat finalised the projection, the government decided to delay the lifting of the production cap until 1 April 2000; the effects in 2000 of this decision on oil and gas production and on total GDP are around –2 and –¼ per cent, respectively. However, those effects are probably partly offset by production below target in the second half of 1999, as indicated by third quarter national account figures. Low mainland GDP growth in 2000 largely reflects a drop in investment offset by an increase in private and public consumption. Petroleum investment is expected to contract dramatically after the completion of a number of large investment projects on the Norwegian shelf. In addition, mainland business investment will continue to drop due to weak domestic demand and profitability. High house

Table 8. **Short-term projections**[1]

Percentage change, constant prices

	1996 Current prices NOK billion	1997	1998	1999	2000	2001
Private consumption	490.3	3.7	3.1	1.7	1.7	1.9
Government consumption	206.8	2.8	3.7	2.1	2.0	2.0
Gross fixed investment[2]	216.2	15.1	8.1	–9.4	–10.9	0.6
of which:						
Oil sector[2]	51.3	15.6	25.7	–10.0	–29.0	0.0
Mainland business sector	100.1	12.3	2.8	–11.3	–7.5	0.0
Residential construction	27.2	8.2	–0.6	–9.0	0.5	1.8
Public sector	31.6	18.1	3.4	–4.0	0.0	1.7
Stockbuilding[3]	15.8	0.3	0.9	–0.1	0.0	0.0
Total domestic demand	929.2	6.4	5.4	–1.2	–1.3	1.6
Exports of goods and services	414.4	5.7	0.5	1.9	7.5	3.5
of which:						
Non-manufactured goods (including energy)	176.2	2.7	–2.8	3.8	15.0	3.0
Imports of goods and services	327.1	12.0	9.1	–2.2	–2.4	3.5
Foreign balance[4]	87.4	–1.5	–3.1	1.7	4.4	0.4
GDP	1 016.4	4.3	2.1	0.6	3.3	1.8
of which: Mainland	833.0	4.4	3.3	0.0	0.1	1.6
Memorandum items:						
GDP deflator	..	2.7	–0.5	6.0	5.0	1.7
Private consumption deflator	..	2.5	2.6	2.3	2.1	2.0
Employment	2 131.3[5]	2.9	2.3	0.1	–0.5	0.1
Unemployment rate (level)	4.8	4.1	3.2	3.2	3.8	3.9
Private sector compensation per employee	..	4.8	7.1	5.1	4.3	3.8

1. Projections published in the OECD *Economic Outlook* No. 66 (December 1999).
2. Includes platforms under construction, crude oil production, oil drilling and pipeline transport.
3. Contribution to GDP growth, excluding platforms under construction.
4. Contribution to GDP growth.
5. In thousand persons.

Source: OECD Secretariat.

prices, in combination with lower interest rates, will, however, lead to a recovery in residential investment. Increased unfilled construction orders confirm the nascent recovery (Figure 14). Private consumption, underpinned by firm consumer sentiment and steady, although lower, increases in disposable income, is projected to make a substantial positive contribution to GDP growth. Exports of manufactured goods are expected to pick up due to stronger world demand but the further deterioration of competitiveness will lead to continuing losses in market share on the world market.

With production growth low for the second consecutive year, employment is projected to decline by ½ per cent in 2000, before stabilising in 2001. The unemployment rate is expected to increase only moderately from 3¼ per cent in 1999 to close to 4 per cent in both 2000 and 2001. Labour supply should remain almost constant until 2001, after having expanded by an annual average rate of 1¾ per cent over the 1993-98 period. With unemployment rising somewhat, wage inflation (on a national accounts basis) should ease gradually, from 5 per cent in 1999 to 3¾ per cent in 2001. Despite the rise in excise taxes, consumer price inflation should edge down further from 2¼ to 2 per cent in 2001, which is rather close to the projected 1¾ per cent for the euro area. The surging oil production and high oil prices, accompanied by the sharp drop in imports of investment goods, will boost the current account surplus to above 10 per cent of GDP. The favourable development of the offshore oil sector will also be reflected in a sharp increase in the general government surplus during the projection period.

There are considerable downward and especially upward domestic risks to the projection. Third quarter mainland GDP was stronger than expected at the time of finalising the projection. Forecasting petroleum investment has proved to be difficult in the past and the drop of almost 30 per cent projected for 2000 could be smaller given the recovery of the oil price. Moreover, the impact on the mainland economy is also difficult to estimate since it depends on the ability of supply industry to readjust their activity to other markets. Private consumption could be somewhat higher than projected, because consumers could reduce their high savings in reaction to lower disposable income growth. Moreover, private consumption could be more buoyant due to less wage moderation (see next section). Government revenues, and therefore transfers to the Petroleum Fund, are strongly influenced by oil price movements: a permanent drop of the oil price of US$1 per barrel would reduce revenues by around ⅔ per cent of GDP.

Macroeconomic policy in the coming years: challenges lie ahead

The wage negotiations starting in early 2000 will be crucial for short-term macroeconomic developments and determine, to a major extent, whether or not additional macroeconomic policy measures are necessary. Policy concessions to get a moderate central wage agreement could, therefore, have clear short-term benefits. Nevertheless, wage moderation is in the interest of employees and employers – and the economy as a whole – and should therefore not be "bought" by policy concessions. Moreover, the short-term benefits of wage moderation could easily be outweighed by heavy long-term costs. This is, for instance, the case with the reduction in the early retirement age by two years agreed in 1997[32] and it is potentially the case for the *Competence Reform*. Moreover, as policy concessions are part of the central income agreements, the policy measure can be seen

as a long-term acquired right (*acquis*) which has been "paid" for by wage moderation. Reneging on such long-term commitments therefore poses difficulties and can even endanger further wage moderation.

Even with a moderate wage settlement, tight labour and product markets could lead to increasing wage drift and higher profit margins in the sheltered sector[33] There may therefore be a need to tighten fiscal policy in the Revised Budget for 2000, to be sent to Parliament in May 2000. At the least, any spending overruns should be offset by expenditure cuts elsewhere. Not sufficiently tight fiscal policy should have consequences for monetary policy. To strengthen budgetary discipline, Norway should enhance its medium-term framework, for instance by setting more explicit medium-term expenditure targets – while retaining flexibility for short-term counter-cyclical action if that proves necessary. A number of countries (the United Kingdom and Australia, for example) have recently strengthened their medium-term fiscal frameworks, but it remains to be seen whether their new framework can live up to the initial ambitions.

There will also be new challenges when the economy starts to pick up in 2001, as the slack in the economy will be limited. For 2001, the Secretariat projects an output gap of –0.8 percentage point which is very small compared with the gap of –4.2 percentage points in 1992 at the eve of the previous recovery. This could imply that labour market bottlenecks could reappear quickly. An early tightening of the macroeconomic policy stance may therefore be required.[34]

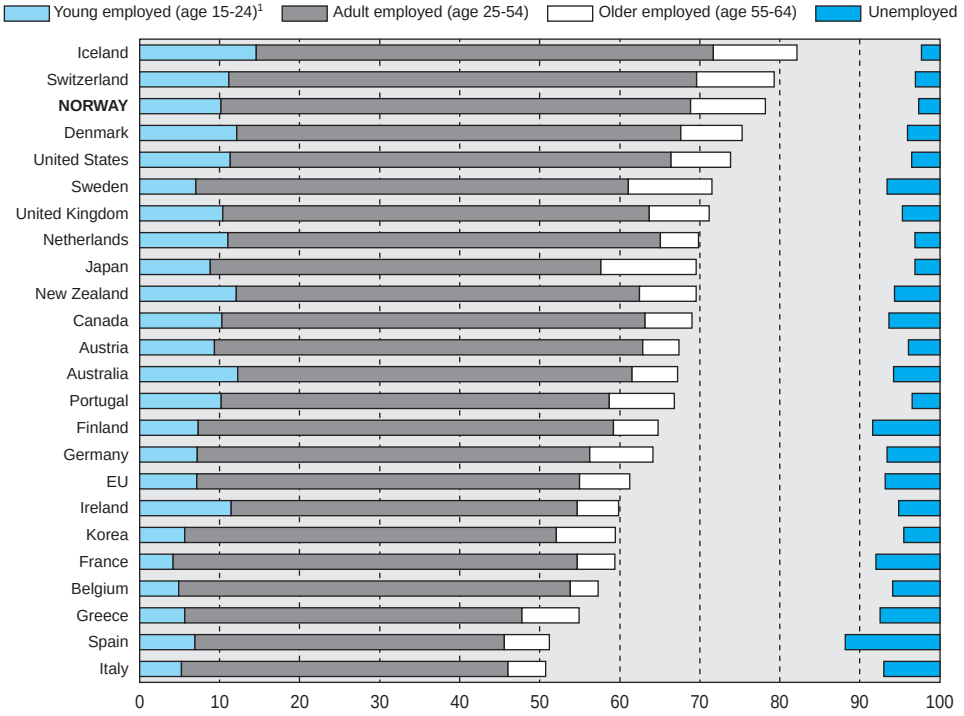
II. Implementing structural reform: a review of progress

With the unemployment rate among the lowest in the OECD and a very high share of the working-age population at work (Figure 15), Norway's labour market performance is outstanding.[35, 36] The labour productivity level of the mainland economy, on the other hand, is considerably below that of the best-performing OECD countries, despite the well-educated labour force, while there are indications that productivity growth has been below the OECD average and has lagged the other Nordic countries by even more (Figure 16).[37] The weak productivity performance is probably largely related to product market characteristics. Public involvement in product markets has remained high, not only through public ownership but also through regulation and high subsidies, especially to the agricultural sector. Moreover, public sector employment is very high and productivity of the public sector is much lower than in the private sector. Against the backdrop of a high income per capita, low unemployment and a strong national asset position, structural reform can easily be perceived as a low priority in Norway. However, given the estimated productivity gap of the mainland economy with the best performing OECD countries, Norway could certainly benefit from structural reform initiatives. Moreover, policy needs to prepare the economy for the period of sharply falling oil revenues, starting in one or two decades. Structural reform has a major role to play to smooth the transition to a less resource-based economy. This chapter reviews the progress made recently in implementing such reforms and analyses the scope for further action to increase productivity and the standard of living.

Recent initiatives could undermine the strong labour market performance

Norway has succeeded in avoiding the shift to persistent high unemployment, which has been experienced by most other European OECD countries since the late 1970s. It is among the few countries that have recently restored full employment. The labour market has, however, some weak spots, some of which were reinforced by recent policy measures. As emphasised in the 1998 and 1999

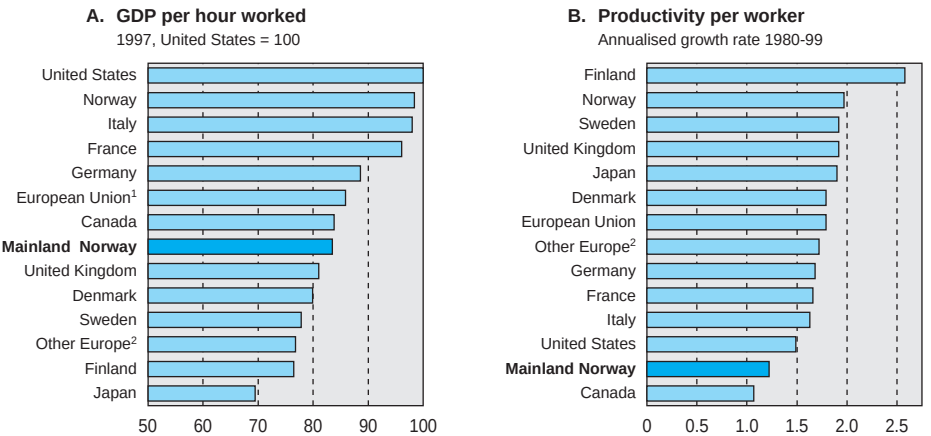
Figure 15. **Employment rates in the OECD**
As a per cent of total working age population, 1998



1. Age 16 instead of 15 for Norway, Spain, Sweden, United Kingdom and United States.
Source: OECD (1999), *Labour Force Statistics, 1978-1998,* Part III (forthcoming).

Surveys, disincentives in the form of high replacement ratios exist for people on public unemployment income support to seek and accept work, which are partly offset by strict job seeking requirements. At the same time, a steep trend rise has occurred in the number of persons on disability pensions and in early retirement schemes.[38] Other weaknesses are the high unemployment among immigrants[39] and limits to the economy's adaptability to change implied by relatively strict labour market regulations, especially concerning fixed-term contracts. The following paragraphs discuss these drawbacks and the way they could be overcome.

Figure 16. **Labour productivity in the OECD**



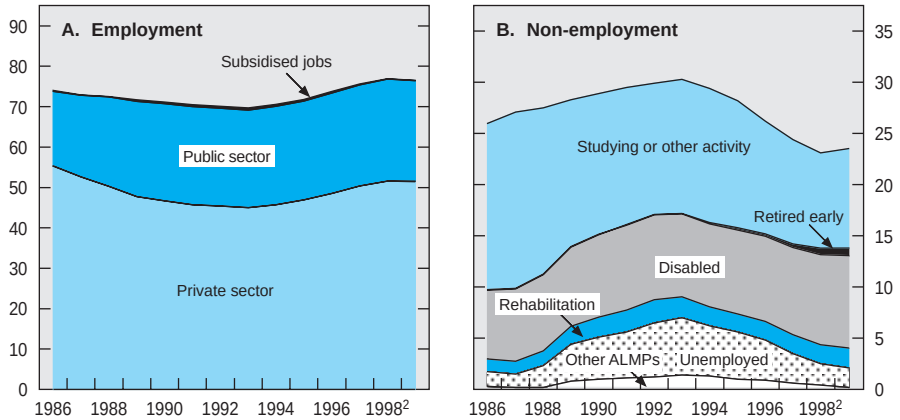
1. Excluding Luxembourg.
2. Other Europe is the weighted average for the following countries: Austria, Belgium, Greece, Ireland, Netherlands, Portugal, Spain and Switzerland.

Source: OECD Secretariat.

The retirement age drops due to disability pensions and early retirement

The effective retirement age at the end of 1999 is estimated slightly above 62 years, around a year lower than in 1994 due to the steep rise in the number of persons on disability pensions and in early retirement schemes.[40] The number of disability pensioners surpassed a quarter of a million in 1998 and is estimated to have risen to 269 000 in 1999 (Figures 17 and 18), 9.3 per cent of the working-age population.[41] This is one of the highest rates in the OECD area.[42] Low and medium-skilled persons are over-represented, as are those previously working in the manufacturing and construction industries. As a result, general government outlays on disability pensions have risen to around 3 per cent of GDP and without measures to restrain access, the number of disability pensioners is projected to increase to 11.6 per cent of the working-age population in 2005. Earlier in the 1990s, this number fell temporarily, which may be partly due to the 1991 reform of the disability scheme that tightened the eligibility rules and led to stricter enforcement (requiring a second opinion from neutral physicians and a more rigorous interpretation of the medical requirements). Moreover, occupational rehabilitation efforts were stepped up. The number of persons in rehabilitation programmes indeed increased significantly, almost doubling to 40 000

Figure 17. **Composition of the working-age population**
Population aged 16-74,[1] per cent



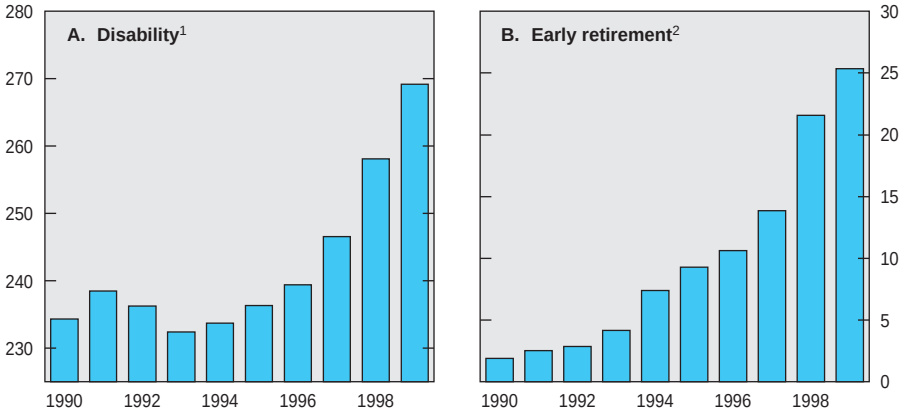
1. Excluding persons registered as retired (of standard retirement age).
2. Estimate for 1999 based on average for first three quarters.

Source: Ministry of Labour and Government Administration, and OECD Secretariat.

in 1998, but evaluations of the programmes are inconclusive concerning their effectiveness in improving the employability of the disabled. A comprehensive expert evaluation of the vocational rehabilitation reform will be published in 2000. The renewed increase in disability pensions since 1993 is due to the rising share of older persons within the working-age population and to some softening of qualification requirements in 1995. In the draft 2000 Budget, the government stressed the need to curb the upward trend given the increase in the dependency ratio (the ratio of retired people to employed persons) in the coming decades. However, no specific measures were announced.[43]

The recent drop in the effective retirement age has also been caused by a rapid increase in early retirement beneficiaries (Figures 17 and 18). In the 1997 tripartite wage negotiations, it was agreed to reduce the entry age for early retirement in two steps, from 64 years in 1996 to 62 years in 1998 (the standard retirement age is 67 years). As a consequence, the number of those retiring early increased steeply from less than 2 000 in 1990 to more than 20 000 in 1998 (3/4 per cent of the working-age population). In parallel, the labour force participation rate of those between 64 and 66 years old has fallen to around 30 per cent from 50 per cent in 1980.[44] The difference between the effective retirement age and the standard pension age is also partly caused by the option for certain occupational groups to retire earlier with a full pension. A third of government employees

Figure 18. **Disability pensioners and early retirement beneficiaries**
Total number of persons, in thousands



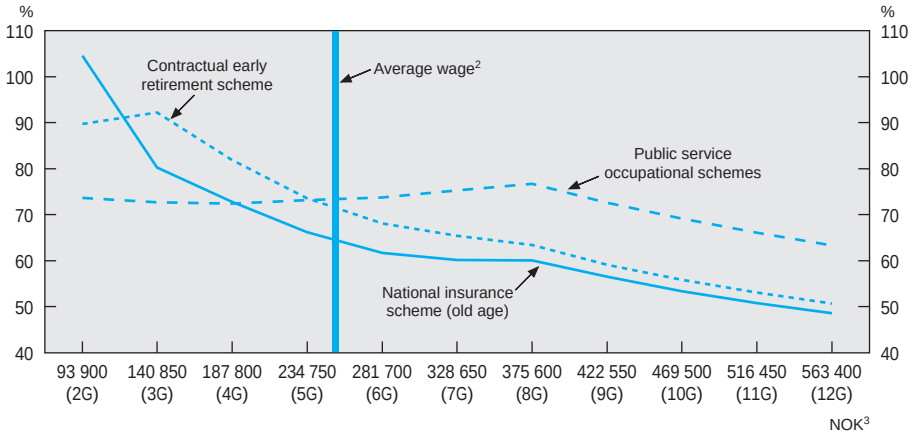
1. December estimate for 1999.
2. Data at 30 September for 1999.
Source: Ministry of Finance.

(though less than 10 per cent of private sector employees) have this option. Moreover, all public sector employees have the possibility to retire three years earlier if the sum of age and period of work amounts to 85 years or more. Finally, the part-time early retirement scheme also reduces the effective retirement age.[45]

The private sector early retirement schemes, which are part of collective agreements binding all employers, are not actuarially neutral. They are financed by employers collectively, with substantial tax concessions and transfers from the government. The collective financing makes the scheme attractive for individual employers, as it reduces restructuring costs. The scheme is also very attractive for older workers due to high benefit levels, with an after-tax replacement rate of around 80 per cent for low-paid workers (Figure 19). As a consequence of tax exemptions, some part-time early-retired persons even receive an after-tax income that is higher than the remuneration from work. Compared with other European countries, the effective retirement age is still high – in 1995 it was three years higher than in the European Union, while the disincentives to continue to work are probably still lower than in the European Union.[46]

The labour unions see the recent lowering of the entry age in the early retirement scheme as a first step towards a gradual reduction in the retirement age to levels seen in other western European countries. The government,

Figure 19. **Replacement rates after tax**[1]
1999



1. Based on the assumption that the person has acquired a full pension and is single.
2. The annual wage of the average production worker in 1998 was NOK 252 600.
3. G is the so-called "basic amount" for social insurance and income tax purposes. It equaled NOK 46 950 in 1999.

Source: Ministry of Finance.

however, has underlined in the 2000 draft Budget that the trend increase in early retirement cannot persist given the ageing of the Norwegian population in the coming decades. Ageing, in combination with higher average pension benefits, is estimated to increase government outlays on old-age and disability pensions from 8 per cent of GDP in 1998 to 15 per cent in 2030. At the same time, health outlays are projected to increase as well, while petroleum revenues will drop substantially due to falling production. To reduce the pension burden and promote activity by older workers, the government is considering an end to pension accruals during the period of early retirement and a reduction of the favourable tax treatment of early-retirement pensions. It has already proposed to Parliament to make part-time early-retirement schemes less attractive.

Childcare cash benefits have lowered labour supply

The introduction of the childcare cash benefits has led to some reduction in the labour supply. The benefits were introduced, for 1 year-olds in 1998 and extended to 2 year-olds in 1999, to allow parents to care for their children at home after the termination of maternity leave. They are paid to all families with children who do not use the publicly-funded childcare system. Such benefits are high (NOK 3 000 per month in 2000), approximately one-quarter of an average full-time

salary, not means-tested and not taxed. Costs in 1999 are estimated at NOK 2.4 billion (0.2 per cent of GDP), and around 8 per cent higher than budgeted due to stronger than expected demand. At the end of 1999, parents received cash benefits for three quarters of the 1 and 2 year-olds (90 000 out of the total of 120 000). Around 40 per cent of mothers receiving the cash benefit have reduced the number of hours worked and one out of two has reported that this is due to the cash benefit (Reppen and Rønning, 1999). No effect on the labour supply of fathers could be detected.[47] The labour supply effect will probably increase as the benefit rises by almost a third in 2000.

The *Competence Reform* enables adults who have not completed upper-secondary education to take study leave. It also facilitates certification of "non-formal" training in the workplace. The reform is part of a tripartite effort to raise competencies of all individuals, both inside and outside the labour force, in line with recommendations by the OECD (1996). In early 1999 during the wage negotiations, the government promised to spend NOK 400 million on "competence programmes" in the coming two to three years if wage growth in Norway develops in line with trading partners. In the 2000 Budget, NOK 180 million on measures related to the Competence Reform are included, half of which are earmarked for loans to adult students participating in vocational and other training. In 1999, Parliament enacted legislation that gives all employees an individual right to up to three years study leave if they meet certain criteria. In 2000, government will present a bill to Parliament to extend the right to those who have not completed upper secondary education. Moreover, a bill will be sent to Parliament concerning the right to free basic education and the duty of local governments to provide both lower and upper secondary education to all adults upon demand.

Some aspects of labour market regulation remain strict

In 1999, Parliament adopted some reforms of the employment protection legislation (EPL), which are largely in line with the *OECD Jobs Strategy* recommendations, while further reforms are under review. It includes a complete liberalisation of temporary work agencies but only a limited liberalisation of hiring-in rules. Hiring-in from temporary work agencies will only be allowed if approved by elected union representatives or in the limited cases the law allows for temporary contracts.[48, 49] As it becomes somewhat easier to hire on temporary contracts, the main economic advantage is probably during the first phase of an economic recovery when firms are still hesitant to hire permanent staff. The Ministry of Labour will supervise the temporary work agencies and can set additional rules for their operation. Given its close link with the Public Employment Service (PES), this supervisory role could lead to conflicts of interest. The risk of conflicts of interest would be even stronger if a competing state-owned temporary work agency were to be created.

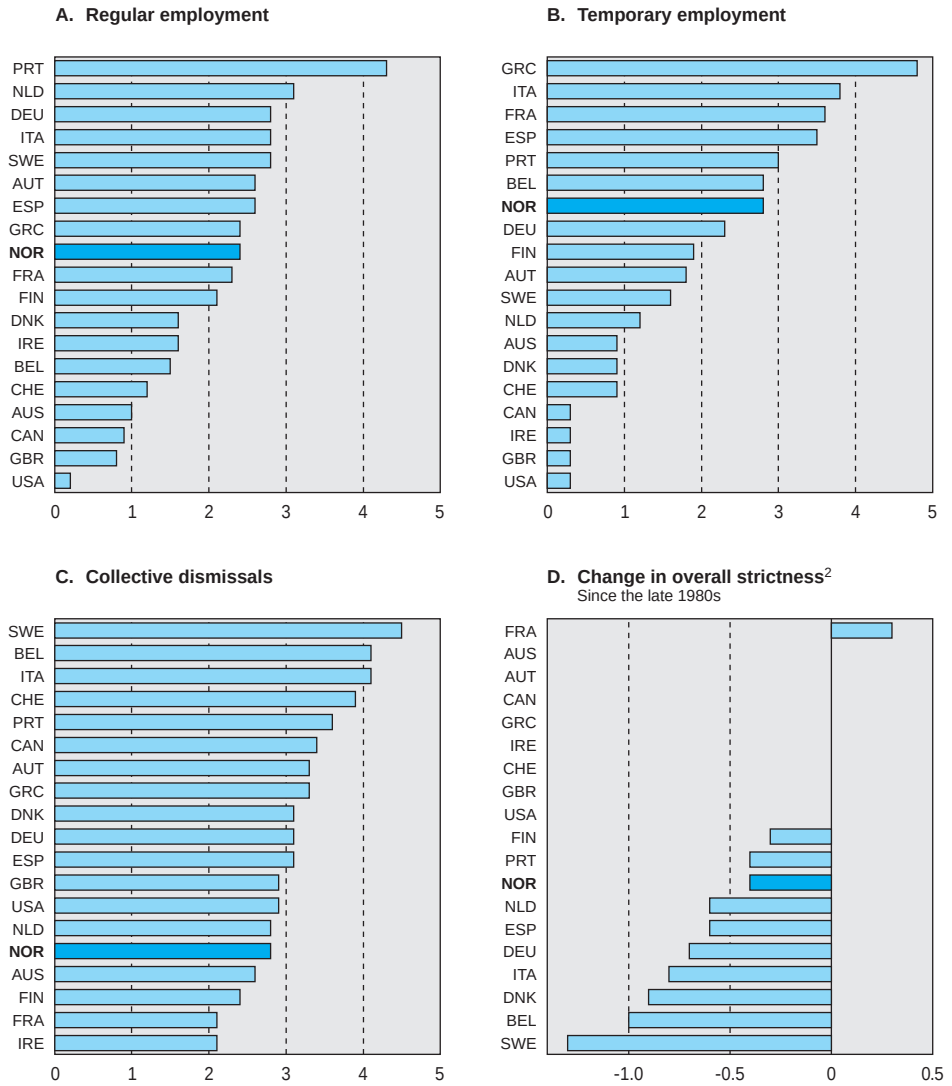
After the implementation of the reform measures, the restrictiveness of Norwegian employment protection legislation remains close to the European average (Figure 20). For regular employment, procedural inconveniences are relatively limited, notice periods in the case of individual dismissals are relatively short, and there are no severance payments required by law. Collective agreements in the private sector, however, may require lump-sum payments for long-serving staff who have reached the age of 50 to 55 years or where the dismissal arises from company reorganisation. Moreover, dismissals for economic reasons are considered unfair if the permanent employee could have been retained in another capacity and reinstatement is frequently ordered by the courts, as well as compensation of up to 6 month's pay plus back-pay for the duration of the court case. No changes are envisaged at the moment in the rules governing permanent contracts. In its report published at the end of 1999, a government appointed committee (*Colbjørnsen Committee*) was divided on the need to change those rules. However, it recommended unanimously to reduce the restrictions on overtime.

Strong increase in placements by the Public Employment Service

Brokerage activities of the Public Employment Service have risen strongly since 1997 (Arbeidsdirektoratet, 1999). The number of placements increased by half in the first half of 1999 compared with a year earlier. This was mainly due to agreements with private enterprises, schools and hospitals to provide adequately-qualified short-term substitutes in case of absence of regular employees on sick or maternity leave, or otherwise. This facility is appreciated by enterprises as it reduces the administrative costs and as the service is free of any charge. In the first half of 1999, one in four PES placements were for short-term substitutes. The increased activities of the PES may be the first beneficial effect of the proposed loss of the job placement monopoly and the proposed increased opportunities for temporary work agencies. To level the playing field, the Ministry of Labour has proposed to Parliament to let the PES charge for extended placement services, including the placement of short-term substitutes. These proposals go, however, considerably less far than recent changes in other OECD countries, especially Australia (OECD, 2000). Moreover, private placement is hampered by the relatively strict rules on fixed-term contracts.

The tight labour market has implied that the PES could step up tests on the work availability of the unemployed. Job seeking requirements are quite strict in Norway; in 1998, 5 per cent of the benefit recipients had their benefit temporarily reduced after a refusal to accept an offered job, one of the highest sanction rates in the OECD area. The willingness of the PES to sanction job refusals offsets, at least partly, the fact that the benefit structure does not give strong incentives for job search (see previous *Survey*).[50]

Figure 20. **Employment protection legislation**[1]
Late 1990s



1. Indicators of strictness of legislation. Scores can range from 0 to 6 with higher values representing stricter legislation.
2. Average of indicators for regular and temporary contracts.

Source: OECD (1999), *Employment Outlook*.

Substantial rise in active labour market programmes in 2000

The improvement in the labour market situation has led to a drop in participation in active labour market programmes (ALMP) from 53 000 in 1994 to 8 000 in 1999. The effectiveness of ALMPs is assessed annually. In its most recent review, Statistics Norway reported that participation in ALMPs increased the chance of finding a job by 11 per cent in the six months after finishing the programme. Given the expected rise in unemployment, the major restructuring problems in the shipbuilding and other petroleum-related mainland industries, the number of ALMP slots will be increased by 4 000 in 2000. Special training courses for petroleum-related industries will be set up to maintain key competencies during the period of low oil investments.

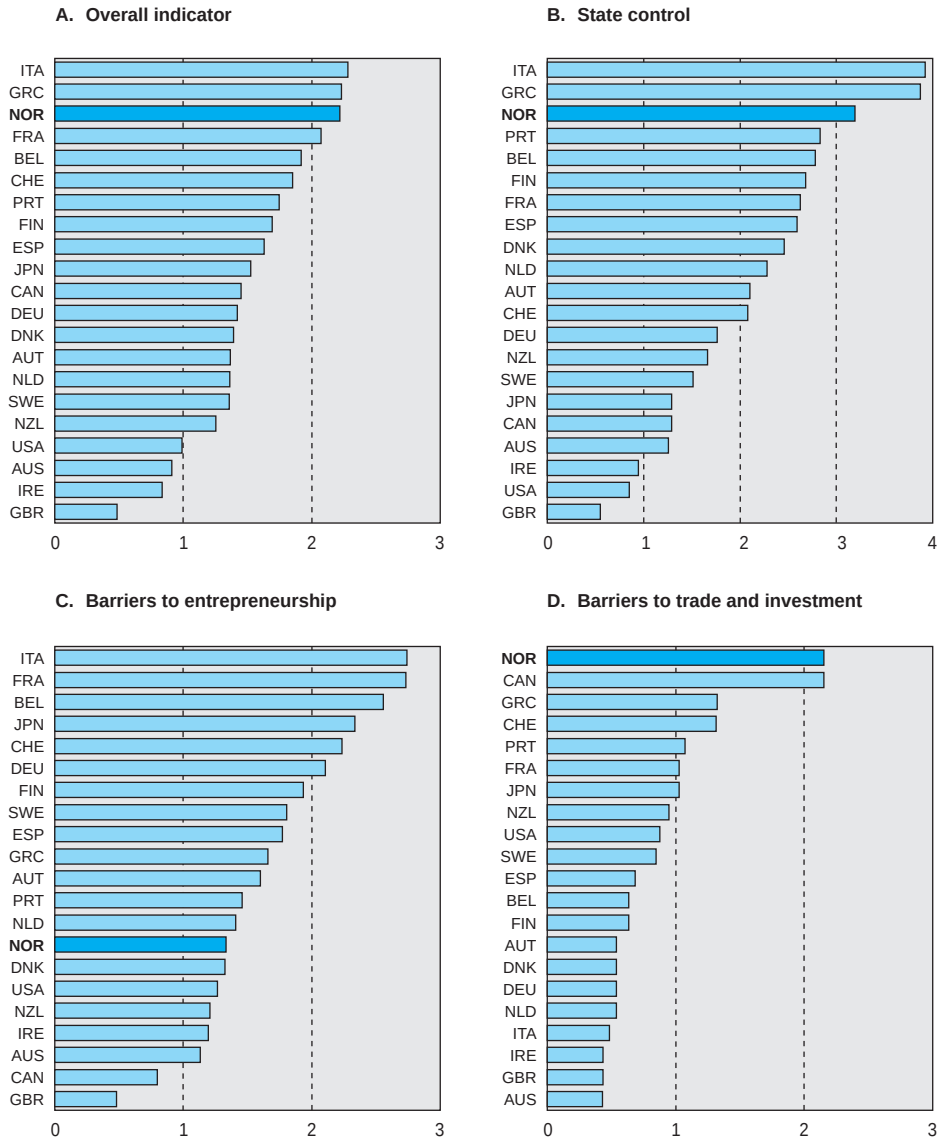
Government concerned by rising inequality

In its Equitable Redistribution White Paper (Ministry of Health and Social Affairs, 1999), the Norwegian government expressed the view that the trend increase in income and wealth disparity should be reversed. Income inequality has risen in the 1980s and 1990s, by somewhat more than in most OECD countries, but has still remained fairly limited in international comparison (Oxley *et al.*, 1999). A governmental working group will review the issue of introducing a guaranteed minimum income. The government will adjust taxation to make the income distribution more equal (the top rate of the central government income surtax was increased in 2000 by 6 percentage points raising the overall top marginal rate to over 50 per cent, see also Box 3 and Chapter III). In this context, the *OECD Jobs Strategy* has underlined major trade-offs: sufficient pecuniary incentives are important to realise a high level of labour supply and to improve human skills, while too high minimum wages or income transfers could have a detrimental effect on the employment of low-skilled workers.

Product markets are still heavily regulated

Some Norwegian product markets remain strongly regulated despite the rapid deregulation of the telecommunications market, the exemplary liberalisation of the electricity market and the membership of the European Economic Area. The cross-country rankings in Figure 21[51] present indicators of the relative friendliness of regulations to market mechanisms, that do not weigh the costs and benefits of the regulatory regime; moreover, they are only partial. Competition policy indicators, for instance, are not included. Having these caveats in mind, the indicators suggest that the overall regulatory environment in Norway is restrictive in international comparison. Regulation is important to promote policy objectives such as safety, health, environment and universal services. Moreover, it is

Figure 21. **Product market regulation**[1]
1997-98



1. Indicators of strictness of legislation. Scores can range from 0 to 6 with higher values representing stricter regulation.

Source: OECD Secretariat.

essential for the proper functioning of market economies by providing a reliable and stable business environment, for instance in network industries. However, regulation often stifles competition, impedes innovation, reduces economic efficiency, prevents a high standard of performance or creates barriers to trade. Regulation can have particularly detrimental effects at times of profound and rapid changes.

Norway's high overall score in the regulation rankings is due to restrictive outward-oriented policies (high barriers to trade caused by high tariffs) and to substantial public ownership in some sectors. This is partly offset by low barriers to entrepreneurship and a low administrative burden.[52] The high average tariffs are mainly the result of the extremely high tariffs on agricultural products (Table 9). Tariffs on other products are comparable with those of the European Union and the United States. The average collected tariff rate is much lower (0.7 per cent in 1998) than the average most-favoured-nation (MFN) rate reported in Table 9. Tariffs on agricultural products are prohibitive so that they do not yield any revenue. Moreover, trade is tariff-free, except for agricultural products, with the other European Economic Area countries (80 per cent of all manufactured imports come from these countries).

Table 9. **Tariff barriers**

Production weighted average applied MFN tariff rates, as a per cent of value added, 1996

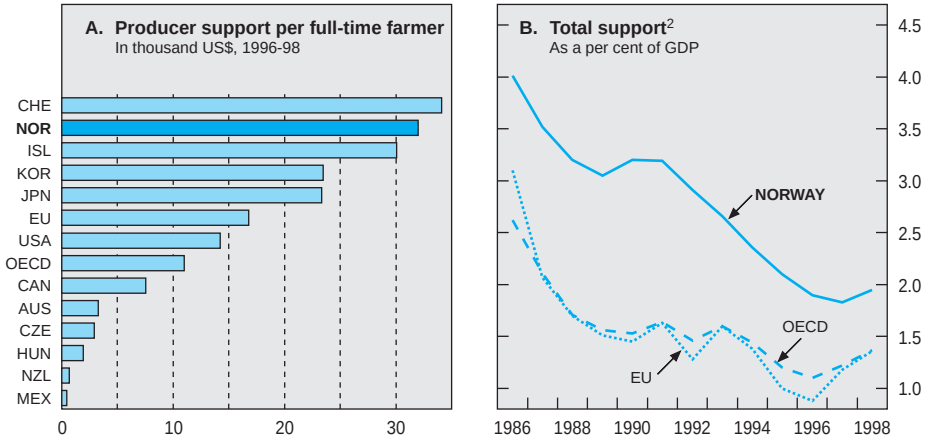
ISIC	Description	Norway	European Union	United States	Japan
1	Agriculture, forestry, fishing	60.3	10.7	7.9	5.0
2	Mining, quarrying	3.0	0.6	0.2	0.3
22	Crude petroleum	3.1	0.9	0.1	..
3	Manufacturing	33.4	7.7	5.4	3.3
31	Food, beverages and tobacco	135.1	32.5	15.9	18.9
32	Textiles and apparel	12.0	9.8	11.3	10.1
33	Wood and wood products	1.8	3.4	3.5	3.6
34	Paper and paper products	2.3	4.7	1.8	1.2
35	Chemicals, petroleum products	4.5	5.3	4.4	3.2
36	Non-metallic mineral products	1.9	3.9	4.5	1.5
37	Basic metal industries	1.1	3.6	3.7	3.0
38	Fabricated metal products	4.0	4.3	3.2	0.3
39	Other manufacturing	3.3	4.2	4.8	2.5
Total all products		**22.3**	**7.7**	**5.2**	**3.4**
Simple average applied tariff rate (%)		30.3	9.5	6.2	6.7

Source: OECD (1997), *Indicators of Tariff and Non-tariff Trade Barriers.*

Government support to agriculture has remained extremely high

Support to the agricultural sector, already the highest in the OECD after Switzerland, has risen in 1998 and in 1999 (Figure 22). This was mainly due to a fall in world prices resulting in higher market price support and to a lesser extent due to increases in administered prices. The high subsidies reflect the aim of maintaining a level of agricultural income comparable with that in the rest of the economy. Moreover, Norway with its arctic and sub-arctic conditions is a low-yield area. In dairy and cereal production, costs in Norway are four to seven times above those in the most efficient producer countries (Ministry of Agriculture, 1999). In 1999, government outlays for agricultural support were NOK 12½ billion (1.0 per cent of GDP). Producer support per farmer was US$33 000 (NOK 250 000) in 1998, compared with US$19 000 and US$11 000 for the European Union and the OECD, respectively (OECD, 1999).[53] Only 30 per cent of total gross farm receipts consisted of earned income, with 70 per cent accounted for by support payments (direct support and market price support derived from border protection), by far the highest rate in Europe and more than twice the OECD average. Norwegian consumers are paying 110 per cent above world market prices for agricultural goods. Meat prices are 50 per cent higher than in neighbouring Sweden (Myran and Sandberg, 1999), leading to significant purchases of agricultural goods by Norwegian consumers in Sweden.

Figure 22. **Agricultural support**[1]



1. Provisional data for 1997-98.
2. EU 12 up to 1994, EU 15 from 1995. OECD excludes the Czech Republic, Hungary, Korea, Mexico and Poland.

Source: OECD (1999), *Agricultural Policies in OECD Countries, Monitoring and Evaluation.*

Agricultural support has undesired side effects.[54] For instance, subsidies, instead of supporting farmers, could support land owners insofar as aid is capitalised into land prices (OECD, 1998). This could even lead to unintended effects in Norway where the owner farms 80 per cent of the land, as intergenerational transfers may move wealth outside the farm sector.[55] Furthermore, as described in the special Chapter of the 1999 *Survey* on Sustainable Development, output-related support may lead to environmentally-harmful increases in agricultural production. Environmental objectives would be better promoted by support that is not linked to output.[56] Finally, agricultural policy is not a very effective way to achieve regional objectives. The slow trend towards increased market orientation seen elsewhere in the OECD is hardly discernible in Norway, where the drop in agricultural support has been the smallest in the OECD countries over the past decade.[57] On the other hand, Norway has the lowest share of market price support in total support of all OECD countries. In the case of unchanged policy, total agricultural support may even rise in the medium-term due to an increase in market price support caused by some drop in world prices resulting from the agricultural reform of the European Union agreed on in March 1999.

The Norwegian agricultural policy aims at a more competitive agricultural sector in Norway and a reduction of support but, at the same time, attaches great value to non-trade concerns (NTCs) such as food security, the viability of rural areas, the agricultural landscape and agro-biological diversity (Ministry of Agriculture, 1999). These NTCs are considered as public goods or positive externalities related to agricultural activity that are to a large extent provided as joint products by agricultural production itself. According to the Norwegian agricultural authorities, without border protection and output-related subsidies, the agricultural sector could provide this multifunctionality only at unacceptably high administration and control costs. But, as suggested in the previous paragraph, there are arguments that run counter to this assessment and new international trade negotiations are likely to increase the pressure for structural reform in the Norwegian agricultural sector.[58]

Subsidies to the manufacturing and service sectors have increased in 1999

The downward trend in government support to the manufacturing and service sectors was interrupted in 1999 and support rose by an estimated 6 per cent in real terms (Table 10). This rise is entirely due to the rise in shipbuilding subsidies to the maximum allowed rates under the European Economic Area (EEA) Agreement.[59] Subsidies to the shipbuilding sector are estimated to have risen to NOK 2.0 billion (in 1998 prices), but are expected to drop to a quarter of that in 2000, as under the existing EEA regulations all operating support for shipbuilding contracts has to be ended on 1 January 2001.[60] The ending of the shipbuilding subsidies may raise the pace of restructuring in this sector.[61]

Table 10. **Support programmes**

Amount of subsidies budgeted, in million NOK, 1998 prices[1]

	1995	1996	1997	1998	1999[2]	2000[3]	% of GDP 1999[3]
Total	20 397	19 912	19 142	18 543	18 767	17 419	1.69
Industry	6 112	6 079	5 501	5 249	5 621	4 304	0.50
Sectoral	2 635	2 778	1 979	2 295	2 720	1 378	0.24
of which: Shipyards	1 339	1 305	878	1 347	1 960	523	0.18
Regional	1 531	1 299	1 530	1 302	1 347	1 398	0.12
Research and development	719	811	857	726	675	739	0.06
Small and medium-sized enterprises	499	527	751	523	511	383	0.05
Export promotion	337	358	276	261	231	264	0.02
State enterprises	391	306	108	142	137	142	0.01
Agriculture, forestry and fishing	13 756	13 445	13 133	12 992	12 980	12 928	1.17
Labour market programmes	279	222	152	106	39	48	0.00
Environment and energy	171	131	174	148	89	104	0.01
Other	79	35	182	49	38	36	0.00

1. Using the GDP deflator for mainland Norway.
2. Estimate.
3. Projection.

Source: Ministry of Finance (1999), *St.meld. nr. 1, Nasjonalbudsjettet 2000.*

Regional considerations continue to motivate a major part of public support to the manufacturing and service sectors, reflected in the expected total increase in regional subsidies of 7 per cent in real terms in 1999-2000, the strongest cumulative rise in support.

An ambitious target has been set for R&D expenditures

Despite their importance for productivity increases in the medium and long term, research and development (R&D) expenditures are relatively low in Norway. In 1997, privately-financed R&D expenditures were 1.0 per cent of GDP compared with 1.5 per cent on average in the OECD and 1.9 per cent on average in the Nordic countries (OECD, 1999*b*).[62] Publicly financed R&D expenditures, 0.7 per cent of GDP, are comparable with levels elsewhere in the OECD. The authorities consider that the low R&D intensity is largely due to the industrial structure and the predominance of small and medium-sized industrial enterprises that traditionally carry out little research.[63] Nevertheless, in its White Paper on R&D of June 1999, the government set the goal of reaching the OECD average for total R&D expenditures in five years (Ministry of Education, Research and Church Affairs, 1999). To reach this target, in 1999 the government established a Research and Innovation Fund with a capital of NOK 3 billion (1/4 per cent of GDP), financed

by privatisation proceeds. The impact will, however, be small as only the yield on the investments of the fund will be spent on R&D. Furthermore, in 2000, the government will consider tax incentives and other measures to stimulate research by Norwegian enterprises after receiving the report of a committee analysing the options. The tax treatment of R&D is at the moment one of the least favourable in the OECD area (OECD, 1999*c* and Guellec *et al.*, 1999).[64] Finally, the government intends to change the funding of universities, with budgets based to a greater degree on research activity and to a lesser extent on the number of students. Concrete steps in this direction have, however, not yet been taken.[65]

Public telecommunication operator to be partially privatised

The OECD's product market indicators presented in Figure 21 describe the regulatory environment in 1998. Privatisation thereafter has reduced government involvement somewhat in Norway, as in many other countries.[66] In 2000, a partial privatisation is planned of the Norwegian telecommunication company *Telenor*.[67] The listing of Telenor will increase the public stake in companies listed on the Oslo stock exchange. In December 1999, public ownership was 17 per cent of the market value of the Oslo stock exchange, 8 percentage points below the 1994 record level.[68]

Norway will be one of the last OECD countries that privatises its public telecommunication operator (PTO). Despite the difficulty of finding clear empirical evidence of the negative effects of public ownership on performance of the telecommunication sector, most countries started to privatise their PTO early in the 1990s. On average in the OECD, PTOs were already privatised for slightly less than 50 per cent (unweighted average) in 1998 (OECD, 1999*d*).[69] Technological developments and deregulation in the EEA are likely to increase competition further in the coming years. Speedy privatisation would contribute to levelling the playing field, nationally and internationally.[70]

Is the oil sector on the eve of a major reform?

The White Paper on petroleum activities, to be published in early 2000, may lead to a wide-ranging reform.[71] The major reasons for a comprehensive policy assessment of state ownership and competitiveness are the wave of mergers and acquisitions taking place in the international oil industry, the maturing production on the Norwegian continental shelf, the changes in the EU gas industry and the increased international competition due to the opening up of new production areas.[72] In early 1999, the government selected external advisers who will contribute to the evaluation of the state involvement in the sector and also asked the two major Norwegian oil companies, *Statoil* (fully state-owned) and *Norsk Hydro* (51 per cent state-owned in early 1999) for their views concerning future reforms. Both companies suggested a partial privatisation of Statoil and a reduction in

government involvement through the SDFI (State Direct Financial Interest).[73] In 1999, the consolidation wave also hit Norway and led to the takeover of *Saga Petroleum*, the third Norwegian oil company, by Norsk Hydro, the second Norwegian oil company, after a takeover bid by the French oil company *Elf* failed.[74] The takeover reduced the government stake in Norsk Hydro from 51 to 44 per cent.

Furthermore, the government considers allowing production in smaller fields and the residual production of matured major fields (currently undertaken by oil companies) to be carried out by oil-sector supply companies. Such a change would increase competition on the Norwegian shelf and could increase the overall recovery rate of oil reserves. To improve the competitive position of the Norwegian shelf, the government already proposed in its draft 2000 Budget the gradual phasing out of royalties on oil fields approved for development before 1986 and a 20 per cent cut in the carbon dioxide (CO_2) tax for the offshore oil sector.

The government also seeks to clarify the consequences of the 1998 EU Gas Market Directive on the functioning of the *Gas Negotiation Committee* (GFU) before incorporating the directive in the Norwegian EEA Agreement. Norwegian gas accounts for around 10 per cent of total gas consumption in western Europe and exports are likely to increase in the coming decade. The directive, to be implemented in national legislation of EEA countries by August 2000, aims at increasing competition in the European natural gas market. Competition for Norwegian gas is limited as customers cannot negotiate with individual license holders but have to negotiate with the GFU, the sales agent of Norwegian gas. The committee only includes the Norwegian oil companies Statoil and Norsk Hydro. Gas sales agreements are not field specific. The government, advised by the GFU and the GSC (the *Gas Supply Committee*, representing the eleven largest licensees on the Norwegian shelf), allocates sales contracts to a specific field. In the government's view, this set-up is essential for reaching a high overall recovery rate, low flaring and economies of scale in the transport of gas. The European Commission recently asked foreign oil and gas companies to submit their view on the implications of the GFU for competition.[75]

Government policy concerning petroleum activities has been fairly successful in the past in extracting the resource rent. Estimates by Statistics Norway indicate that about 80 per cent of the cumulative oil and gas rent accrued to the government between 1980 and 1995. Moreover, the industrial policy targets to develop major oil companies and an oil-supply sector have been achieved. The state involvement in the sector helped to gain experience and know-how in oil and gas policy matters. However, in the current circumstances, partial privatisation of Statoil and more room for new entrants are likely to improve investment incentives for future resource developments, with positive consequences for oil production and the government's oil revenues. Moreover, it is likely to facilitate

the further internationalisation of Norwegian oil companies and oil-supply enterprises, a necessary development as Norwegian production nears its peak and new oil provinces are opening up elsewhere.

The position of the Competition Authority has been somewhat strengthened

In 1999, the government proposed to Parliament to strengthen the position of the Norwegian Competition Authority (NCA) by giving it the authority to prohibit mergers and acquisitions (M&As) pending the NCA's analysis of the competition consequences. In the past, speedy integration of companies made it virtually impossible for the NCA to block some M&As. Government also proposed to raise the NCA's possibilities for bilateral co-operation with other national competition authorities. With the increase in international mergers, there is a growing need for bilateral co-operation and exchange of confidential information. For cross-border mergers, the NCA co-operates with the EFTA Surveillance Authority (ESA) and the European Commission, the competition authorities of the European Economic Area. A review of the impact of the EEA agreement on the Norwegian competition policy is under way. This is part of a broader review of competition policy, legislation and supervision. It may lead to an improved demarcation between competencies of the NCA and sector-specific agencies. At the moment there is a need to negotiate the division of labour in the case of overlapping (and sometimes conflicting) competencies. Australian telecommunications are often cited as an example of the feasibility of combining competition policy and regulatory functions within the national competition agency.[76] The broad review may also lead to changes in the appellate process. Currently, decisions of the NCA may be appealed to the Ministry of Labour and Government Administration. The ministry has occasionally overruled decisions for reasons other than enforcing competition, thereby eroding the credibility of the NCA.[77]

Parliament recently granted the NCA greater flexibility in operating its regional offices network, improving the effectiveness of the NCA.[78] Although recently some ministries have moved regulatory functions to separate entities, the major weakness of the Norwegian supervisory system is that ministries combine ownership, policy oversight and regulatory functions in the relevant sectors. This could produce significant conflicts of interest, hamper the transparency of supervision, and lead to insufficient action against anti-competitive activities, with negative consequences for economic efficiency.

Financial markets: state ownership of banks drops somewhat but remains substantial

The government's involvement in the banking industry dropped somewhat in 1999 as it reduced its stake in *Christiania Bank/Kreditkassen* (the second

largest commercial bank; see Table 11) from 51 to about 35 per cent by a public offering. State involvement in the banking sector was further watered down by the transformation of *Kommunalbanken* into a limited company and the sale of 20 per cent of the shares to *Kommunal Landspensjonskasse* (KLP), the pension fund of local government employees. The objective is to provide the bank with more flexibility. In the coming years, the remaining shares will gradually be sold to KLP and to local government authorities. In 1997, Parliament decided to keep at least one third of the shares in *DnB* (the largest bank of Norway) and Kreditkassen in state hands. The Swedish/Finnish *MeritaNordbanken* made a takeover bid for Kreditkassen in September 1999 but the bid was turned down with a reference to the 1997 parliamentary decision. In December 1999, a majority in Parliament agreed that the government should join its interests as an owner in one entity and that the government should own at least one third of this entity. Based on this, the Ministry of Finance has asked the *Government Bank Investment Fund* to consider the options of structural solutions for DnB and Kreditkassen with the objectives to establish a strong financial solution with stable national ownership and to keep main strategic functions (headquarters) in Norway. State ownership in the banking sector is therefore likely to remain substantial. In addition to its stake in DnB and Kreditkassen, the government has full ownership of the traditional state banks: SND, the Housing Bank (providing around 50 per cent of the housing loans for new dwellings) and the State Education Loan Fund. In 1999, the traditional state banks received NOK 1.9 billion from the central government for interest subsidies and NOK 1.1 billion to cover working expenses and losses.[79]

Further concentration and rationalisation occurred in 1999 in the Norwegian banking sector. DnB and Postbanken agreed to merge. As a consequence of the merger, state ownership in DnB increased from 51 to about 60 per cent as

Table 11. **Market shares of banks and other financial institutions**

As a per cent of total assets, at 30 June 1999

	Banks	Finance and mortgage companies	Mutual funds	Non-life insurance	Life insurance	% of total market
DnB Postbanken	26	6	20	..	18	21
Gjensidige NOR	15	13	23	27	16	16
CBK (Christiania/Kreditkassen)	14	18	8	..	3	12
Sparebank 1/Samarb.sp.b./VÅR	14	2	10	5	3	11
Storebrand/Finansbanken	2	0	7	23	32	8
Fokus/Den Danske Bank	4	0	2	..	..	3
Handelsbanken/Bergensbanken	3	1	0	..	..	2
Total	79	39	71	56	73	72

Source: Kredittilsynet.

Postbanken was fully state-owned. Furthermore, *Fokus Bank* (the sixth largest bank) was taken over by *Den Danske Bank* (one of the rare major cross-border mergers and acquisitions in the European financial sector) and *Bergensbanken* was acquired by the Swedish *Handelsbanken*.[80] Finally, the Swedish *Föreningssparbanken* (Swedbank) formed an alliance with *The Sparebank 1 Group* and acquired a 25 per cent stake in it. The number of savings banks has remained high, at 133, but most of them are part of three alliances that have intensified their co-operation. In October 1999, the government announced that – based on the present market situation – it would not approve a merger of DnB and Kreditkassen, given the importance attached to competition in the financial markets. Concentration has also risen in the insurance industry as *Storebrand* agreed to merge non-life insurance activities with the Swedish *Skandia* and the Finnish *Pohjola*, creating the biggest Nordic insurance company with a market share of 20 per cent in the Nordic countries as a whole and 40 per cent in Norway.[81] The trend towards the formation of financial conglomerates led to the announced formation of *Gjensidige NOR*, the merger of *Sparbanken NOR* (Union Bank of Norway) with the mutual insurance company *Gjensidige Insurance*, the merger of The Sparebank 1 Group and the banking/insurance company *VÅR Gruppen*, and to the takeover of *Finansbanken* by *Storebrand*.

The international restructuring and consolidation in the financial service industry is likely to continue.[82] Norway will not be immune to this trend. The reconfiguration will remain driven by technological advances, liberalisation, international consolidation and excess capacity. With the increased institutionalisation of savings and the associated growing role of professional fund managers, maximising shareholder value will become ever more important. This will give a strong incentive to improve efficiency. Poor performers will be driven from the market. An ownership structure sensitive to market forces may therefore be important. In general, partial state-ownership hampers the raising of capital on favourable terms if private domestic and foreign investors assume that a major shareholder may have other objectives than maximising shareholder value. It may also have negative consequences for the borrowing conditions on the interbank lending market. With the ratio of deposits to loans falling for Norwegian banks, they may be obliged to obtain more funds abroad, making favourable borrowing conditions crucial. The international trend in the banking sector is towards greater size. However, Norwegian banks are relatively small, even in a regional context, with the biggest Norwegian bank half the size of the average of the three major Nordic banks. The introduction of the euro may strengthen the trend of increased competition in the euro area, with repercussions for the rest of the EEA. Already, most major Norwegian companies have one foreign bank as one of their core banks. In the future, this will probably also hold for medium-sized companies with, as a consequence, a likely loss of market share for Norwegian banks.

The Banking Law was changed in 1999 to make mergers of mutual insurance companies and savings banks possible, as was proposed in 1998 by the Banking Law Commission. The government is still considering other proposals of the commission such as allowing mutual savings banks to convert into limited companies, if they wish to, and the relaxation of the strict ownership regulations which make it necessary for a takeover bid to be accepted by 90 per cent of the shareholders.[83] Such legislative changes would facilitate acquisitions and create scope for further integration of commercial banks and savings banks. After the takeover bid for *Kreditkassen*, Parliament changed the Act on Financial Activities and Financial Institutions to allow private ownership stakes in banks of up to 25 per cent in case of strategic co-operation and alliances. The maximum stake was 10 per cent, except for the government.

At the moment, the Norwegian financial sector is in a healthy state with relatively high profitability and sound balance sheets (Table 12). Bank profitability increased slightly in the first three quarters of 1999 despite the economic slowdown (Kredittilsynet, 1999). Capital adequacy ratios improved somewhat for commercial banks, while they dropped a little for the savings banks, although from a high level.[84] Non-performing loans increased marginally but remained very low.[85] The loan growth of the three major banks (3.3 per cent compared to the corresponding period of 1998) stayed behind that of other commercial banks (11.5 per cent) and of savings banks (12.0 per cent). Loan growth of the three major banks already fell in 1998 when the Banking, Insurance and Securities Commission of Norway issued warnings on low capital equity ratios due to vigorous bank lending (Kredittilsynet, 1999*b*). Credit from foreign sources continued to increase steeply in 1999; the share of foreign sources in total credit has risen to 23 per cent from 15 per cent in 1990. The introduction of the euro in January 1999 had up to now no major direct effect on the Norwegian financial sector but, as in other smaller European countries, the introduction has probably increased the liquidity premium in the bond market (Financial Times, 1999).

Shares on the Oslo Stock Exchange recovered sharply in 1999 due mainly to the oil price recovery, falling interest rates, takeovers and buy-outs. The index increased 46 per cent between end 1998 and end 1999, the second-strongest rise in western Europe, after having been the worst performer in 1998. Efficiency and transparency of the Oslo stock exchange improved in 1999 as electronic trading for shares and bonds was introduced. The Oslo Stock Exchange agreed to participate in NOREX, the alliance of the Swedish and Danish exchanges. The three bourses will create a common electronic share trading system by 2001, allowing brokers to trade in shares listed on the two other stock exchanges without extra charges. The government has not yet reacted to the 1998 report of the *Committee on the Stock Exchange*. The committee proposed to move supervision of the stock exchange to the *Banking, Insurance and Securities Commission* and to convert the Oslo Stock Exchange into a limited company.

Table 12. **Banking resources in selected OECD countries**

	1985	1990	1995	1996	1997	1998[1]
	Per 10 000 inhabitants					
Branches						
Norway	4.82	4.23	3.58	3.71	3.61	3.49
Finland	5.99	5.66	3.16	2.75	2.54	..
Sweden[2]	1.72	1.57	2.53	2.49	2.45	..
France	..	4.60	4.58	4.51	4.50	..
Germany	6.37	6.26	5.39	5.32	5.24	..
Italy	2.06	2.59	3.68	3.79	3.93	..
United Kingdom[2]	2.40	2.26	1.81	2.05	1.97	..
Employees						
Norway	72	74	63	56	55	52
Finland	75	92	60	53	50	..
Sweden[2]	27	29	44	44	44	..
France	..	78	70	69	68	..
Germany	89	105	89	87	88	..
Italy	56	58	60	58	57	..
United Kingdom[2]	60	71	65	71	70	..
	As a per cent of the average balance sheet total					
Profits before tax						
Norway	0.55	–0.66	1.36	1.23	1.06	0.85
Finland	0.39	0.40	–0.37	0.41	0.86	..
Sweden[2]	0.34	0.22	1.33	1.30	0.62	..
France	..	0.36	0.15	0.18	0.28	..
Germany	0.72	0.48	0.57	0.53	0.47	..
Italy	0.97	1.00	0.36	0.50	0.33	..
United Kingdom[2]	1.09	0.70	1.17	1.15	1.15	..

1. Provisional data.
2. Commercial banks only.
Source: OECD (1999), *Bank Profitability.*

Sustainable development

The previous *Survey* contained a special chapter on environmentally sustainable economic growth. Norway has traditionally held a leading position in the international discussion of sustainable development and stands out for its valuable efforts to stimulate international co-operation on environmental and resource management issues. Substantial initiatives have been undertaken, for instance the development of an integrated institutional framework, and government policy succeeded in reducing the emissions of a large number of pollutants. Direct

regulation has remained an important policy instrument but since the late 1980s tax instruments, including a carbon dioxide (CO_2) tax, have played an increasingly important role in providing incentives for cleaner production and consumption patterns. The *Survey* concluded that there is scope for realising greater cost-effectiveness of environmental measures by increasing the use of economic instruments, such as taxes, subsidies and tradable permits.

Recently, the use of economic instruments, notably taxes, for environmental purposes increased somewhat. The taxes on consumption of electricity, heating oil and diesel oil have been raised in 2000. The diesel oil tax was changed to better reflect sulphur content, while the annual tax on heavy vehicles became dependent on the vehicle's emission standard, and the tonnage tax on ships on environmental aspects. Furthermore, a tax on hazardous substances (trichlorethene and tetrachlorethene) was introduced and the tax on pesticides was increased. However, for competitiveness reasons, the CO_2 tax for the offshore petroleum sector was reduced. Furthermore, in 1999 the government ended the CO_2 and sulphur emission tax on international air traffic, as it may be in conflict with bilateral agreements on international flights.

At the end of 1999, a commission of experts published its report on how to establish a domestic trading system for greenhouse gas emissions based on the Kyoto Protocol. It recommends to introduce in 2008 domestically tradable certificates that give the right to emit a certain quantity of CO_2-equivalents covering close to 90 per cent of Norway's total emissions. It argues that the necessary institutions (exchanges and brokers) can be established without government intervention. The domestic system envisaged will be compatible with the use of Kyoto mechanisms. The commission recommends that all participants will be given direct access to these mechanisms. The government will lose the revenues from existing CO_2 taxes (NOK 8 billion in 1999) when the emissions trading system is introduced. Based on a price of NOK 125 per tonne of CO_2-equivalent, the Norwegian tradable certificates are worth NOK 33 billion. The majority of the commission recommend that the government refrains from issuing certificates free of charge.

Scope for further action

The Norwegian economy could benefit from further structural reform. Table 13 provides a summary of recent reforms, a follow-up to the recommendations of previous surveys, and a synthesis of options for further structural reforms. Concerning the labour market, most needed are reforms that reduce the inflow into the disability pension and early retirement schemes. Furthermore, changes in the rigid regulations concerning fixed-term contracts would increase adaptability to change and could increase labour productivity. Concerning product market

reforms, the focus should be on reducing agricultural subsidies, further deregulation and privatisation, especially in the telecommunication and financial sectors, and the strengthening of competition policy. To take advantage of policy complementarities, reforms should be comprehensive, covering a wide range of structural policies. In this case, they will reinforce each other and further enhance the macroeconomic prospects of the economy.

Table 13. **Recommendations for further structural reform**

Based on previous and current *Surveys* and action taken since early 1999

Previous *Survey*	Action taken	Current *Survey*
I. LABOUR MARKET		
A. Reform unemployment and related benefits		
a) Introduce longer waiting period for unemployment benefits and reform part-time unemployment benefits.		Same as previous *Survey.*
b) Limit disincentives to working arising from benefit and tax system.	Childcare cash benefit scheme was broadened in the beginning of 1999 to cover parents with 3-year-old children.	Broadening of the childcare cash benefit is a reform that goes in the wrong direction.
B. Reduce incentives for early retirement		
a) Limit distortions in benefit and tax system leading to early retirement.	In the 2000 Draft Budget, the government stressed the need to curb the inflow into the early retirement schemes. Specific measures were, however, not announced.	End pension accruals during the period of early retirement and favourable tax treatment of early retirement benefits. Reconsider the possibilities in the public sector to retire early with full pension.
b) Introduce more restrictive medical criteria for use of disability pension system in order to reduce the increasing inflow into the scheme.	In the 2000 Draft Budget, the government stressed the need to curb the inflow into the disability pension scheme. Specific measures were, however, not announced.	Same as previous *Survey.*
C. Increase employment flexibility		
A further easing of job protection legislation is needed.	A committee reviewed the rules on fixed-term contracts and working hours.	Ease the rules on fixed-term contracts and working hours.
D. Make active labour market programmes (ALMPs) more effective		
a) Closely monitor labour market developments by occupation and skill level and target training programmes on emerging mismatches.	With the expected rise in unemployment, the number of ALMP slots will be increased in 2000. Special training programmes will be set up for shipbuilding and other petroleum-related mainland industries.	ALMPs should not only focus on preserving existing competencies but should also take into account restructuring needs of the economy.

Table 13. **Recommendations for further structural reform** *(cont.)*

Based on previous and current *Surveys* and action taken since early 1999

Previous *Survey*	Action taken	Current *Survey*
b) Open up job brokerage to private services.	End to the monopoly of the Public Employment Service (PES) on job placement. Complete liberalisation of hiring out but only limited liberalisation of hiring-in rules.	Same as previous *Survey*. Guarantee a level playing field between private temporary work agencies and the PES.
c) Evaluate ALMPs to better identify those which raise human capital of participants in a cost-effective way.	The effectiveness of ALMPs is assessed annually.	Continue the evaluation of the effectiveness of all ALMPs and end programmes with poor results.
E. Improve labour force skills and competencies		
Ensure that education and training meet evolving needs of labour market. Apprenticeship programme functions well but there are some mismatches.	The government decided to allocate NOK 400 million to the Competence Reform, aimed at raising the skill level of active workers and other adults.	Ensure that education and training meet evolving needs of labour market.
F. Increase wage and labour cost flexibility		
Review existing wage formation arrangements in order to enhance employability of young and low-skilled workers other than those enrolled in apprenticeships.	The Holden Committee was appointed to review the role of incomes policy in stabilising the economy and increasing economic growth.	Review the possibilities to decentralise further wage negotiations in order to enhance the allocative role of the wage system. Do not "buy" wage moderation by policy concessions.
II. PRODUCT MARKETS		
A. Improve product market regulation and promote entrepreneurship		
a) Examine opportunities for privatisation of state-owned companies.	Privatisation process continued.	Speed up the privatisation of wholly and partly state-owned enterprises.
b) Significantly reduce the very high support for the agricultural sector. Reassess regional policies with a view to raising efficiency without jeopardising its objectives.	Support to agricultural sector increased in 1998 and 1999.	Reduce agricultural subsidies. Support should become less linked to output and better targeted to reduce overall costs.

Table 13. **Recommendations for further structural reform** *(cont.)*

Based on previous and current *Surveys* and action taken since early 1999

Previous *Survey*	Action taken	Current *Survey*
c)	The government launched "A simpler Norway" project aimed at reducing the administrative burden for enterprises.	Aim at a lighter regulation of product markets.
d) Review opportunities for further promotion of technology diffusion.	The government published a White Paper on research and development, setting the goal of reaching the OECD average for total R&D expenditures in five years.	Same as previous *Survey.*
e) Proceed with efforts to strengthen the SME sector.		Same as previous *Survey.*
f)	A comprehensive policy review of state ownership in the offshore oil sector will be published in early 2000.	Government involvement in the oil sector should be diminished by a partial privatisation of Statoil. Competition on the Norwegian shelf should be enhanced.
B. Enhance competition		
Strengthen competition policy	The government sent a bill to Parliament proposing to give the Norwegian Competition Authority the right to prohibit M&As pending its analysis of the competition consequences. A broad review of competition policy was started.	Strengthen the position of the Norwegian Competition Authority and review the supervisory role of ministries.
III. FINANCIAL MARKETS		
A. Enhance competition		
a) Examine opportunities for privatisation of state-owned companies, especially the two leading commercial banks.	Privatisation process continued, but the government confirmed the decision taken in 1997 that at least one third of the shares of the two major banks has to be in state hands.	Speed up the privatisation of wholly and partly state-owned enterprises.
b)	Banking Law was changed to make mergers of mutual insurance companies and savings banks possible	Relax the strict takeover regulation. Allow the conversion of savings banks in limited companies.

Table 13. **Recommendations for further structural reform** *(cont.)*

Based on previous and current *Surveys* and action taken since early 1999

Previous *Survey*	Action taken	Current *Survey*
c)		Move the supervision of the stock exchange to the Banking, Insurance and Securities Commission. Convert the Oslo Stock Exchange into a limited company.
IV. PUBLIC SECTOR		
A. Raise public sector efficiency		
a) Reduce government outlays to make income tax cuts possible.	1998 and 1999: taxes were increased to finance additional government spending; the top tax rate for labour income has been raised by 6½ percentage points in 2000.	Same as previous *Survey.*
b) Increase economic efficiency of local government services.		Increase economic efficiency of local government services, for instance by outsourcing.
B. Improve health sector efficiency		
		Introduce better incentives for health providers to raise efficiency.
C. Reform the pension system		
	The government is considering the proposals of the Moland Committee to create a public pension fund and to strengthen the role of private pension funds.	Put the pension system on an actuarially sound footing. Prepare the pension system for the ageing of the population in the coming decades.

Source: OECD Secretariat.

III. The tax system: past reforms and future challenges

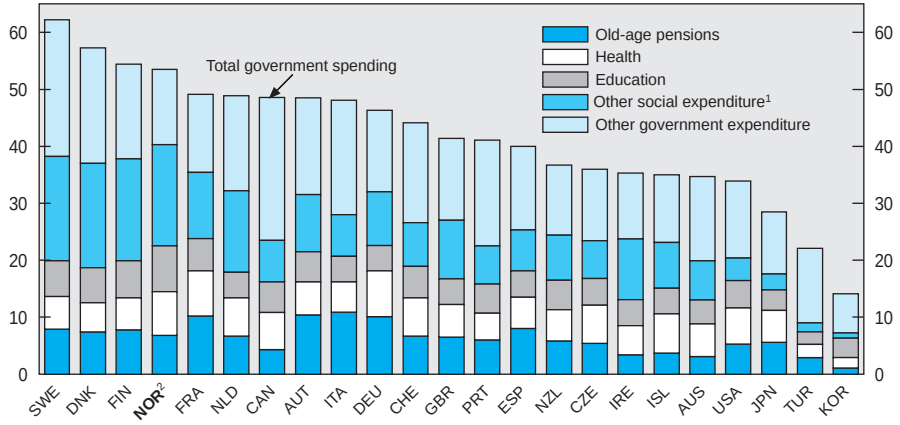
Forces shaping the system

The Norwegian tax system is similar to that observed in the other Nordic countries, owing to a stronger emphasis on income redistribution, a wider social safety net and a broader provision of social services as compared to most other OECD countries. Not surprisingly, reflecting the universal public provision of health care and education services, and high income transfers through the budget, government spending as a share of GDP, and hence the tax burden, is among the highest in the OECD area (Figure 23).

The Norwegian economy is, moreover, characterised by the availability of huge natural resources (oil, gas, hydropower, forestry and fisheries) that in various ways impinge on the tax system.[86] Different tax regimes for corporate income co-exist for mainland industries, shipping[87] and operators on the Norwegian continental shelf. In particular, the "petroleum tax" regime, together with the extraction facilities of the State Direct Financial Interest (SDFI) and the state-owned company Statoil, allow the government to capture the bulk of the oil and gas rent while permitting enterprises to exploit these resources in a profitable and efficient manner. Importantly, the revenues raised in this way have also served to fund the expansion of the welfare state, either directly by sustaining strong growth in public sector employment and social security, or indirectly, by avoiding public debt accumulation and the associated large interest payment obligations most other OECD nations are facing. The shipping tax regime, in contrast, provides an implicit subsidy to the merchant fleet so as to better face the strong international competition prevailing in that industry.

A key to the design of the Norwegian tax system is also the overriding objective to keep remote areas populated, notwithstanding a very low population density overall.[88] To preserve settlement patterns the government aims to keep employment in remote areas and the provision of public services up to the high national standards. This requires a substantial re-allocation of public funds across regions, including huge horizontal transfers of local tax revenues and various

Figure 23. **Government spending**
As a per cent of GDP, 1995



1. Social transfers and services such as unemployment and housing benefits, family services, etc.
2. As a per cent of mainland GDP.

Source: OECD Social expenditure database and OECD (1999), *Education at a Glance*.

forms of differentiated tax treatment, such as regionally-differentiated rates for employers' social security contributions.[89] Moreover, the government deliberately does not capture much of the resource rent from fishery and forestry, in order to promote economic activity in remote areas where these industries are based.

The above features of the Norwegian economy have been conducive to calls for preferential tax treatment of specific sectors or regions, resulting in a tax system that by the late 1980s was blurred by a plethora of special exemptions and allowances. Spurred by developments in other OECD countries, Norway's tax system went through a sweeping base-broadening and rate-cutting reform in the early 1990s. From this reform emerged a fairly lean and rational tax system, even if it still contains some warts in areas where specific regional interests prevail. The centrepiece of the reform was the move towards a pure *dual income tax* in 1992, which strictly separates the taxation of labour and capital income at the level of individual taxpayers. It ensures that capital income is consistently taxed at a low, flat rate, while income from labour is subject to a higher, progressive rate. At the same time, statutory tax rates on both capital and labour income were lowered, tax bases broadened and effective rates harmonised across sources of capital income in order to enhance tax neutrality. Norway is the only country in the OECD

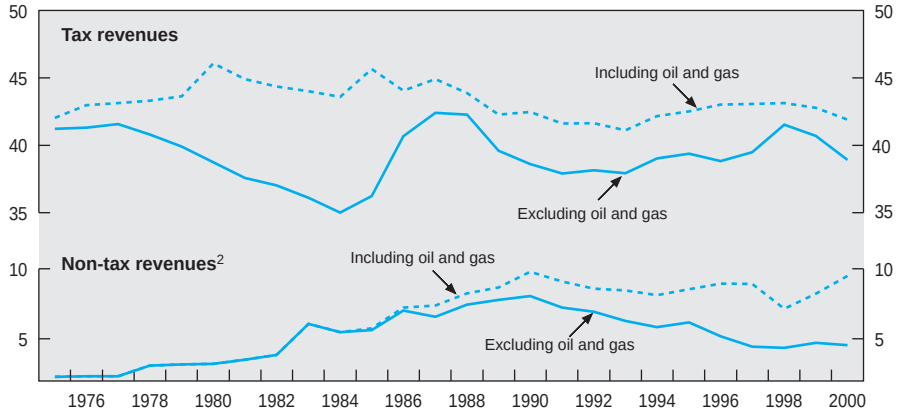
area that has adopted such a pure form of universal dual income taxation. A similar system was introduced in the late 1980s in Denmark and in the early 1990s in Sweden and Finland, and is currently being considered by the Netherlands, but these are less "pure" reform initiatives.[90] Reform efforts since 1992 have focused on "greening" the tax system, by introducing or raising product-specific indirect taxes to improve the environmental impact of economic behaviour, the most prominent example being the CO_2 tax.[91] This has led to an increasing share of green taxes in total tax revenues over the 1990s, which is currently among the highest in the OECD area.

Looking ahead, the Norwegian tax system will, as elsewhere in the OECD, be susceptible to erosion of revenue bases due to the increasing international mobility of financial capital and growing electronic commerce. This development is even more challenging for Norway, since it faces downward pressure on taxes on specific products, such as alcoholic beverages and food, to EU levels in surrounding countries. Meanwhile, as highlighted in Chapter II, human capital investment in Norway needs to better match skills in the face of a tight labour market. The tax system could play a role in this regard, by re-balancing the trade-off between a fair income distribution and ensuring sufficient pecuniary rewards for human capital investment. In the longer run, like most OECD countries, Norway will be confronted with rising pressure on welfare spending as the population is ageing, while public revenues from oil and gas activities are expected to peak soon after the turn of the century. This may require adjustment to both the level and mix of taxation, as well as the removal of tax privileges for retirees.

Key features of the system

Norway taxes mainland activities heavily, despite the substantial budgetary contribution of oil and gas revenues. The latter comprise both tax and non-tax revenues, and have hovered around 8 per cent of GDP over the 1990s (Figure 24). Nevertheless, the share of tax revenues levied on mainland activities amounted to 48 per cent of mainland GDP in 1999, which represents a heavier tax burden than in most other OECD countries. While major distortions are typically associated with such a high tax burden, even more damaging for economic efficiency is the combination of high marginal rates of taxation and narrow tax bases, and differential tax treatment of similar tax bases due to specific characteristics of the taxpayer. In this area Norway has made considerable progress since the early 1990s, although there has been some backsliding recently. Against this backdrop, the sections below examine the key features of the tax system in more detail (for a technical description of the system also see Annex I).

Figure 24. **General government funding**
As a per cent of GDP[1]



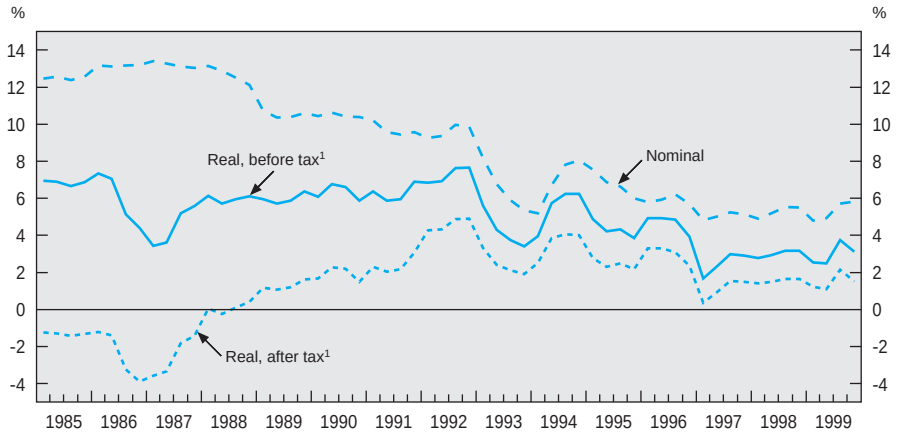
1. Data for 1999 are preliminary and for 2000 projected.
2. Transfers, profits, dividends and interest received.
Source: Statistics Norway and OECD Secretariat.

The legacy of the 1992 reform

The 1992 reform of income taxation was motivated by major tax distortions that were exposed by the liberalisation of financial markets in the early 1980s. In particular, the rapid development of credit markets provided an opportunity for households to take advantage of generous debt-interest tax deductibility, as reflected in negative after tax real interest rates (Figure 25). This induced a debt-financed consumption spree and a housing bubble, which unwound in 1987. With hindsight, the sequencing of reform – financial liberalisation preceding tax reform – was unfortunate. Official committees appointed in the late-1970s and early 1980s foresaw this risk, but it was only when the economy nose-dived that action was taken.

From 1987 onwards the foundation for the tax reform was laid. The tax base of individuals was broadened, the top marginal tax rate against which tax deductions were allowed was gradually reduced and an income-surtax against which no interest deductions were allowed was introduced in 1988. In the same year the government appointed the *Aarbakke Committee*, whose proposal for a broader tax base, lower tax rates and a pure dual income tax system was

Figure 25. **Real interest rates before and after tax**
Government bonds with five-year maturity



1. Deflated by the rate of change of the consumer price index; assuming deduction against the relevant top marginal rate of taxation.
Source: Ministry of Finance.

submitted to Parliament in 1990, where it met little opposition.[92] Since the 1992 tax reform, the thrust of the system has remained broadly unchanged. However, there has been "erosion" due to demands for preferential tax treatment by interest groups, such as the special regime for shipping that was introduced in 1996.

Personal income tax: a "dual" approach

The dual income tax system taxes all forms of capital income at a rate of 28 per cent while labour income is taxed at a higher progressive rate. When the system was implemented in 1992, the top marginal rate of labour income was set at around 50 per cent, but has been raised by 6 percentage points with the introduction of an additional income tax bracket in 2000 (see Box 4). Importantly, work-related and interest expenditure are deductible against the capital income tax rate of 28 per cent only. The dual principle also applies to self-employed and active shareholders in limited companies, for which a "split model" has been designed to separate the labour and capital income components.

The separate taxation of capital and labour income of individuals may be justified in various ways. First, it facilitates the acceptance of relatively low, uniform taxation of capital income – although due to the taxation of net wealth, the effective rate of taxation of capital is somewhat higher than the low statutory

Table 14. **Statutory tax rates by government level and income source**[1]

As a per cent of relevant taxable base

	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Corporate income										
Central	29.80	17.00	→	18.50	19.75	20.75	21.25	28.00	→	→
Local	21.00	11.00	→	9.50	8.25	7.25	6.75	0.00	→	→
Municipalities	13.50	7.00	→	5.50	→	4.75	4.25	0.00	→	→
Countries	7.50	4.00	→	→	2.75	2.50	→	0.00	→	→
Total	50.80	28.00	→	→	→	→	→	→	→	→
Ordinary income[2]										
Central	5.50	7.00	→	7.75	8.75	9.25	9.75	11.00	9.90	10.35
Local	21.00	21.00	→	20.25	19.25	18.75	18.25	17.00	18.10	17.65
Municipalities	13.50	13.50	→	13.00	12.25	11.75	11.50	10.75	11.50	11.20
Countries	7.50	7.50	→	7.25	7.00	→	6.75	6.25	6.60	6.45
Total	26.50	28.00	→	→	→	→	→	→	→	→
Personal income[3]										
National income tax										
1st bracket	7.00	Abolished		..	..	..	..	..	..	..
2nd bracket	14.00	Abolished		..	..	..	..	..	..	..
National surtax										
1st bracket	9.50	9.50	→	→	→	→	→	→	0.00	→
2nd bracket[4]	9.50	13.00	13.70	→	→	→	→	→	13.50	→
3rd bracket[5]	..	..	..	..	..	..	..	..	..	19.50
National insurance scheme										
Wage earners	7.80	7.80	→	→	→	→	→	→	→	→
Self-employed	12.70	10.70	→	→	→	→	→	→	→	→
Pensioners	1.60	3.00	→	→	→	→	→	→	→	→
Memorandum item:										
Top marginal rate on labour income[6]	57.80	48.80	49.50	→	→	→	→	→	49.30	55.30

1. An arrow indicates that there has been no change in the rate over the period covered.
2. Net of allowances. Ordinary income is personal income net of work-related deductions.
3. Gross income from employment, self-employment and pensions.
4. Applies to earnings above NOK 277 800 (class 1) or NOK 329 000 (class 2) as of 1 January 2000.
5. As of 1 January 2000; applies to earnings above NOK 762 700.
6. Tax rate on ordinary income plus top rate on personal income, including National insurance scheme contributions.

Source: Ministry of Finance.

corporate income tax rate suggests. Efficiency considerations strongly favour lower taxation of (mobile) capital rather than of labour in order to minimise tax distortions and to shield the tax system from erosion associated with international mobility of tax bases. Second, the relatively low taxation of personal capital income stimulates personal saving. In particular, with the 1992 reform the average marginal tax rate on net interest income has declined by approximately 5 percentage points. The dual income tax therefore turned out to be a pragmatic way of

Box 4. The dual income tax

Norway adopted considerable changes in its personal income tax system in 1992 (Table 14). Prior to the reform labour and capital income of individuals were both taxed according to a progressive rate structure, with a top marginal rate for labour income of 57.8 per cent (62.7 per cent for self-employed), whereas corporate income was taxed at a flat rate of over 50 per cent. Since 1992, labour income is taxed according to a progressive rate structure whereas capital income of individuals and corporations is taxed at a uniform flat rate. To achieve this, all income of individuals is first taxed jointly at the corporate income tax rate of 28 per cent. The taxable income computed for this purpose is the ordinary income, which is personal income (including realised capital gains and imputed rent income from owner-occupied housing) net of deductions. There are deductions for interest expenses, a standard personal allowance (NOK 26 300 in 1999 for a single earner),[1] a standard deduction of 21 per cent for wage and pension income up to a ceiling of NOK 34 900 in 1999 and standard family allowances. In addition, as of January 2000 employees may opt for a special allowance of NOK 30 600 if that were more favourable to them than the standard deduction (which in 2000 was raised from 21 to 22 per cent with a ceiling of NOK 36 600). Next, personal income, which is not eligible to deductions and comprises labour, self-employed labour or pension income, is subject to a surtax at a rate of 13.5 per cent above a certain threshold (NOK 269 100 in 1999 for a single earner). In addition there is a social security contribution on labour, self-employed labour and pension income of 7.8, 10.7 and 3 per cent, respectively.[2] Accordingly, the top marginal rate amounts to 49.3 per cent for a salaried worker, 52.2 per cent for a self-employed worker and 44.5 per cent for a pensioner. In 2000 a new bracket in the surtax, applied to personal (labour and pension) income of NOK 762 700 and above, has been introduced. The rate is 19.5 per cent, which raises the top marginal rate for personal income by 6 (= 19.5 – 13.5) percentage points.

In order to apply the dual principle to individuals that combine income from labour and capital stemming from a single source (self-employed, shareholders that are actively involved in the day-to-day business of their company), a so-called "split model" has been developed. This divides the assessed income from business activity into ordinary income (total business income less deductions for *e.g.* depreciation and interest payments) and imputed personal income. Ordinary income is taxed at 28 per cent, while imputed personal income is liable to social security contributions and the surtax according to the rate schedule for self-employed individuals. Personal income is computed by subtracting an imputed capital income, augmented by a fraction of salaries paid to employees,[3] from total business income. The imputed capital income is calculated by multiplying the capital stock with an assumed rate of return on capital – which is fixed annually by the Parliament on the basis of the average government bond rate (5 per cent in 1999 and 2000), plus a risk premium (6 per cent in 1999 and 5 per cent in 2000).

For most practical purposes active shareholders and self-employed individuals are treated in the same way within the split model. However, since 1995, shareholders who work less than 300 hours annually in their business are not considered to be active owners and hence all their income is treated as capital income – except for liberal professions for whom the 300 hours rule was again abolished in 1998. On the other hand, active shareholders may escape the personal income surtax if they own

(continued on next page)

(continued)

less than two-thirds of the shares or have the right to less than two-thirds of the dividends. Since most of such "closely held" firms are in fact family-owned, the main owner could escape the personal income surtax on business income by selling shares to family members. To avoid this loophole "identification" rules were established to define which persons may, or may not, be regarded as passive. In particular, an active owner who sold his shares to his spouse would not escape the tax on personal income, as the spouse would have to pay the tax instead. The identification rules were eased in 1997, however, with shares in the hands of the active owner's parents, spouse or partner and children older than 18 years no longer taken into consideration. This measure has again been reversed in 2000 for liberal professions, with, moreover, shares in the hands of the active owner's siblings and relatives of his spouse or partner also taken into account.

The split model contains income thresholds above which personal income is taxed as capital income. These thresholds have been changed several times. Initially the split model contained a ceiling of 34 times the so-called "basic amount" for social insurance purposes (usually denoted as "G"; the annual average was equal to NOK 46 423 in 1999) which is equivalent to roughly 6 times the average production worker's (APW) wage. This left all personal income above that ceiling taxed at a rate of 28 per cent. The authorities have come to realise that the split model is more accurate for liberal professions than for other occupations. The capital base used to calculate imputed capital income is much larger for other occupations than for liberal professions, hence assessment errors have much larger repercussions for the imputed labour income of the former category. Therefore in 1995 separate ceilings for liberal professions and other occupations were introduced – amounting to 75 times G for liberal professions and to 23 times G for other occupations. At the same time, a second ceiling for other occupations was introduced, with income in the range of 75 to 127 times G taxed as personal income. In 1998 the ceiling for liberal professions was entirely abolished while for other occupations the 23 G ceiling was lowered to 16 G and the 75 to 127 G interval was widened to 134 G.

1. Couples can choose between individual or joint assessment, whichever is favourable.
2. Companies also pay an employers' social security contribution on wages and most fringe benefits, from which self-employed workers and pensioners are exempt.
3. Initially this fraction was fixed at 12 per cent, but it was raised to 20 per cent in 1995. The purpose of this allowance is to capture capital income stemming from goodwill not quantified in the balance sheet.

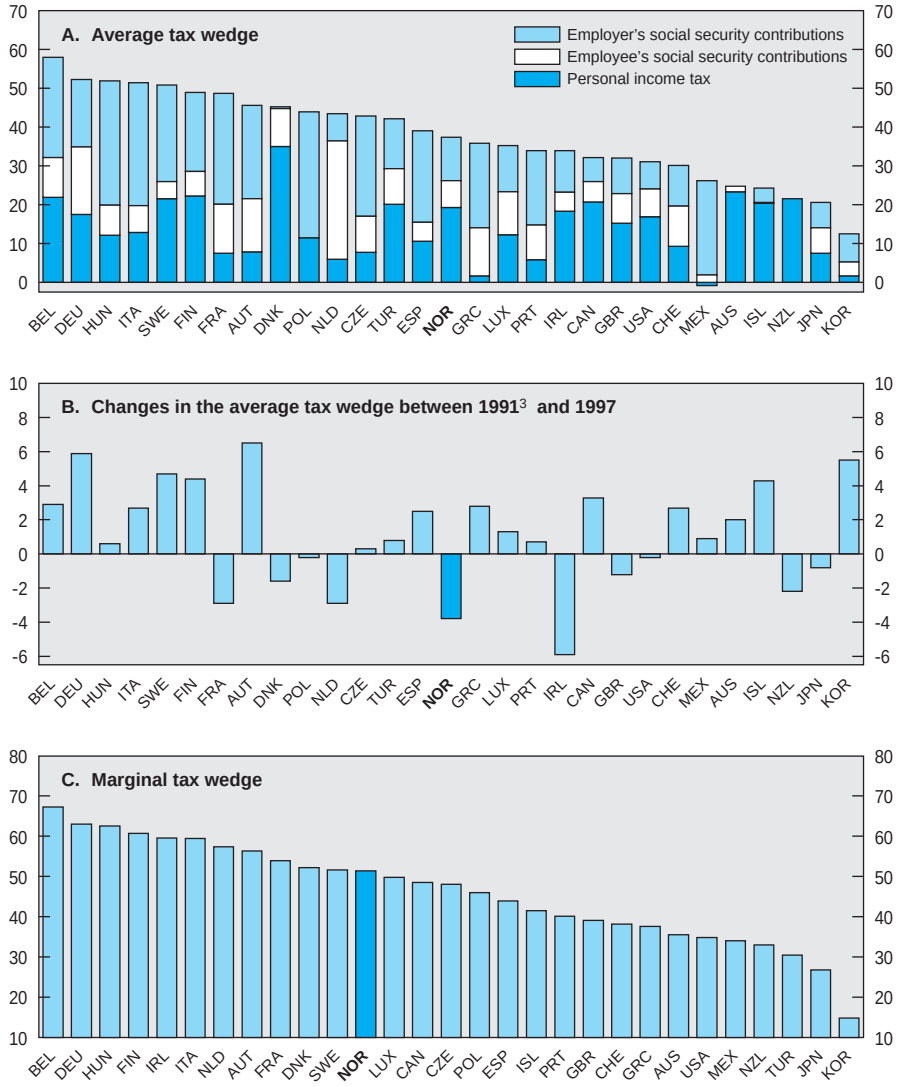
reducing the generosity of the tax treatment of debt-funded owner-occupied housing (although the bias towards this form of wealth formation was not entirely removed; see below). Third, by facilitating the symmetric treatment of capital income across sources, the dual income tax removes arbitrage gains from accumulating returns in a low-taxed corporation while deducting interest payments against a high personal tax rate.

Looking at the taxation of labour, it is striking that the average tax wedge declined substantially with the tax reform and, at around 35 per cent of average labour cost, is below the levels prevailing in continental Europe (Figure 26, panels A and B). Moreover, the marginal tax wedge, which is an important indicator for work incentives, is also somewhat lower than in most continental European countries (Figure 26, panel C). This relatively favourable picture in part reflects the comparative bias toward indirect taxes (see below), which are not included in the reported tax wedges. Interestingly, part-time workers without secondary earnings from a partner face a much smaller marginal effective tax rate in Norway than in most other OECD countries, owing to the combination of a low personal tax allowance and a relatively flat rate structure (Table 15). This may help explain why part-time and temporary work, which represent an important source of "flexible" labour supply, increased substantially in the recent upswing. Unfortunately, a change in the personal income tax system in 2000, whereby workers below a certain income threshold may opt for a higher standard personal allowance with a more progressive rate structure, will partly undo this favourable property (see Box 4).

The taxation of firms: focus on neutrality

From the 1992 reform a tax system emerged that is neutral with regard to sources of capital income (distributed and non-distributed corporate profits, other business income, interest income and capital gains).[93] A key element of the reform was the introduction of one single statutory tax rate for all capital income, of 28 per cent, against which interest expenditure and depreciation allowances can be deducted. At the same time, tax depreciation rates were lowered and the practice whereby corporations could defer taxes indefinitely by allocating profits to a so-called consolidation fund – thereby reducing the statutory tax rate from 50.8 to 39.1 per cent – was abolished. Finally, double taxation of dividends and capital gains was removed.[94] In particular, the personal tax liability on dividends was reduced by a tax credit corresponding to the corporate tax rate, in line with the *imputation principle*. Since in the Norwegian case the tax rate on capital income of individuals is the same as the corporate tax rate, the Norwegian shareholder is entitled to a full tax credit on dividends received.[95] To avoid also double taxation of retained profits to the extent these are reflected in capital gains on shares, a so-called RISK method (or opening value adjustment method) was introduced.[96] This method assumes that retained profits are in fact deferred dividends that are reflected in the share values, hence the associated capital gains are already taxed at the company level. Therefore, only capital gains that exceed the increase in the stock of retained earnings of the company are taxed.[97] As a result, Norway not only went further than other countries in abolishing income allowances, tax deferrals and tax credits, it also went further in terms of cutting the statutory tax rate

Figure 26. **Tax wedge on labour**[1]
As a per cent of gross labour costs,[2] 1997



1. Single individual at the income level of the average production worker.
2. Gross wage plus employers' contributions.
3. 1993 for the Czech Republic and Poland, 1994 for France, 1995 for Hungary and Korea.

Source: OECD (1999), *The Tax/Benefit Position of Employees, 1997*.

Table 15. **Marginal effective tax rates on additional income for different family types**[1]

1997

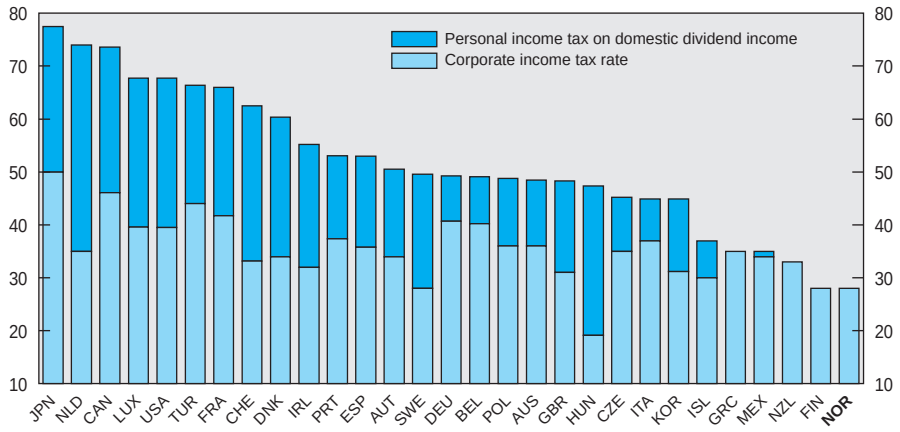
Principal earner	Part-time employed	Full-time employed			Unemployed	
Secondary earner	Non-employed	Full-time employed	Part-time employed	Non-employed	Full-time employed	Part-time employed without benefit entitlements
Poland	58	19	17	48	19	17
Australia	60	29	15	78	78	60
France	69	28	38	76	29	30
Norway	**77**	**47**	**43**	**77**	**42**	**35**
Spain	77	23	19	78	23	19
Sweden	79	37	42	88	43	42
Ireland	83	32	25	68	20	38
Denmark	84	50	48	84	55	61
Italy	84	33	25	63	37	19
Netherlands	90	39	37	89	45	52
United Kingdom	93	28	20	72	60	55
Switzerland	96	30	27	85	20	15
United States	102	19	11	68	20	0
Greece	104	30	30	54	66	118
Canada	105	37	33	75	34	29
Hungary	106	29	12	73	34	23
Belgium	109	57	61	68	43	25
Germany	115	51	50	80	31	19
Finland	117	36	23	88	48	23
Korea	129	13	9	55	2	1
Japan	133	12	10	60	10	7
Austria	135	30	21	76	32	43
Iceland	139	44	56	63	27	49
Czech Republic	162	19	2	85	22	15
Portugal	174	21	13	79	14	11
Luxembourg	198	30	14	87	26	12

1. Marginal effective tax rate = 1 – (net income in work – net income out of work)/change in gross income. Countries are ranked according the rate for a part-time employed principal earner with a non-employed secondary earner. Part-time employment corresponds to 16 hours or two days each week, and total earnings are 40 per cent of the average production worker level of earnings. Earnings from full-time employment correspond to average production worker earnings.

Source: OECD (1999), *Benefit Systems and Work Incentives.*

Figure 27. **Combined corporate and personal income tax wedge on distributed profits**

Resident top earner individuals, per cent, 1998



Source: OECD Tax Data Base.

and double taxation of corporate income (Figure 27).[98] This has led to a reduction in tax planning and investment in tax-sheltered activities, which probably explains part of the increase, in cyclically-adjusted terms, in the corporate-tax take as a per cent of GDP since 1992.[99]

Since 1992, the Norwegian tax system incorporates essential features of neutrality across ways of doing business, sources of finance, etc. In particular, it avoids a number of pitfalls that were inherent to the pre-1992 system and that distorted saving and investment decisions. There were three main sources for distortions prior to the reform: *i)* the weak link between taxable and actual income due to deferred funds and accelerated depreciation allowances; *ii)* the generally low taxation of real capital gains, notably in the shipping and airline industries; and *iii)* the differential taxation across business sectors.[100] Hence, tax administration was cumbersome and tax planning highly lucrative. Moreover, the system tended to "lock in" retained profits due *inter alia* to the double-taxation of dividends. The current system taxes dividends and retained earnings equally and hence investment is more geared to equalise pre-tax returns across economic activities. The (almost) negligible variation in marginal effective tax rates (METRs) across sources of funding and types of investment that are implied by the current system illustrate this (Table 16). METRs gauge the *ex ante* tax wedge between the pre and after-tax rates of return on investment and savings for different

Table 16. **Marginal effective tax wedges on physical investment, R&D and human capital**[1]

	Manufacturing, 1998							1996			
	Standard deviation	Sources of financing[2]			Physical assets[3]			R&D[4]		Human capital	
		Retained earnings	New equity	Debt	Machinery	Building	Inventories	Short lived	Long lived	Training[5]	Tertiary studies
New Zealand	0.3	1.8	1.8	1.8	1.7	1.5	2.3	0.7	0.3	0.0	..
Germany	0.3	1.4	0.9	1.3	1.1	1.7	1.3	0.0	0.0	–0.2	–0.4
Norway	**0.3**	**1.3**	**1.3**	**1.3**	**1.0**	**1.2**	**1.9**	**0.1**	**0.1**	**0.0**	**..**
Australia	0.4	2.6	2.5	2.5	2.1	2.7	3.4	–6.0	–0.9	0.9	–0.6
Spain	0.5	3.9	3.2	2.6	3.1	3.6	3.7	–7.1	–0.8	2.0	–0.1
Mexico	0.5	1.5	1.5	0.2	0.9	0.9	1.3	–0.3	–0.3	..	..
United Kingdom	0.6	2.2	2.8	1.8	1.7	2.1	3.1	0.8	0.8	0.8	..
Denmark	0.6	2.4	3.2	2.9	2.2	2.6	3.8	–1.7	0.6	1.6	..
Ireland	0.7	1.9	3.4	2.4	1.8	2.1	3.1	0.8	0.8	0.8	–0.8
Finland	0.8	2.5	1.0	1.0	1.4	1.9	2.9	0.7	0.7	0.7	–0.7
Sweden	0.8	2.4	3.4	1.0	1.7	2.1	2.5	1.1	1.1	1.0	–1.8
Greece	0.9	1.8	1.8	–0.1	0.9	0.5	2.4	–0.6	–0.6	–0.6	..
Italy	1.0	2.2	2.5	0.6	1.0	1.8	3.1	0.3	0.3	0.0	–0.1
Iceland	1.0	2.2	2.7	–0.1	1.0	1.6	2.3	1.3	1.3	1.0	..
Luxembourg	1.1	4.0	2.7	1.8	2.4	3.0	4.7	1.7	1.7	1.6	..
Switzerland	1.2	0.5	4.1	2.2	1.3	1.6	1.6	0.5	0.5	0.4	–0.3
Austria	1.3	0.9	3.2	0.1	0.0	1.1	2.6	–2.4	–0.8	–0.1	–0.8
Portugal	1.4	2.1	4.4	0.0	1.5	1.5	2.0	–0.2	–0.2	–0.3	–0.7
United States	1.5	2.0	5.7	1.7	1.7	3.0	2.6	–3.8	–0.2	1.0	0.0
Belgium	1.5	1.6	3.0	–0.7	0.1	0.8	3.1	–0.5	–0.5	–0.5	..
Canada	1.7	4.7	5.7	1.4	2.5	4.3	5.5	–4.0	–0.4	1.1	–0.7
Netherlands	2.0	0.6	6.3	2.9	1.8	2.3	2.0	–3.6	–0.1	1.0	–0.5
France	2.6	4.4	8.5	0.8	2.6	4.1	4.8	–1.1	0.1	0.5	0.0
Japan	2.7	4.5	7.4	–0.3	1.8	5.1	3.7	0.2	0.6	0.5	0.7
OECD[6]	1.1	2.3	3.5	1.2	1.6	2.2	2.9	–1.0	0.2	0.6	–0.5
EU[6]	1.1	2.3	3.4	1.2	1.6	2.1	3.0	–0.8	0.2	0.6	–0.6

1. These indicators show the degree to which the personal and corporate tax systems scale up (or down) the real pre-tax rate of return that must be earned on an investment, given that the household can earn a 5% real rate of return on a demand deposit. See OECD (1991), *Taxing Profits in a Global Economy: Domestic and International Issues,* for discussion of this methodology. Calculations are based on top marginal tax rates for the personal income tax and a 2% inflation rate.
2. The weighted average uses the following weights: machinery 50%, buildings 28%, inventories 22%.
3. The weighted average uses the following weights: retained earnings 55%, new equity 10%, debt 35%.
4. The weighted average uses the following weights: machinery 5%, buildings 5%, current expenditure across assets 90%, and weights in footnote 3 for financing.
5. The weighted average uses weights in footnote 3 for financing.
6. Simple average across available countries.

Source: OECD Secretariat.

hypothetical marginal investment projects. A tax system where different investment projects and savings instruments yield the same pre-tax return for a given after-tax return can be called *neutral*, which is indeed a key feature of the Norwegian system. A similar degree of neutrality is only found in Germany and New Zealand (see the first column of Table 16), which also employ a full imputation system and, moreover, do not tax capital gains on shares. Nevertheless, neutrality has not been achieved in all respects, in particular since owner-occupied housing and shipping activities continue to receive a favourable tax treatment.[101]

Welfare gains from the reform

Statistics Norway estimates suggest that welfare increased by ¾ per cent due to the reform.[102] This should be considered as a minimum estimate, however, since it incorporates only the effects of the reallocation of labour and capital across various industries, thus ignoring possible dynamic effects on overall capital accumulation and labour supply.[103] The favourable performance of the Norwegian economy after the reform was adopted suggests that such dynamic effects may have been important. In particular, the reform has helped to bring about a return of the household saving ratio to levels that prevailed prior to the mid-1980s boom/bust cycle, owing to weaker incentives to borrow. As a result, the household savings ratio moved up rapidly, thus strengthening households' balance sheets and contributing to a shift from consumption to capital formation. Although this development is unlikely to be fully attributable to changes in the tax code, evidence from Sweden, which went through a similar tax reform to Norway, suggests that this might be the case to some extent.[104] Labour supply may have been stimulated by lower marginal effective tax rates, in particular for part-time labour (see above). Labour supply has indeed soared since 1992, although cyclical factors have probably been predominant.

There is little evidence that the tax reform would have contributed to a widening in the after tax income distribution – in fact it remained narrow by international comparison. In effective terms, the progressivity of the personal income tax system was little affected by the reform, as rate cuts at the upper-end of the pay scale were offset by smaller tax rebates (in particular for mortgage lending).[105] As a result, the "vertical" after-tax distribution of wage income was little changed by the reform.[106] On the other hand, the recorded overall vertical income distribution, including capital income received by households, has become more skewed (Table 17). However, this mainly reflects the economic upswing and lower interest rates since 1993, which boosted profits, dividends and realised capital gains, the bulk of which typically flows to high-income groups. Moreover, to the extent that the reform prompted previously unreported earnings to surface and retained profits to be converted into distributed profits, the recorded widening in the income distribution is merely a statistical artefact.

Table 17. **After-tax income distribution**

Per cent

Decile	1986	1990	1992	1996
1	4.1	4.0	3.8	3.7
2	6.1	6.0	5.9	5.6
3	7.2	7.2	7.0	6.9
4	8.2	8.2	8.1	7.9
5	9.1	9.1	9.0	8.8
6	9.9	9.9	9.9	9.7
7	10.9	10.9	10.9	10.6
8	12.1	12.0	12.1	11.7
9	13.8	13.7	13.8	13.4
10	18.6	19.0	19.6	21.7
Gini coefficient	0.222	0.228	0.237	0.257

Source: Ministry of Finance.

One of the aims of the tax reform of 1992 was to ease the administrative burden for the taxpayers and tax authorities. According to the authorities the tax reform represented, on balance, a move in the right direction. The previous income tax system required a very complex and detailed assessment due to many special provisions and exceptions, most of which have been removed by the reform. One of the most onerous elements of the previous system, the tax assessment of partnerships with joint liability or unlimited *pro rata* liability, has been considerably simplified. However, some tax rules introduced by the reform lead to more complexity, in particular the split model and the RISK method for adjusting capital gains on shares.

Differential tax treatment of specific sectors or regions

Taxing natural resource rents

The Norwegian government faces the challenge of ensuring that it captures the important natural resource rents in a non-distorting way. A natural resource levy corresponding to the resource rent is, in principle, not distorting. Indeed, it merely transfers pure economic rent from the natural resource sector to the government while leaving the allocation of labour and capital resources unaffected.

The management of the oil and gas reserves, which represents by far the most significant natural resource in Norway, involves fairly extensive public decision-making, including the allocation of oil and gas acreage.[107] In addition, a special "petroleum" tax regime has been designed so as to ensure that the bulk

of the natural resource rent is captured by the government without compromising, in principle, the production area's efficiency and competitiveness. The present tax regime includes the standard 28 per cent corporate tax net of depreciation allowance and a special surtax of 50 per cent on those profits minus a special allowance (uplift).[108] Through these taxes, the operating surpluses from its own production facilities on the continental shelf (SDFI) and the state-owned company Statoil,[109] the government extracts around 80 per cent of the natural resource rent. The effective tax burden on the continental shelf is not out of line with that observed in other gas and oil production areas in the world that apply similar tax systems,[110] but it is high compared to the mainland. On the other hand, the special depreciation allowances (16⅔ per cent per year), the "uplift" and the deductibility of interest payments against the standard corporate and the surtax create incentives for investments in less profitable fields. As a result, the degree to which the current tax system captures the whole rent has been subject to some debate, and a majority in the Green Tax Commission (1996) recommended a reassessment of the tax rules on the continental shelf.[111] A government commission appointed to this end is expected to present its recommendations by June 2000.

Among the other main natural resources in Norway, only hydropower is subject to a rent tax. Following recommendations by the *Rødseth Committee* in 1992, a rent tax on hydro power stations at a rate of 30 per cent was implemented in 1997.[112] The rent is calculated as net income less the "normal" return and taxed at a rate of 27 per cent. The "normal" return is equal to a market rate of interest multiplied by the value of invested capital as assessed for tax purposes. There are, in contrast, no rent taxes on fishery and forestry activities, nor are rents in these sectors extracted by other means such as the auctioning of quotas or acreage.

Preferential tax treatment

Norway maintains several special tax schemes to support business it considers essential for economic activity in remote areas, or where similar preferential schemes abroad exist. Most importantly, a special tax regime for the shipping industry was adopted in 1996 in an effort to respond to similar moves by other seafaring nations. In particular, shipping companies are exempt from the corporate income tax on retained profits, thereby effectively postponing the tax payment until the profits are distributed.[113] This measure partly restored a facility that existed before the 1992 reform, but that was abolished *inter alia* to remove a strong incentive for investors on the mainland to reduce their tax liability by investing in a shipping company. These incentives have thus been reintroduced to some extent, but in the current set-up investors are not allowed to deduct expenses or losses stemming from shipping companies against taxable profits in other sectors – a facility that existed prior to the 1992 tax reform. Nonetheless, the

introduction of such a special regime for shipping threatens to exacerbate tendencies of international tax base erosion and could prompt demands for special tax treatment by other industries. Calls from the fishery industry to make fishing vessels liable to the same special tax regime have, however, not been heeded to date. Meanwhile, self-employed fishermen benefit from a more lenient interpretation of the split model, with the bulk of business income *de facto* taxed as ordinary (capital) income. In addition, a special tax deduction of 30 per cent (up to a maximum of NOK 70 000) and favourable depreciation rules apply.[114] A somewhat more lenient tax treatment also exists for farmers and forest owners.[115]

Other features: high indirect and low property tax

Looking at the composition of tax revenues by source, the high share of taxation on goods and services stands out (Table 18). This feature, which Norway has in common with the other Nordic countries, is even more marked when the large (corporate and indirect) tax revenue from oil and gas activities offshore is excluded. Excluding oil and gas taxes, the shares of corporate, personal income and social security tax in (mainland) GDP are broadly in line with those found in the European Union, whereas (mainland) taxes on goods and services yield over 5 percentage points more. The differences with the United States and Japan, where taxes on goods and services represent less than 5 per cent of GDP and 20 per cent of total tax revenues (compared to 38 per cent in Norway), are even more marked (Figure 28). Also striking is the much higher share of property taxes in these countries compared to both Norway and the EU.

The large indirect tax take reflects a combination of factors. At 23 per cent, the value-added tax (VAT) – which comprises two-thirds of indirect tax revenues – is high by international comparison, with higher rates found only in Denmark, Iceland and Sweden. Moreover, Norway does not have a reduced VAT rate, but a narrower group of service activities is taxed than in other countries.[116] Norway also has a rather unique investment tax of 7 per cent on capital goods purchases.[117] Finally, Norway levies high excise taxes on specific products. Excises on alcohol and tobacco represent almost a tenth of total indirect tax revenue, while approximately one-fifth of the revenue from indirect taxes consists of environmental ("green") taxes. They consist mainly of vehicle excises, CO_2 and sulphur taxes. At around 3½ per cent of GDP, the environmental tax take is among the highest in the OECD area, and well above the area-average of 2½ per cent.

The strong emphasis on indirect taxation has a number of advantages. Indirect taxes are relatively neutral with respect to saving and investment decisions while the cost of tax collection and administration is relatively low. Green taxes are expected to internalise the adverse external effects of consumption and production on the environment, even though their efficacy could be improved.[118] It is remarkable that a country attaching a large weight to equity has adopted such

Table 18. **Tax shares in GDP**

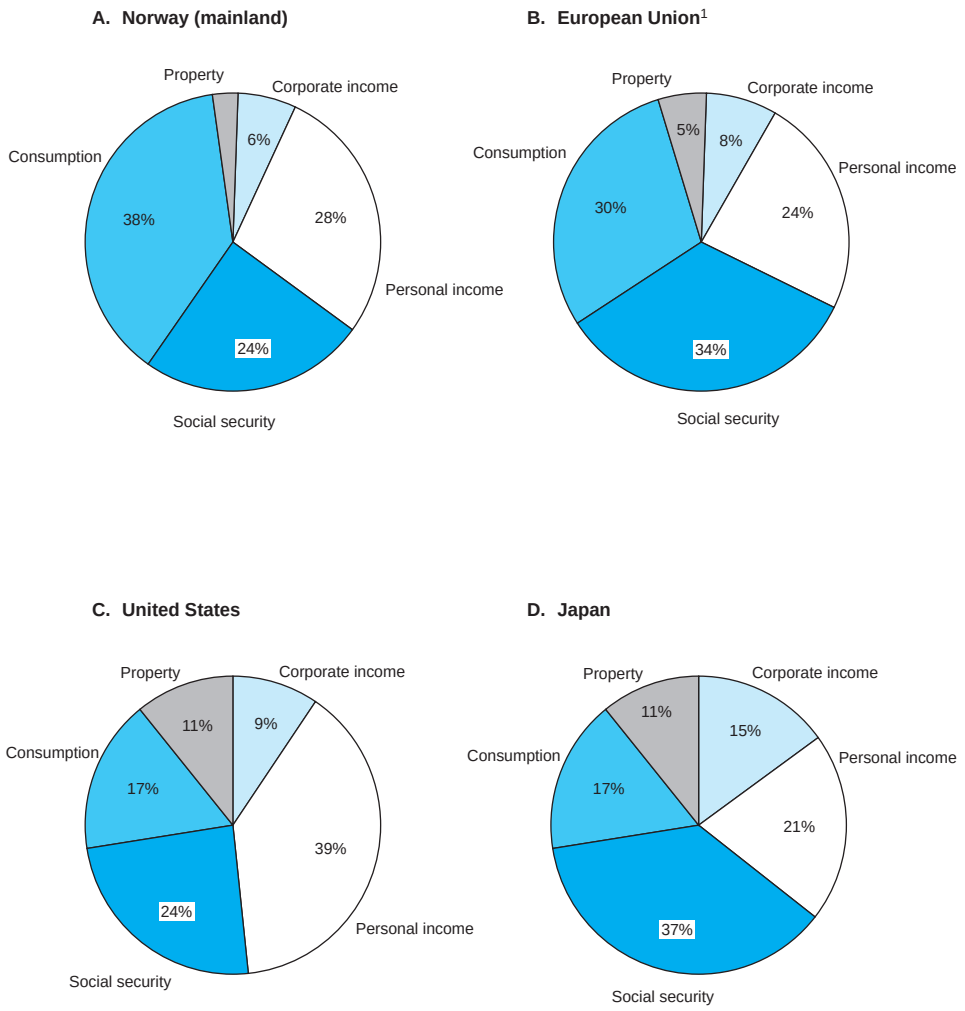
Per cent, 1997

	Corporate income	Personal income	Social security	Goods and services	Property and wealth	Total
Norway	**5.2**	**11.0**	**9.6**	**15.8**	**1.1**	**42.6**
Norway (mainland)[1]	**3.1**	**13.4**	**11.6**	**18.1**	**1.4**	**47.6**
EU[2]	3.5	11.6	11.9	12.6	1.8	41.4
OECD[2]	3.0	9.5	9.3	8.0	2.5	32.4
United States	2.7	11.0	6.9	4.7	3.0	28.3
Japan	4.3	5.9	10.6	4.7	3.1	28.6
Germany	1.5	8.9	15.5	10.3	1.0	37.3
France	2.6	7.4	18.3	12.6	2.4	43.3
Italy	4.2	11.6	14.9	11.5	2.3	44.4
United Kingdom	4.3	8.8	6.1	12.3	3.8	35.3
Canada	3.8	15.1	5.0	9.1	3.7	36.7
Australia	4.5	14.9	..	8.5	2.8	30.7
Austria	2.1	13.5	15.2	12.5	0.6	43.8
Belgium	3.4	14.3	14.6	12.3	1.3	46.0
Czech Republic	3.3	5.2	16.9	12.6	0.5	38.6
Denmark	2.6	27.4	1.6	16.4	1.7	49.6
Finland	3.7	15.5	11.7	14.3	1.1	46.3
Greece	2.1	5.8	10.7	13.8	1.3	33.7
Hungary	1.9	6.7	14.1	15.3	0.6	38.6
Iceland	0.9	10.6	2.8	15.3	2.6	32.2
Ireland	3.3	10.7	4.2	13.0	1.6	32.8
Korea	2.2	3.8	1.9	9.7	2.9	20.5
Luxembourg	8.6	9.5	11.8	12.6	3.6	46.0
Mexico[3]	..	4.2	2.5	9.3	0.3	16.3
Netherlands	4.6	6.8	17.9	12.2	2.0	43.4
New Zealand	3.9	18.1	..	12.7	2.0	36.6
Poland	3.2	9.2	13.2	14.4	1.2	41.2
Portugal	3.7	6.2	8.9	14.5	0.8	34.2
Spain	2.6	7.4	11.8	9.7	2.0	33.6
Sweden	3.2	19.9	15.2	11.6	2.0	51.8
Switzerland	2.0	10.6	12.5	6.2	2.6	33.8
Turkey	1.6	6.0	4.0	10.3	0.8	22.8

1. As a per cent of mainland GDP.
2. Weighted average.
3. 1996 data.

Source: OECD (1999), *Revenue Statistics.*

Figure 28. **Tax mix by source**
Per cent share of total revenue, 1997



1. Weighted average.
Source: OECD (1999), *Revenue Statistics.*

high indirect taxes, which are often considered to widen the real after-tax income distribution – at least in a "static" sense, not considering the distribution of lifetime income. However, in Norway this is deemed to be less of a concern since indirect taxes are tilted towards goods that are considered to be "luxury" goods (purchase and use of passenger cars, alcoholic beverages), and therefore have a progressive thrust, or carry serious health risks (alcoholic beverages, tobacco products). On the other hand, the investment tax entails high administrative costs both for the government and business and is distortive, not in the least due to the many exemptions that have been introduced over time. Although several committees, like the Green Tax Commission and the Aarbakke Committee, have recommended its abolition, the investment tax has been maintained for revenue-raising purposes.

By contrast, the tax take on property and wealth is relatively small, not only in comparison with the United States and Japan, but also relative to some of the larger European countries (France, Italy and the United Kingdom). The municipalities are allowed to levy a property tax and they are free to set the tax rate subject to a ceiling.[119] A net wealth tax is also levied on (both real and financial) net assets but it is split between levels of government, with the respective rates determined by the central government as part of the annual budget process.[120] The valuation of owner-occupied houses for net-wealth and income tax purposes is around 25 per cent of the market value. This is low by international comparison, in particular in comparison with the other Nordic countries.[121] Concerning the net wealth tax, non-listed shares are assessed at 65 per cent of book value, listed shares at full market value and bank deposits at face value, while financial assets accumulated in occupational pension funds escape the tax net.[122] A drawback of this set-up is that very mobile tax bases with extensive avoidance opportunities (listed shares) are taxed at a high effective rate and immobile ones (housing, occupational pensions) at a low or zero effective rate. Moreover, since the tax authorities use the valuation of owner-occupied houses (including secondary residences) to compute the imputed rent income (of 2.5 per cent of the assessed value up to NOK 451 000 and 5 per cent of the value exceeding that threshold), the low assessment of property spills over into the income tax system.[123] As a result, there may be a considerable dead-weight loss associated with the property and net wealth tax. Countries that tax property and wealth at higher rates but employ a rate structure that better reflects the relative mobility of tax bases achieve higher revenues at a lower social efficiency loss.

A unique feature of the Norwegian system is that the employers' social security contributions are used as an important vehicle for maintaining settlements in remote areas. Social security in Norway is administered by the comprehensive National Insurance Scheme (NIS), which provides universal coverage for old age, disability and survivors pensions, as well as individual medical expenses and income compensation for unemployment and sickness. It is run by the central

government and co-funded by employees' and employers' contributions. The former is levied as a flat rate on personal income, with different rates applied to *e.g.* labour and pension income. Employers' contributions are levied on the companies' wage bill according to a regionally-differentiated rate depending on the permanent residence of the employees; rates vary between 0 and 14.1 per cent, with the lowest rates applied to employees who live in the northern part of Norway. In 1998 the EFTA Court ruled that Norway should discontinue the rate differentiation in its present form as it was seen to conflict with EEA regulations. After the Court dismissed a Norwegian appeal in May 1999, the government pledged to abolish the rate differentiation in certain industries (notably steel products and shipbuilding).

A salient feature of the tax system in Norway is that local governments (municipalities and counties) are automatically entitled to a percentage of the tax collected in their jurisdiction. Such an arrangement exists for net wealth taxes (see above), and, more importantly, also for taxes on ordinary income (capital, labour and pension income net of deductions) of individuals (Table 14). Around half of local governments' budgets are financed by locally collected taxes, the bulk of which consists of tax on ordinary income.[124] This is reflected in a substantial local tax take (almost 10 per cent of mainland GDP and 25 per cent of total tax revenues excluding social security; see Tables 19 and 20), akin to levels found in federal countries like the United States, Germany and Switzerland. However, unlike federal states, local governments in Norway have practically no tax autonomy, as the central government determines both the tax base and the rate structure.[125] Such a combination of tax sharing between levels of government and the virtual absence of local tax autonomy is rather unique in the OECD.[126]

Table 19. **Funding of local government**

1999

	Billion NOK	Per cent
General revenues	139	70.2
Tax revenue	90	45.5
Block grants	49	24.7
Earmarked grants	30	15.2
Charges and fees	25	12.6
Other revenues	4	2.0
Total revenues	198	100.0

Source: OECD (1999), *Tax Policy Studies No.1, Taxing Powers of State and Local Government.*

Table 20. **Tax revenues by level of government**

1997

	As a per cent of GDP			As a per cent of total revenue[1]		
	Federal or central government	Social security funds	State and local governments	Federal or central government	Social security funds	State and local governments
Norway	25.8	9.0	7.8	60.5	21.2	18.3
Norway (mainland)[2]	27.1	11.0	9.5	56.8	23.1	20.0
United States	12.5	6.9	9.0	44.1	24.2	31.7
Japan	10.9	10.6	7.1	38.2	36.9	24.9
Germany	10.8	15.5	10.7	29.0	41.6	28.8
France	19.4	20.3	4.8	43.0	44.9	10.5
Italy	26.8	14.9	2.6	60.4	33.5	5.8
United Kingdom	27.1	6.1	1.4	76.9	17.2	3.9
Canada	15.5	5.0	16.7	41.7	0.0	44.9
Australia	23.8	0.0	6.8	77.7	0.0	22.3
Austria	23.0	12.3	8.8	52.0	27.8	19.8
Belgium	16.4	16.1	13.0	35.5	35.0	28.2
Czech Republic	17.1	16.9	4.5	44.4	43.9	11.7
Denmark	32.3	1.6	15.5	65.2	3.2	31.2
Finland	23.9	12.0	10.3	51.5	25.9	22.3
Greece	23.0	10.1	0.4	68.3	30.0	1.2
Hungary	24.6	12.9	1.6	62.9	33.0	4.1
Iceland	24.6	0.0	7.5	76.6	0.0	23.4
Ireland	28.0	3.8	0.7	85.2	11.5	2.1
Korea	15.4	1.9	4.1	72.2	8.8	19.0
Luxembourg	29.9	11.3	2.7	66.9	0.0	6.0
Mexico	14.2	2.7	0.0	83.9	16.1	0.0
Netherlands	23.8	17.9	1.3	54.7	41.0	3.0
New Zealand	34.6	0.0	2.0	94.6	0.0	5.4
Poland	22.9	12.5	3.6	58.8	32.1	9.1
Portugal	22.9	9.3	2.0	66.6	27.1	5.7
Spain	16.4	11.7	5.4	48.8	34.6	16.1
Sweden	24.0	11.4	15.8	46.2	21.9	30.5
Switzerland	9.4	12.5	11.9	27.9	36.9	35.2
Turkey	20.4	4.0	3.5	73.1	14.5	12.4

1. Where the sum of the per cent of total revenues for any country does not add up to 100, the residual covers the supranational sector.
2. The per cent of GDP data is calculated using mainland GDP.

Source: OECD (1999), *Revenue Statistics.*

Since lower levels of government in Norway possess virtually no tax autonomy, tax competition between local governments is not an issue. Meanwhile, a system of block grants from the central to lower governments (based, *inter alia*, on local per capita income and demographic features) ensures that per capita local government revenues are equalised across regions. As a result, the share of grants in municipal revenues varies between 25 and 55 per cent and in county revenues between 30 and 65 per cent.[127] This feature, together with a system of centralised planning, regulation and supervision of locally provided health care, education and other public services, ensures that provision levels are also equalised across regions – even if some competition between jurisdictions on the provision of public services exists.[128] At the same time, employers' social security contributions are differentiated by region (see above) while residents of the northern part of Norway benefit from several tax expenditures, among which reduced rates of personal income taxes (tax on ordinary income and the surtax) levied by the central government. The purpose of this set-up is to attract or maintain employment in remote areas, and is part of a broader overall framework of regional policy that includes agricultural support, support schemes for small and medium-sized enterprises located in rural areas and soft housing loans (see Chapter II).

Weaknesses of the system and options for reform

The 1992 reform of the system of corporate and personal income tax has had favourable effects on macroeconomic efficiency and performance. However, the reform left some unfinished business, in particular concerning the taxation of wealth and the preferential tax regimes for agriculture and forestry. In addition, the "split model" to separate labour and capital income of self-employed and small business owners proves to be susceptible to tax planning and loopholes, while recent initiatives unduly increase the progressivity of the personal income tax at the upper and lower ends of the pay schedule. Moreover, the special tax regimes offshore (shipping, oil and gas) have unintended externalities on mainland taxation. Finally, the tax-sharing arrangements between the central and local governments have, at times, led to a lack of local spending discipline. The sections below examine these weaknesses in more detail and suggest some options for reform; a synopsis of these options is provided in Box 5.

The taxation of property and wealth is distorting

Important unfinished business left by the 1992 tax reform relates to the distortions that stem from the property and the net wealth tax. Taxation of wealth and property should in principle serve to supplement the taxation on income and consumption by widening the overall base of the tax system. It is also deemed to

Box 5. Synopsis of the options for future tax reform

Enhance the neutrality of capital taxation

- Broaden the tax base for property and income tax (on imputed rent) concerning owner-occupied housing, to level the tax playing field between housing capital and other forms of capital.
- Move away from the net wealth tax to an extended property tax to strike a better balance between the taxation of immobile and (internationally) mobile forms of capital. In a similar vein, remove or, at least, reduce the preferential tax treatment of occupational pension funds.

Remove incentives for tax planning in the dual income tax system

- Weigh a rise in the rate of capital income against lower double-taxation relief of distributed or retained profits accruing to domestic investors through the imputation and RISK systems.
- Reconsider the introduction early in 2000 of the additional income tax bracket for top wage earners.

Limit the incentives for tax shifting from the mainland to the petroleum tax regime

- Reduce interest deductions stemming from investments on the mainland against income on the continental shelf, *e.g.* by changing the rules for allocating interest expenditure between the mainland and petroleum tax regimes or by adopting a tighter "thin capitalisation rule".

Improve the mix of indirect taxes

- Widen the VAT base to include various services, in particular passenger transportation.
- Remove exemptions from green taxes granted to several industries, such as heavy metals, fisheries and shipping.
- Abolish the investment tax.

Adjust the inter-regional allocation of resources through the tax system

- In order to raise incentives for local spending restraint, further move away from tax sharing to block-grant funding and provide local government with a stable own tax base – *e.g.*through greater reliance on the property tax – to ensure budget flexibility at the margin.
- Reconsider the amount of regional redistribution through the tax system in order to minimise efficiency losses that could emerge from taxing earnings in remote areas at lower rates.

be an important instrument of income distribution policy since it typically affects higher-income groups most (44 per cent of the net wealth tax is paid by the upper income decile). However, the diverging tax assessment of wealth components jeopardises "horizontal" equity, as taxpayers with comparable levels of net wealth may face widely different tax burdens.

As already indicated, the lenient taxation of housing generates a strong bias towards accumulation of wealth in the housing stock. The tax authorities, as noted, typically assess house values at around 25 per cent of the market price, while they calculate the imputed rent income at 2.5 per cent of the assessed value for all but the most expensive houses – a rate much below the normal return on other types of investment. This way of assessment considerably reduces the tax base for owner-occupiers with respect to income and net wealth taxation, and represents an implicit subsidy discriminating against other instruments of capital formation. The authorities estimate the revenue loss associated with the lenient taxation of owner-occupied housing at NOK 22.4 billion, or almost 3 per cent of mainland GDP.[129] The potential welfare gain resulting from a move towards neutral tax treatment of owner-occupied housing *vis-à-vis* other forms of capital may be as large as the gain achieved by the 1992 tax reform.[130] Unfortunately, a coherent approach to the taxation (including tax assessment and valuation) of property, net wealth and imputed rent has so far met formidable political obstacles. This issue should nevertheless remain on the agenda with a view to eventually abolishing the net wealth tax while enhancing the role of property taxation in line with developments in other countries. Such a shift from internationally mobile to immobile capital tax bases would have the additional advantage of removing incentives to expatriate financial capital. A similar reasoning suggests the removal or easing of the preferential tax treatment of occupational pension funds (an immobile tax base) as opposed to other forms of saving.

The "dual" income tax prompts tax planning

Due to the wide difference in tax rates between capital and labour, the so-called split model to separate capital and wage income for self-employed and owners of small incorporated businesses has been found to produce strong incentives for tax planning in various areas:

- Incentives for small corporate business owners to exploit tax loopholes are strong. For example, small businesses that are subject to the split model have an incentive to inflate the assets of the company since this will lower their tax bill by raising the income share that is taxed at the 28 per cent rate.[131] A generous provision in the split model, moreover, is that losses may be indefinitely carried forward whilst there is no loss offset against current income from other sources. This generates incentives to buy small corporations that accumulated losses that can be used to reduce the taxable profits of the acquirers. Finally, analysis of individual tax statements suggests that tax incentives must play an important role when companies move from active to passive ownership.[132] Such incentives have become even stronger with the introduction of the new upper bracket in the income surtax, in particular for liberal professions.

– Incentives for liberal professions – *e.g.* lawyers, dentists and accountants – to incorporate in order to exploit the above tax loopholes, have become stronger over time. As discussed in Box 4, in 1995 the Parliament raised the income threshold for liberal professions above which personal income is taxed as capital income and in 1998 it went as far as to entirely abolish the threshold. As a result, all personal income of liberal professions in the split model is henceforth taxed as labour income. In parallel, the taxation of other occupations under the split model (including farmers) was reduced. Moreover, the introduction, in 2000, of a new upper bracket in the national income surtax widened the differential between the (top) marginal tax rates on labour and capital income in the split model for liberal professions by as much as 6 percentage points. The resulting incentives for liberal professions to incorporate and subsequently qualify as a "passive" company owner in order to avoid the labour income tax are strong, although these incentives have been contained somewhat by measures in 2000 to tighten the "identification rules" (see Box 4).

All considered, the taxation of income from small business is the Achilles heel of the dual income tax. Not surprisingly, the tax authorities report a high administrative burden associated with the split model, even if it applies in effective terms to only a minority of self-employed workers and to around 50 000 limited companies that in most cases have just one employee, the "active owner". The policy debate has so far moved in the direction of exploring the possibilities of reducing the gap in taxation between capital and labour income economy-wide while preserving the neutrality of capital income taxation across funding options and ways of doing business. An official *Tax Reform Committee*, which reported its findings in 1999,[133] has examined the welfare impact of a small increase in the capital income tax rate combined with a reduction in the income surtax. The reported effects are generally small, but positive. Unfortunately, while the report of the *Committee* remains on the political agenda, the marginal tax rates on wage income were, as noted, raised further early in 2000, both at the lower and higher ends of the pay schedule. This will provide further incentives for tax avoidance by liberal professions (who may incorporate), small company owners in liberal professions (who may move toward passive ownership) and low-paid workers (who may withdraw from the labour market). The impact on lower-paid workers may be particularly damaging in the longer run as it tends to discourage the maintenance or formation of human capital.

While it is likely that the government will maintain the split model, there are pros and cons associated with this choice. On the one hand, by preserving the split model, Norway will continue to achieve a high degree of neutrality between ways of doing business and funding investment, thereby avoiding an important source of inefficiency. But, on the other hand, this advantage comes at a

potentially high price. In particular, the arrangements that undo double taxation of dividends and retained earnings (imputation and RISK systems) have certain disadvantages. Dependent upon the bilateral tax treaties in place, foreign investors in Norwegian corporations may not fully benefit from such double taxation relief. This feature may restrain foreign capital inflows and therefore slow down the potential growth rate of Norway's mainland economy. Moreover, the dual income tax system applies a uniform income tax by all forms of capital, mobile and immobile capital alike, and therefore carries an important dead-weight loss. The efficiency advantages stemming from Norway's dual income tax may have more than outweighed this drawback so far, but an increase in the rate of capital taxation from its present level of 28 per cent, an issue which has remained on the political agenda, would tend to accentuate the disadvantages. All considered, the positive aspects of the dual income approach could be strengthened by extending the use of property taxation and raising the rate of imputed rent income of owner-occupied housing, both to ensure that immobile capital (housing) is taxed at a higher effective rate than is presently the case and to provide room for cutting high marginal tax rates in areas that are susceptible to distortions.

The special tax regimes entail unintended externalities

The presence of three separate regimes for corporate taxation for the mainland, the shipping and petroleum sectors is a special feature of the Norwegian tax system. There are clear rationales for some of these special arrangements, notably those that have been designed to capture natural resource rents by the government. However, these differences in tax treatment may produce incentives for tax planning by companies or investors that are in a position to choose between two or more of these regimes. While such behaviour may create inefficiencies, it also heightens the risk that some of the natural resource rents that policy aims to collect as government revenue in fact remain in the hands of the private sector. In particular, the "thin capitalisation rule" may not be sufficiently rigorous (see Box 6), and could be tightened in order to reduce distortions associated with tax planning. A straightforward alternative measure would be to tighten the tax rules for allocating interest expenditure between the mainland and petroleum tax provinces. Similar considerations apply to the shipping sector. Companies have also sought to avoid corporate tax by establishing holding companies in countries where dividend received from foreign subsidiaries is not taxed (Singapore, Denmark).

High indirect taxes encourage cross-border tax avoidance

Given that taxes on goods and services in Norway are among the highest in the OECD area, it is not surprising that these are particularly prone to

Box 6. Incentives for tax shifting

Although the evidence of tax shifting across regimes is scant, the incentives are clear. The larger operators on the continental shelf have stakes in the shipping and mainland traditional industries, and are likely to aim for an optimal regime mix of assets and liabilities from a tax-planning point of view. In particular, while a "thin capitalisation rule" limits the amount of debt that the operators on the shelf can claim interest deductions for under the petroleum tax regime, they may attempt to maximise debt/equity ratios within that constraint.[1] Indeed, a mainland investment which benefits from the possibilities of tax shifting towards the petroleum tax regime yielding a 6.5 per cent pre-tax rate of return would need to yield 12.5 per cent to be equally profitable without that advantage.[2] The incentives to avoid corporate tax by minimising the debt/equity ratio for tax purposes under the shipping tax regime are also clear. To limit the risks in this regard, the authorities have adopted a "thick capitalisation rule" which caps the allowed debt/equity ratio – at 50 per cent in 2000 (raised from 30 per cent in 1999).

A related concern is that the petroleum tax regime may give rise to incentives for investment in marginal, less productive fields. Under certain conditions the petroleum tax system provides a net subsidy. To yield a 9 per cent nominal after-tax rate of return, a 6 per cent nominal pre-tax rate of return may suffice owing mainly to the "debt subsidy" associated with interest deductions against the special surtax of 50 per cent.[3] On the other hand, unless they are small, such low-yielding investment projects are unlikely to be undertaken since they will normally not qualify for an investment permit from the competent authorities. But the incentives are strong and it may not always be easy for the authorities to disentangle low-yield and high-yield projects.

1. The thin capitalisation rule allows interest deductions for debt of up to 80 per cent of the investment. Under this rule, debt is defined to include the deferred tax liability associated with the return on the investment, hence in effective terms less than 80 per cent of the project is allowed to be debt financed. According to the petroleum tax act, interest payments are split between the mainland and petroleum tax provinces according to the company's net income in each province. In practice the latter rule does not act as a binding constraint since net income offshore is boosted by the natural resource rent.
2. The following assumptions have been made. An oil company provides capital to a subsidiary on the mainland, 80 per cent of which is raised via a loan. The associated interest expenditure is deducted against the company's net income on the continental shelf. The latter assumes that initially net income of the mainland subsidiary is zero or negative – if not, part of the interest expenditure would have to be allocated to the mainland tax regime in proportion to the company's mainland income. Income generated by the investment is taxed at 28 per cent. Dividends or other profit transfers from the subsidiary to the mother-company are not taxed under the petroleum tax regime.
3. This assessment is based on the following assumptions: the present value of tax depreciation, uplift and "debt subsidy" amounts to 79.4 per cent of the investment; the nominal after tax required rate of return is 9.1 per cent; the rate of inflation is 2.5 per cent; the rate of economic depreciation is 10 per cent; and the economic lifetime of the investment is 15 years.

avoidance through cross-border shopping, smuggling and international tax loopholes. Cross-border shopping has historically been common among residents that live along the border with Sweden, and mainly concerns alcoholic beverages, tobacco, meat and petrol. The reduced VAT rate on food products in Sweden provides additional incentives for cross-border shopping. Smuggling is related mostly to alcoholic beverages, for which import restrictions are particularly tight. A future decline in excises on alcohol and tobacco in Sweden to comply with EU rules may put pressure on Norway to move in the same direction – with first steps possibly on the agenda in 2000. By contrast, the high tax on vehicles does not result in cross-border tax evasion due to registration requirements, but it does prompt evasion through registration of mini vans as business cars. Meanwhile, the airline industry has the possibility to avoid (non-reimbursable) VAT on fuels by combining domestic flights with international flights – a distortion that would be removed by including transportation services in the VAT net.

The scope for moving away from taxation of labour and capital towards taxation of goods and services – which is usually recommended as a remedy for tax base erosion in the face of globalisation and population ageing – is thus very limited in Norway. Nevertheless, there are ways to further raise the yield from indirect taxes. The VAT net could be widened to include service activities such as legal services, broadcasting and, importantly, passenger transportation – akin to most OECD countries. The implicit subsidies to these activities could then be converted into cash expenditure to improve transparency, or abolished. Moreover, and for the same reasons, exemptions from green taxes, including the CO_2 tax, that have been granted to a number of industries, such as metals, fisheries and shipping, could be removed.[134] This would also contribute to lowering the overall cost of abatement of environmentally-harmful emissions, especially if tax rate differentiation better reflected the content of polluting substances in products and activities. Alternatively, the government could opt for a tradable quota system covering the bulk of greenhouse gas emissions along the lines of the Kyoto Protocol, as proposed by an expert commission in late-1999 (see also Chapter II). However, such a system would be cost-effective only if, as advocated by a majority in the expert commission, the quotas were freely tradable, both domestically and internationally. On the other hand, since there is no rationale for maintaining the investment tax, its abolition could offset a widening of the VAT and, possibly, the green tax net.

Vertical tax sharing generates spending pressures at lower levels of government

The system of local government funding in Norway has been designed to offset differences in income and cost levels, resulting in limited competition between local jurisdictions on levels of tax and public service provision. A key

element of public funding arrangements in Norway is the sharing of major tax bases (personal income, net wealth) between administrative levels, with the rates largely determined at the central level. The literature emphasises that this feature could produce unintended externalities whereby policies implemented at one level affect the tax yield at the other.[135] In Norway, in contrast, such externalities are deliberate, in the sense that the central government actively uses the tax-sharing arrangement to steer local government budgets. A recurrent problem associated with this set-up in the 1990s has been that the annual projections on which the central/local split of income tax revenues is based often proved to be too conservative, thus leaving local governments with tax windfalls. This has, at times, created unintended incentives for local spending increases and thus contributed to tendencies of macroeconomic overheating. In an attempt to deal with these problems the government has gradually reduced the local governments' share in income taxes (see Table 14). However, local governments nevertheless succeeded to "lock in" the higher levels of local expenditure.

Being aware of these risks, the government appointed a special committee (the *Rattsø Committee*), whose recommendations led to the abolition of the local component of the corporate income tax in 1998 (see above).[136] A consideration in abolishing corporate rather than personal income tax sharing was that corporate tax is relatively volatile and therefore particularly prone to projection errors and, more generally, less suited for jurisdictions that face a balanced-budget constraint. Cutting down the still existing sharing arrangements for other taxes and thus further shifting towards block-grant funding could yield additional benefits for budget planning and control at the local level. Meanwhile, providing local governments with more autonomy to exploit stable local tax bases, notably property, to a greater extent, could also help to strengthen their expenditure discipline. Indeed, greater local tax autonomy implies that local governments that opt for raising spending programmes also face the fiscal consequences, provided that central government grants are used to internalise spill-over benefits to neighbouring jurisdictions rather than to compensate for local funding deficits. Property taxes are best suited for this purpose since they tax a tax base that is least susceptible to avoidance, and therefore provide stronger incentives for expenditure restraint through voting behaviour of citizens.[137]

Moreover, the local government's funding and tax system (including the differentiation of employers' social security contributions and the preferential tax treatment of self-employed people in agriculture and fisheries; see above) is an important element of regional policy, and the question arises whether the current set-up is cost-efficient in achieving regional policy goals. As discussed in Chapter II and in the 1999 *Economic Survey*, Norway already maintains high support for remote areas in the form of support for agricultural activity through administered prices, import quotas and tariffs, alongside the regional re-distribution of resources through the grant system and the favourable tax treatment of remote

economic activities (agriculture, forestry, fisheries, long-distance public transportation). The government should re-consider the amount of such regional redistribution in order to minimise efficiency losses that could emerge from regional policies. As noted in Chapter II, an overall assessment of such productivity losses associated with regional policies is long overdue. This is all the more important since increases in mainland productivity are needed in order to offset the loss in per capita income associated with the expected decline in petroleum production in the longer run.

Notes

1. The very strong expansion was partly caused by cost overruns on the Åsgard offshore oil and gas project.
2. Household purchases of cars, as measured in the quarterly national accounts, are influenced by the technical assumption that passenger cars bought by the business sector are sold to households after three years. The sharp increase in the first quarter of 1999 therefore reflects the high level of car sales in 1996. This accounting practice pushes up consumption by around 0.4 percentage point in 1999.
3. The government's decision to reduce production by 200 000 barrels per day from its prognosis of 3 million barrels presented in the 1999 revised budget was a contribution to balance the oil market and to raise the oil price, which had fallen below US$10 per barrel (Brent blend) in December 1998 from US$20 in October 1997. Due to better compliance by OPEC members to agreed quotas, and helped by the economic recovery in Asia, the effort was much more successful than past attempts and the oil price recovered to more than US$25 in December 1999.
4. For a description of the Norwegian salmon farming industry see Box 1 of the 1997 OECD *Economic Survey* on Norway.
5. The competitiveness indicator is based on Norwegian unit labour costs in the manufacturing sector relative to the other OECD countries and twelve major non-OECD countries (see Durand *et al.*, 1997).
6. Temporary jobs were 10.5 per cent of total jobs in the first half of 1999, compared with 11 per cent in 1998 and 12 per cent in 1997.
7. The inflow into early retirement reduced the labour force by ¼ per cent in 1999, while 40 per cent of the mothers (but none of the fathers) receiving the childcare cash benefit have reduced working hours.
8. Seasonally-adjusted registered unemployment increased from 55 000 in May 1999 to 62 000 in December and was 59 500 on average in 1998. Including participants in labour market programmes, the number of unemployed went up from 64 000 in May to 71 000 in December and was 71 000 on average in 1998. However, unemployment on a labour force survey basis has not yet risen. It was 67 000 (2.9 per cent of the labour force) in the third quarter of 1999, compared with 74 000 (3.2 per cent) in 1998.
9. The Norwegian authorities estimate the NAWRU as between 3½ and 4 per cent. See Holden (1996) for a critical analysis of the NAWRU concept in the Norwegian context.
10. Unions and employers' organisations agreed on settlements increasing wages by 1.0 per cent in 1999. As wages were increased from 1 April onwards or later, the contribution to wage growth in 1999 was, however, limited to 0.4 percentage point and the carry-over into 2000 is 0.6 percentage point. The agreed wage rise was smallest for the manufacturing sector (0.4 per cent; with a contribution to wage

growth in 1999 of 0.3 percentage point) and largest for central government and the education sector (1.6 per cent; with a contribution of 0.6 percentage point). Wage rises including wage-drift are estimated to have ranged from 4.4 per cent in the wholesale sector to 5.5 per cent in the banking sector.

11. This committee, with high level representatives of the social partners and the government, was appointed to prepare the 1999 wage settlements.

12. Consumer price inflation decelerated from 2.5 per cent (twelve-month-change) in May 1999 to 1.9 per cent in August but rose again to 2.8 per cent in November and December, mainly due to higher electricity prices and a rise in prices of clothing. The consumer price index in the twelve months to December compared with the previous twelve months was 2.3 per cent, unchanged from December 1998. On a national accounts basis, the consumer price deflator dropped steadily, from 2.9 per cent (four-quarter change) in the final quarter of 1998 to 2.3, 2.0 and 1.9 per cent, respectively in the first three quarters of 1999.

13. Underlying consumer price inflation (CPI) excludes the impact of indirect taxes and electricity prices on the consumer price index. It is the most closely watched price index by Norges Bank. Regrettably, the index is not published in the monthly CPI press release of Statistics Norway. Due to high reservoir levels and increased competition, electricity prices restrained overall consumer price inflation by 0.1 percentage point in 1999. Changes in indirect tax rates did not influence consumer price inflation.

14. The academic empirical literature is inconclusive on the impact of the Solidarity Alternative. Rødseth and Nymoen (1999) found some wage moderation due to the Solidarity Alternative and limited effects of unemployment on wages. Other studies, for instance Lindquist and Naug (1996), found high unemployment to be the major explanation of wage moderation in the early 1990s. Some studies point to the drawbacks of the Solidarity Alternative; for instance Røed (1998) concludes that the egalitarian wage distribution due to the centralised negotiations led to higher unemployment among low-skilled workers.

15. The negotiations for the two-year wage deal are expected to start in March for the manufacturing sector followed later in the year by negotiations for the sheltered sector.

16. Moreover, of the currencies presented in Figure 10, the standard deviation of the daily percentage changes is the lowest for the krone/Deutschemark rate (0.4 per cent).

17. This was not only spelled out in the Annual Address of the Governor in February and the quarterly Inflation Reports but also in an article of the Governor appearing in the newspaper Aftenposten (Gjedrem, 1999).

18. To increase transparency further, Norges Bank decided in June to publish the dates of the monthly Executive Board meetings well in advance, to have an extensive discussion on monetary policy during every monthly board meeting and to clarify any interest rate decision taken immediately. This is in line with initiatives of other central banks to increase the transparency of monetary policy.

19. Norges Bank did this in 1997 (Norges Bank, 1997); it advised further fiscal tightening to prevent overheating. Moreover, it recommended a revision of the guidelines for monetary policy to provide for greater flexibility in the conduct of monetary policy, but without changing the operational goal of exchange rate stability.

20. Somewhat surprisingly, the upward pressure on the krone in 1999 is, *ex post*, not reflected in increased capital inflows. Portfolio investment (including other financial investment excluding loans to the central bank) in Norway dropped to NOK 49 billion in the first nine months of 1999 from NOK 87 billion in the corresponding period of 1998. However, in the three last months of 1998, there was a considerable reduction in portfolio investment in Norway. Nevertheless, the international reserves increased by NOK 30 billion after dropping by NOK 14 billion. The main cause was the increase in net direct investment in Norway to NOK 37 billion from NOK 17 billion. Other reasons were the drop in portfolio investment abroad to NOK 57 billion from NOK 71 billion and the improvement of the current account surplus to NOK 19 billion from NOK –0.1 billion.
21. The increase in capital inflows in 1998 is also somewhat surprising in the light of the downward pressure of the exchange rate. Portfolio investment (including other investment) in Norway increased from NOK 56 billion in 1997 to NOK 81 billion. The rise in purchased bonds more than offset the drop in deposits. Appreciation expectations and an expected drop in long-term interest rates may be the reasons behind the strong foreign demand for Norwegian bonds.
22. Its other key rate, the rate on central bank overnight loans, was raised from 5.5 to 10 per cent.
23. Although it has the same impact, fiscal policy is to be preferred in the light of the long-term goal of maintaining the industrial base as it dampens consumption relatively more and investment relatively less. See also Eika and Moum (1998).
24. As on other financial markets, fears of liquidity shortages had a discernible effect on short-term interest rates. Due to the Y2K effect, the forward rate for December was a good percentage point higher than in the preceding and subsequent months, with consequences for the three-month interbank rate.
25. The underlying growth excludes central government expenditure on petroleum activities, interest payments and unemployment benefits. The under-prediction for total government consumption volume was 1½ percentage points on average in the 1990s.
26. With substantial net government debt in the euro area and substantial net government assets in Norway, expenditures including interest payments are lower in Norway than in the euro area.
27. Norway is governed by a small minority coalition, headed by the Christian Democrats. The three coalition parties have only 42 of the 165 seats in Parliament, while the main opposition party, the Labour Party, has 65 seats. The coalition parties reached an agreement with the Labour Party on the budget for 2000. The 1998 and 1999 budgets were approved by Parliament with the support of the Conservatives and the Progress Party.
28. The allocation is less than the general government surplus due to the surplus in the National Insurance Scheme Fund, the inclusion of interest income of the Petroleum Fund in the general government surplus and the definitional differences between the national accounts and the central government account.
29. The Fund is expected to continue to swell in the coming two decades, peaking above 100 per cent of GDP.
30. Apart from these assets, the government owns the so-called State Direct Financial Interest (SDFI, participations by the Norwegian state in oil fields), valued at US$50 billion (around 30 per cent of GDP).

31. Making the same assumptions on economic growth and interest rates as made in a recent EU generational accounting study, a somewhat smaller tightening, of 0.3 per cent of GDP, is required. A tightening of 4.2 per cent would be required for the EU countries. The main reasons for the more limited need to adjust in the case of Norway are the expected high petroleum revenues in the coming years and the smaller increase in the age-dependency ratio.

32. Moreover, this had the short-term disadvantage of reducing labour supply in a very tight labour market situation.

33. According to calculations by Norges Bank, a 1 percentage point higher wage rise in 2000 would increase consumer price inflation by 0.2 percentage point in 2000 against the baseline and by 0.4 percentage point in 2001.

34. The United Kingdom, which also experienced a pick-up in activity following a mild growth pause, started to increase its repo rate in September 1999, less than a year after it started to cut the rate.

35. A further substantial increase in the participation rate is unlikely as the non-active part of the working-age population consists almost completely of students, disabled and early-retired. In 1998, the potential entrants (those outside the labour force and not studying or disabled, retired or early-retired) were only 5 per cent of the population aged 16-74 years, one of the lowest rates within the OECD area.

36. However, the hours worked per person employed are relatively low. In 1997, the annual average was 1 427 hours. This is the lowest in the OECD after the Netherlands and 8 and 17 per cent less than in the European Union and the United States, respectively.

37. When comparing productivity at the international level, a number of measurement issues arise. While such measurement problems can be identified, it is often difficult to give a clear indication about their size and, sometimes, even their direction. Therefore, results of international comparison should be interpreted with the necessary care. Measurement problems are even more pronounced at the sectoral level, especially for the government sector. Level numbers presented in Figure 16 are productivity per hour worked but growth rates are per worker, as time series on hours worked are not available for many countries.

38. For a detailed description of public income support see Ministry of Health and Social Affairs (1999*b*).

39. Registered unemployment among first-generation immigrants was 6.3 per cent in the second quarter of 1999, almost three times as high as for the entire population. The immigrants from Africa had the highest registered unemployment rate, with 12.6 per cent, followed by the immigrants from eastern Europe and Asia with an unemployment rate of nearly 9 per cent. These rates are high but not as high as in most other western European countries. For instance, the unemployment rate of first and second-generation immigrants in the Netherlands was 18 per cent in 1995-97.

40. This is the expected effective retirement age of 50-year-olds. The expected retirement age of 16-year-olds is more than three years less, around 59 years, due to the possibility of becoming disabled before the age of 50 years.

41. In this case working-age population is defined as the age group 16-67 years.

42. The relatively high disability rate may be partly because alternative possibilities to retire early were limited until recently.

43. A government appointed committee will report in mid-2000 on possible measures to curb the rise in disability and sickness allowances. Some minor measures have already been taken recently: the requirement for vocational training before disability certification was reinforced to reduce the inflow into the disability scheme; the income test was relaxed to make it more attractive for disability pensioners to work partially; the retention right for entitlement during work trials was raised from 1 to 3 years to stimulate re-entrance. In its White Paper on Equitable Redistribution (Ministry of Health and Social Affairs, 1999), the government proposed a five-year trial of permanent wage subsidies for persons with reduced work capacity. This could reduce the number of disability pensioners. Other proposals, however, could have the opposite effect. The government proposed higher benefits for disabled women previously working part-time. Furthermore, it proposed a three-year trial period with a lower degree of occupational disability (30 per cent) as a condition for receiving an invalidity pension.
44. The employment rate of workers aged 55-64 years was 67 per cent in 1998, 20 percentage points higher than on average in the OECD area (Moe, 1999).
45. There were 3 400 part-time early-retired in October 1998.
46. With regard to schemes other than the disability pension, the implicit average tax rate in 1995 for those working after 55 years was 15 per cent in Norway, but 24 per cent in the European Union. Recent measures in Norway, such as the lowering of the entry age for early retirement, and in the European Union (where some countries have curtailed access to early retirement) have probably reduced this gap in implicit tax rates. With regard to the disability scheme, the implicit tax rate for those working after 55 years is much higher in Norway (65 per cent) and even higher than the average rate of 50 per cent in the European Union (Blöndal and Scarpetta, 1999 and 1998, OECD 1998*b*).
47. The government proposed in the draft 2000 Budget to introduce an independent right to paternity benefits of four weeks of paid leave.
48. A temporary employment contract is only allowed for work that differs from what is ordinarily performed in the enterprise, for work as a trainee or temporary replacement, for persons participating in labour market schemes, for chief executives and in professional sports.
49. Hiring-in from companies not established with the purpose to hire out (*e.g.* a construction firm) is somewhat less restricted. These companies are permitted to hire-out workers who are on a permanent contract, although consultation with the labour union representatives is necessary. The hiring-out enterprise can only do this for work within the hiring-out enterprise's main activity and limited to half of its permanent staff. Approval by union representatives in the hiring-in company is necessary in case the number of workers hired in exceeds 10 per cent of the workforce and the hiring-in lasts for more than one year.
50. In 1997, the maximum duration in the unemployment insurance scheme for those becoming unemployed was limited to three years. This means that some unemployed may exhaust their benefit right in 2000. For those already unemployed in 1997, the benefit period expired at the end of 1998. This was the case for 5 000 persons, of which, probably due to the favourable labour market situation, only 500 applied for the transitional "last resort benefit" programme.
51. The indicators are taken from the OECD International Regulation Database based on the reply of member countries to the Regulatory Indicators Questionnaire supplemented by other sources. The indicators describe the regulatory environment in

product markets in 1998 (Nicoletti *et al.*, 1999). The answers to the Questionnaire, together with the other information, were used to construct cardinal summary measures. The Questionnaire asked for information on more than 1 500 different regulatory provisions concerning economy-wide and sector-specific laws, regulations and administrative procedures. Its focus was exclusively on the relative friendliness of these regulations to market mechanisms and there was no attempt to assess the quality of the regulations or their aptness in achieving their goals. The country rankings should be interpreted with caution, especially when absolute differences are small, since many indicators lack precision.

52. In 1999, the government launched its "A simpler Norway" project which aims to reduce the administrative burden of enterprises, especially in the service sector, and to step up the regulatory reform process. It is too early to say whether it will have a major impact on red tape and paperwork for private enterprises.

53. This extraordinary support level is probably an important reason why Norway is one of the few countries with a rise in registered agricultural land. Agricultural land use increased by 8 per cent between 1980-81 and 1992-93 while it decreased by 2 per cent in the other OECD countries. The proportion of agricultural land in total, however, has remained the lowest of the OECD countries (3 per cent).

54. For a detailed description of agricultural policies, see the Norwegian case study in OECD (1998*c*).

55. However, farms are valued at (low) "reasonable" prices in case of succession. To assist young farmers to cope with the financial burden, an additional subsidy for young farmers was introduced in 1998 while the existing installation grant was increased.

56. The greater stress on environmental conditions when linking support to output can be seen as a move in the right direction.

57. Producer support dropped by 1 percentage point of total gross farm receipts between 1986-89 and 1996-99, compared with 7 and 8 percentage points in the European Union and the OECD, respectively. Turkey was the only country with a rise in producer support, of 9 percentage points.

58. WTO members are committed to talks on agriculture under the 1996 Marakkesh Agreement.

59. In December 1994, the European Union, Norway, Japan, Korea and the United States signed the so-called *OECD Shipbuilding Agreement* establishing normal, *i.e.* subsidy and dumping-free, competitive conditions in the shipbuilding industries of those countries. The Agreement was originally scheduled to enter into force in July 1996 providing a level playing field for nearly 80 per cent of the world shipbuilding industry. However, the US Congress did not pass the legislation which would allow the United States to ratify the Agreement.

60. Other support given to the sector is through export guarantees. For instance, in 1999 the *Norwegian Guarantee Institute for Export Credit* (GIEK) provided a loan guarantee of NOK 1 billion for a luxury cruise ship ordered by a Norwegian shipowner.

61. Using a disaggregated intertemporal Computable General Equilibrium Model, Fæhn and Holmøy (1999) found a drop in employment in the shipbuilding industry of 16 per cent due to the abolishment of the subsidy.

62. Due to the economic slowdown and the specific problems in the offshore sector, R&D expenditures were probably lower in 1999. Investment in knowledge (R&D expenditure, public spending on education and expenditure on software) was 8.8 per cent of GDP in 1995, 1.1 percentage point lower than in the three other Nordic countries but 0.9 percentage point above the OECD average. R&D expenditures are relatively low for low-technology and medium-technology industries while they are somewhat above the OECD average for high-technology industries (OECD, 1999*c*).

63. See Sandven and Smith (1998) for an analysis of the effects of differences in industrial structure and country size on R&D.

64. Most R&D investments are fully deductible at the time of investment. However, unlike many other OECD countries, there are no additional special tax provisions.

65. Moreover, in its White Paper, the government announced an evaluation of the Research Council of Norway, created in 1993 to fund research.

66. In 1999, the liquor company Arcus and the grain import company Statkorn (the former grain import monopoly) were partly privatised.

67. At the end of 1999 the proposed merger between Telenor and the Swedish fully state-owned telecommunication company Telia collapsed. An initial public offering in the merged company was envisaged for 2000. However, the two governments agreed to maintain a total stake of at least 51 per cent till 2015.

68. Taking the two major currently non-listed companies, Telenor and Statoil, into account public ownership is much higher, close to a half. Moreover, the government owns the SDFI, through which the Norwegian state participates in oil and gas exploration and development of fields. The SDFI is operated on the state's behalf by Statoil and is valued at around US$50 billion (around 30 per cent of GDP).

69. In 1992, the private stake in PTOs was already almost 25 per cent.

70. In December, Telenor made a hostile takeover bid for an Irish telecommunication company. It was outbid by *British Telecom*.

71. For a detailed description of the Norwegian oil and gas sector see Chapter III and Annex I of the 1999 *Economic Survey* of Norway (OECD, 1999*e*) and Ministry of Petroleum and Energy (1999).

72. In 1998, Exxon merged with Mobil, BP merged with Amoco and Total merged with Petrofina. This was followed in 1999 by the merger of BP Amoco and Arco, Chevron and Texaco, and TotalFina with Elf.

73. Statoil suggested merging with the SDFI to form the fourth biggest oil company in the world while Norsk Hydro suggested the gradual transfer of SDFI licenses to commercial companies.

74. Norsk Hydro sold a quarter of Saga Petroleum to Statoil.

75. Statoil recently made the suggestion to engage foreign companies as members of the GFU.

76. Britain provides another example of combining competition policy and regulatory functions, but in that country it is the sector regulators who are charged with both tasks although the competition authorities retain concurrent power to enforce competition law. New Zealand has gone even further in the direction of unifying competition policy and regulation. It has largely abolished sector specific regulation and instead relies heavily on judicious application of competition policy.

77. For instance, in 1998 on the basis of cultural policy considerations, the ministry overruled the decision of the NCA to prohibit fixed retail prices on school books.

78. The minimum number of employees in regional offices was lowered while differences in size between regional offices were allowed.

79. These are gross payments. The banks borrow from the central government at rates above those on central government bonds.

80. The competitive bid of Den Danske Bank for Fokus Bank upset the announced merger of Kreditkassen, Postbanken and Fokus Bank. Moreover, Den Danske Bank outbid the Swedish Handelsbanken.

81. The Ministry of Finance, the Banking, Insurance and Securities Commission of Norway and the European Commission approved the merger under certain conditions. The main condition was the divestment of *Vesta*, the Norwegian subsidiary of Skandia. Without this divestment, the market share would have been almost 60 per cent. The Danish banking group *Unidanmark* acquired Vesta in December 1999.

82. See for an analysis of the current merger wave BIS (1999) and Boot (1999).

83. The commission proposed to reduce the acceptance rate from 90 per cent to two-thirds.

84. The tier I capital ratio of the three largest commercial banks increased marginally to 7.3 per cent in the third quarter of 1999, from 6.4 per cent in 1998. The tier I capital ratio of savings banks declined to 10.3 per cent from 10.5 per cent in 1998.

85. The share of non-performing loans for commercial banks and savings banks was about 1.6 per cent in the third quarter of 1999. In 1993, it peaked at about 8 per cent for commercial banks and 5½ per cent for savings banks.

86. Norway is the world's second largest oil exporter, after Saudi Arabia, and also a major gas provider to western Europe. In addition, Norway is the second largest seafood exporter in the world, and an important producer and exporter of hydropower.

87. The Norwegian merchant fleet is the second largest in the OECD area, after Greece, and the seventh largest in the world.

88. With 13 inhabitants per square kilometre, Norway is one of the least populated countries in the OECD area (only Canada, Australia and Iceland have lower population densities).

89. In July 1998 the EFTA Surveillance Authority decided that the system of regionally-differentiated employers' contributions should be considered as state aid and that parts of it were incompatible with EEA regulations. The EFTA court confirmed this decision in May 1999. The Authority has approved the changes in the system the Parliament has subsequently adopted.

90. Cnossen (1997), Sørensen (1998) and the 2000 *Economic Survey* of the Netherlands.

91. Two policy documents prepared the ground for this reform: "Towards more cost-effective environmental policies in the 1990s: Principles and Proposals for Better Pricing of the Environment" by the Environmental Tax Commission in 1992 and "Policies for a Better Environment and high Employment" by the Green Tax Commission in 1996; see Ministry of Finance (1992) and Green Tax Commission (1996). The 1999 OECD *Economic Survey* of Norway contained an extensive discussion of environmental policies.

92. The reduction of statutory tax rates was welcomed on the presumption that loopholes and deductions would be eliminated and the redistributive goals of the system better served. The cut in statutory rates was seen to strengthen the international competitiveness of Norwegian industry and stem capital outflows.

93. See Aarbu (1997) and Sollund (1991).

94. Already between 1986 and 1991 it became gradually more profitable to take income as dividends due to the stepwise decrease of the personal dividend tax. See also Aarbu (1997).

95. Prior to the reform the actual tax rate on dividends depended on individuals' effective marginal tax rate. After the reform foreign owners of Norwegian shares may still face double taxation of distributed profits.

96. RISK stands for *Regulering av Inngangsverdi med endring i Skattlagt Kapital*.

97. The corporate wealth tax was also abolished.

98. Prior to the reform corporate taxes were levied at a statutory rate of 50.8 per cent.

99. Until 1992 cyclically-adjusted corporate tax revenues hovered around 1½ per cent of potential GDP, but since then it has trended up to almost 3 per cent.

100. For example, prior to the reform an investment in a ship or airplane was profitable even with negative pre-tax returns due to very generous depreciation rates.

101. As in most OECD countries, another non-neutrality stems from the deferred taxation of occupational pension saving.

102. Holmøy and Vennemo (1995). Welfare is derived from utility functions for 14 types of households distinguished by socio-economic status embedded in a computable general equilibrium (CGE) model.

103. Statistics Norway is presently working on an evaluation of the tax reform using an updated and extended version of the 1991 CGE-model. The present model incorporates saving behaviour and endogenous labour supply, and may thus be able to give a more coherent picture of the reform.

104. The experience in Sweden has been that reductions in the top marginal tax rate reduce the returns on favourably taxed assets such as housing relative to the opportunity cost of placing money into saving accounts or the cost of debt finance (Agell *et al.*, 1999).

105. Aarbu and Thoresen (1998).

106. The inequality measured by the Gini-coefficient increased by 3.3 per cent from 1992 to 1993. This is however not statistically significant, see Aarbu and Thoresen (1995).

107. See for an extensive discussion the 1999 *Economic Survey of Norway*.

108. The uplift is an extra deduction that tops up the annual depreciation allowance by 5 per cent of the investment (for a period of six years). Moreover, companies pay a royalty of 8 to 16 per cent of gross sales on oil from fields cleared for development before 1986, an acreage charge (levied as a lump sum per square kilometre licensed) and the CO_2 tax.

109. The SDFI is managed by *Statoil* on the state's behalf.

110. Some oil and gas producing countries in the OECD area, in particular the United States (Gulf of Mexico), Canada and Australia, capture the natural resource rent by auctioning-off production licenses.

111. A minority in the Commission, however, noted that it was too early for any firm conclusions in view of the complexity of the offshore tax regime.

112. The tax is levied per power station, not at the company level, in order to ensure that all rent is taxed, including that used to cross subsidise other (less profitable) business.

113. Domestic shareholders receiving dividends are entitled to a full credit under the imputation system.

114. Fisherman are, moreover, liable to the lower 7.8 per cent rate for social security contributions rather than the 10.7 per cent rate that normally applies to self-employed workers. On the other hand, they pay a special tax on the gross operating surplus of a vessel (at a rate of 3.4 per cent in 2000).

115. Income from forestry is assessed according to a five year moving average of past earnings against which the annual forest duty of 8 per cent of the market price of wood is deductible, provided that this credit is used for investment purposes or environmental protection. Realised capital gains on agricultural or forestry property (including mobile assets such as feedstock or milk quota) are tax free under certain conditions. Also under certain conditions, owners of reindeer may claim a deduction of up to 80 per cent of their income. The amount must be deposited in a special bank account, and will be taxed when cashed. The government has recently proposed to introduce a separate tax deduction against ordinary income from farming (up to a ceiling of NOK 36 000 per farm) to compensate for a drop in prices of agricultural products.

116. Exemptions exist for cultural services, education, financial services, insurance, accountancy, brokerage, advisory services, lottery and gambling, medical and dental care, postal services, property letting, hotels, supply of land and buildings, broadcasting, legal services, public passenger transport, public cemetery and sporting events. VAT on books, periodicals, newspapers and electricity in northern parts of the country is zero-rated. The government envisages submitting a proposal for a wider VAT net to Parliament by the spring of 2000.

117. This is settled as a reduction of the VAT refunds. However, manufacturing and agriculture are exempt from the investment tax.

118. See the 1999 *Economic Survey* of Norway.

119. The property tax is collected on hydropower plants, housing and commercial buildings, each comprising about one-third of the total yield, at a rate of up to the legal maximum of 0.7 per cent.

120. Since 1998 municipalities levy 0.7 per cent on net wealth and the central government has two brackets of 0.2 and 0.4 per cent, respectively. In 1998 the combined top rate was lowered from 1.5 to 1.1 per cent, with the ceiling for the sum of net wealth and income tax paid per individual raised from 65 to 80 per cent of ordinary income (capital, labour and pension income net of deductions). The business sector is not liable to the net wealth tax, but it is levied on shares held by households.

121. Denmark assesses property at full market value and Sweden at 75 per cent of market value, see Youngman and Malme (1994).

122. In 1998 the valuation of listed shares was raised from 70 to 100 per cent and for non-listed shares from 30 to 65 per cent of book value.

123. There is a NOK 51 250 tax-free allowance for primary residences.

124. Before 1998 the corporate income tax was also shared between the central and local governments, but it was then converted into a solely central government tax.

125. Local authorities may in principle reduce the income tax rate, but in practice all local jurisdictions use the highest allowable rates.

126. Hungary is the only country that reports a similar arrangement, see OECD (1999*f*).

127. A tax equalisation fund to redistribute local income tax revenues across jurisdictions was abolished in 1996, but central government grants continue to assume this role.

128. Carlsen (1999).

129. The under-valuation of owner-occupied housing for net wealth purposes, for income tax purposes, and the low rate of imputed rent (2.5 per cent) as compared to market rates of return (assumed to be 5 per cent) each explain roughly one-third of the estimated revenue loss.

130. See Holmøy and Vennemo (1995), who estimate the welfare gain stemming from the 1992 tax reform to be 0.75 per cent, noting that the reform only concerned 60 per cent of the physical capital stock – excluding *e.g.* housing.

131. The income tax bill for a small company owner is determined as:
$T = m \cdot Y - (m - t) \cdot k \cdot A - t \cdot i \cdot B$
where: t = tax rate on capital income (28 per cent), k = imputed nominal return on business assets, A = book value of business assets in the tax accounts, i = nominal interest rate, B = stock of business debt, m = effective tax rate on personal income of the self-employed, Y = business profits before interest [see: Hagen and Sørensen (1998)]. On the other hand, by inflating its assets, the company will face a higher net wealth tax bill.

132. A survey carried out by the tax authorities in 1997 indicated that the number of companies that moved from active to passive ownership increased from about 8 per cent of the stock in 1993 to 11 per cent in 1995. These moves may in part reflect a one-off effect from the introduction of the 300 hours rule in 1995.

133. NOU 1999:7 *Flatere skatt.*

134. See the 1999 *Economic Survey* of Norway.

135. See Keen (1998).

136. Ministry of Labour and Government Administration (1996).

137. The ageing of the population would also argue for a move towards property taxes, since retired people typically store the bulk of their wealth in property.

Glossary of acronyms

ALMP	Active Labour Market Programme
APW	Average Production Worker
CGE	Computable General Equilibrium
CO_2	Carbon dioxide
CPI	Consumer Price Index
DnB	Den norske Bank
ECB	European Central Bank
EEA	European Economic Area
EFTA	European Free Trade Association
EPL	Employment Protection Legislation
ESA	EFTA Surveillance Authority
EU	European Union
G	"Basic amount" for social insurance and income tax purposes
GDP	Gross Domestic Product
GIEK	Norwegian Guarantee Institute for Export Credit
GFU	Gas Negotiation Committee
GSC	Gas Supply Committee
KLP	*Kommunal Landspensjonskasse* (the pension fund of local government employees)
LO	Central labour union organisation
M&As	Mergers and acquisitions
METR	Marginal Effective Tax Rate
MFN	Most Favored Nation
NAWRU	Non-Accelerating Wage Rate of Unemployment
NCA	Norwegian Competition Authority
NHO	Confederation of Norwegian Business and Industry
NIS	National Insurance Scheme
NOK	Norwegian krone
NOREX	Alliance of Swedish and Danish stock exchanges
NTC	Non-Trade Concern
PES	Public Employment Service
PTO	Public Telecommunication Operator
R&D	Research and Development
RISK	*Regulering av Inngangsverdi med endring i Skattlagt Kapital* (opening value adjustment method)
SDFI	State Direct Financial Interest
SME	Small and Medium-sized Enterprises
VAT	Value Added Tax
WTO	World Trade Organisation
Y2K	Year 2000

Bibliography

Aarbu, K. O. (1997),
"Some aspects of tax reform and economic performance in Norway – main objectives and some empirical facts", Ministry of Finance, Tax Policy Department (mimeo).

Aarbu, K. O. and Th. O. Thoresen (1995),
"Skattereformen og progressivitet i skattesystemet", *Økonomisk Analyser*, 9/95.

Aarbu, K. O. and Th. O. Thoresen (1998),
The Norwegian tax reform; distributional effects and the high-income response, Economic Research Programme on Taxation, Rapport 51.

Aarbu, K. O. and Th. O. Thoresen (1999),
Income responses to tax changes – evidence from the Norwegian tax reform, Ministry of Finance and Statistics Norway (mimeo).

Agell, J., M. Persson and H. Sacklén (1999),
"Labour supply prediction when tax avoidance matters" in: Economic Research Programme on Taxation, *Skatteforum 1999*, Rapport 55, pp. 5-38.

Arbeidsdirektoratet (Norwegian Department of Labour) (1999),
"Arbeidsmarkedsetatens formidlingsaktivitet I 1997 og 1998" (Public employment service brokerage activities in 1997 and 1998), *Kvartalsrapport om arbeidsmarkedet*, 2/99, pp. 49-57.

BIS (Bank for International Settlements) (1999),
"Restructuring in the global banking industry", *BIS Quarterly Review*, August 1999, pp. 35-47, *<http://www.bis.org/publ/r_qt9908.htm>*.

Blöndal, S. and S. Scarpetta (1998),
"Falling Participation Rates Among Older Workers in OECD countries", *OECD Economics Department working papers*, No. 202, June 1998, *<http://www.oecd.org/eco/wp/onlinewp.htm#1998>*.

Blöndal, S. and S. Scarpetta (1999),
"Early retirement in OECD countries: the role of social security systems", *OECD Economic Studies*, No. 29, 1997/II, pp. 7-54.

Boot, A. (1999),
"Consolidation and Strategic Positioning in Banking with Implications for Sweden", research paper for the Official Government Inquiry into the International Competitiveness of the Swedish Financial Sector, *<http://www.hhs.se/personal/finjn/fmuold/studier/boot.html>*.

Brendemoen, A. and H. Vennemo (1996),
"The marginal cost of funds in the presence of environmental externalities", *Scandinavian Journal of Economics* 98 (3), pp. 405-422.

Carlsen, F. (1999),
"Do local authorities set local fiscal variables to influence population flows?", in: Economic Research Programme on Taxation, *Skatteforum 1999*, Rapport 55, pp. 69-84.

Cnossen, S. (1997),
Dual Income taxation: The Nordic Experience, Research Memorandum Erasmus University 9710, Rotterdam.

Durand, M., C. Madaschi and F. Terribile (1998),
"Trends in OECD countries' international competitiveness: The influence of emerging market economies", *OECD Economics Department working papers*, No. 195, *<http://www.olis.oecd.org/olis/1998doc.nsf/LinkTo/ECO-WKP(98)8>*.

Eika, T. and K. Moum (1998),
"Monetary policy as an instrument in the current economic situation in Norway", *Statistics Norway Economic Survey*, No. 2/98, *<http://www.ssb.no/english/statistics_by_subject/08prices/es/9803/eika.pdf>*.

Fæhn, T. and E. Holmøy (1999),
"Welfare effects of trade liberalisation in distorted economies, a dynamic general equilibrium assessment for Norway", *Statistics Norway discussion paper*, No. 251, March, *<http://www.ssb.no/english/publications/etter_serie/dp/>*.

Financial Times (1999),
"Emu blamed as small debt markets struggle", 1 October.

Gjedrem, S. (1999),
"Monetary Policy Challenges", *Aftenposten*, 3 May, *<http://www.norges-bank.no/english/publications/aftenp3599.html>*.

Green Tax Commission (1996),
Policies for a Better Environment and high Employment – An English Summary, Oslo.

Guellec, D. and B. van Pottelsberghe de la Potterie (1999),
"Does government support stimulate private R&D?", *OECD Economic Studies*, No. 29, 1997/II, pp. 95-122.

Hagen, K. P. and P. B. Sørensen (1998),
"Taxation of income for small business: taxation principles and tax reforms in the Nordic countries", in P.B. Sørensen (1998), *Tax policy in the Nordic Countries*, pp. 28-71, London.

Holden, S. (1996),
"The unemployment problem – A Norwegian perspective", *OECD Economics Department Working Paper*, No. 172, *<http://www.oecd.org/eco/wp/onlinewp.htm>*.

Holden, S. (1999),
"Wage setting under different monetary regimes", Department of Economics of the University of Oslo memorandum, No. 12/99, *<http://www.sv.uio.no/sosoek/memo/memopdf/memo1299.pdf>*.

Holmøy E. and H. Vennemo (1995),
"A general equilibrium assessment of a suggested reform in capital income taxation", *Journal of Policy Modeling*, Vol. 17, pp. 531-556.

Holmøy, E. and T. Hægeland (1999),
"Effective rates of assistance for Norwegian industries", *Review of Income and Wealth*, 45, No. 1, March 1999, pp. 99-116.

International Bureau of Fiscal Documentation (1999),
European Tax Handbook 1999, Amsterdam, pp. 399-412.

Keen M. (1998),
"Vertical tax externalities in the theory of fiscal federalism", *IMF Staff Papers*, Vol. 45, No. 3, September, pp. 454-485.

Konkurransetilsynet (Norwegian Competition Authority) (1999),
Annual Report 1998 on Competition Law and Policy in Norway, *<http://www.konkurransetilsynet.no/kunder/konkurransetilsynet/index.nsf/d0b85697890d0e6ac12565970076aaf2/0e8ab34c24e25378c12567df0042d002?OpenDocument>*.

Kredittilsynet (Banking, Insurance and Securities Commission of Norway) (1999),
"Rapport for finansinstitusjoner 1-3 kvartal 1999", November, *<http://www.kredittilsynet.no/pressemelding/meldinger/Forside399-off.htm>*.

Kredittilsynet (1999*b*),
Annual Report 1998, *<http://www.kredittilsynet.no/aarsmeld98/index.html>*.

Lindquist, K-G. and B. Naug (1996),
Statistics Norway's Macroeconometric models and International Competitiveness, Statistics Norway.

Ministry of Agriculture (1999),
"Appropriate Policy Measure Combinations to Safeguard Non-Trade Concerns of a Multifunctional Agriculture", Informal Process of Analysis and Information Exchange of the WTO Committee on Agriculture, September 1999, *<http://www.landbruk.dep.no/landbruksdepartementet/multifunctionality/assets/images/Paper_on_policy_measures.final.doc>*.

Ministry of Education, Research and Church Affairs (1999),
"Research at the beginning of a new era", Report to the Storting No. 39 (1998-1999), June 1999, *<http://odin.dep.no/kuf/publ/1999/fakta/stmeld39e.html>*.

Ministry of Education, Research and Church Affairs (1999*b*),
"The Competence Reform in Norway", *<http://odin.dep.no/kuf/publ/1999/competence/>*.

Ministry of Finance (1992),
Towards More Cost-effective Environmental Policies in the 1990s – Principles and Proposals for Better Pricing of the Environment (Report of the Environmental Tax Commission to the Minister of Finance), Oslo.

Ministry of Finance (1992*b*),
Norwegian Tax Reform 1992, Proposal to the Parliament, April 1991 (English summary, mimeo).

Ministry of Finance (1999),
"An indicator of the fiscal stance", *<http://www.finans.dep.no/engelsk/3/test3/oar/engelsk_dokumentasjon_ORA.pdf>*.

Ministry of Finance (1999*b*),
"Kreditmeldinga 1998" (The development of the financial markets and credit institutions in 1998), Report to the Storting, *<http://www.finans.dep.no/6/nye_publ/st_meld/stmeld11_99/>*.

Ministry of Health and Social Affairs (1999),
"The equitable redistribution White Paper", English summary of NOU 50 (1998-1999), October, *<http://odin.dep.no/shd/publ/1999/utjamningsmeldinga/engelsk/>*.

Ministry of Health and Social Affairs (1999*b*),
"The Norwegian Social Insurance Scheme, A Survey", January, *<http://odin.dep.no/shd/publ/1999/norsk_trygd/eindex.html>*.

Ministry of Labour and Government Administration (1996),
Et enklere og mer rettferdig inntektssystem for kommuner og fylkeskommuner (Simpler and fairer revenue system for municipalities and counties), NOU 1996:1.

Ministry of Petroleum and Energy (1999),
"1999 Fact sheet – Norwegian petroleum activity", *<http://195.1.89.90/innhold/fp/publikasjoner/hefter/faktahefte99/pdf-engelsk/wv519.pdf>*.

Moe, T. (1999),
"Ageing and pensions: challenges and policies in OECD countries", Keynote speech at the Conference on "Ageing and Pensions" of the Norwegian Savings Bank Association, October.

Myran, K. and L. Sandberg (1999),
"Matvarepriser i Norge, Sverige og Danmark" (Food prices in Norway, Sweden and Denmark), Statistics Norway, *Økonomiske analyser* 7/99, pp. 16-20, *<http://www.ssb.no/emner/08/05/10/oa/9907/myran.pdf>*.

Nicoletti, G., S. Scarpetta and O. Boylaud (1999),
"Summary indicators of product market regulation with an extension to employment protection legislation", *OECD Economics Department Working paper*, No. 226, *<http://www.oecd.org/eco/wp/onlinewp.htm>*.

Norges Bank (1997),
"Norges Bank's submission on the Government's economic programme for 1998", *Norges Bank Economic Bulletin*, No. 4/97, pp. 370-383.

Norsk Hydro (1999),
"Norsk Hydro's view on future of the State's participation in the Norwegian petroleum activities", report presented to the Minister of Petroleum and Energy by Norsk Hydro's board of directors, *<http://www.hydro.com/INET/OSLO1742.NSF/All/4A8125D682E9F925412567DF004733EE?OpenDocument>*.

OECD (1996),
Lifelong learning for all: Meeting of the Education Committee at Ministerial Level, 16-17 January, Paris.

OECD (1998),
Adjustment in OECD Agriculture, Reforming Farmland Policies, Paris.

OECD (1998*b*),
"The retirement decision", *Economic Outlook 63*, pp. 179-192, June.

OECD (1998*c*),
Agricultural Policy Reform and the Rural Economy in OECD Countries, Paris.

OECD (1998*d*),
Science, Technology and Industry Outlook 1998, Paris.

OECD (1999),
Agricultural Policies in OECD Countries, Monitoring and Evaluation 1999, Paris.

OECD (1999*b*),
Main Science and Technology Indicators, No. 1, Paris.

OECD (1999*c*),
OECD Science, Technology and Industry Scoreboard 1999, Benchmarking Knowledge-based Economies, Paris.

OECD (1999*d*),
Communications Outlook 1999, Paris.

OECD (1999*e*),
Economic Survey of Norway 1999, Paris.

OECD (1999*f*),
Revenue Statistics 1965-1998, Paris.

OECD (1999*g*),
Employment Outlook 1999, June, Paris.

OECD (1999*h*),
The Tax/Benefit Position of Employees 1996-1997, Paris.

OECD (1999*i*),
Taxing powers of state and local government, OECD Tax policy studies – No. 1, Paris.

OECD (2000),
Economic Survey of Australia 1999-2000, Paris.

Oxley, H., J. Burneaux, T. Dang and M. Mira d'Ercole (1999),
"Income distribution and poverty in 13 OECD countries", *OECD Economic Studies*, No. 29, 1997/II, pp. 55-94.

Reppen, H.K. and E. Rønning (1999),
"Barnefamiliers tilsynsordninger, yrkesdeltakelse og bruk av kontantstøtte våren 1999" (Cash-benefit, daycare and employment in 1999), Statistics Norway Reports 1999/27, *<http://www.ssb.no/emner/03/04/30/kontantstotte/rapp_1999-27/rapp1999-27.pdf>*.

Rødseth, A. and R. Nymoen (1999),
"Nordic wage formation and unemployment seven years later", Department of Economics of the University of Oslo memorandum, No. 10/99, *<http://www.sv.uio.no/sosoek/memo/memopdf/memo1099.pdf>*.

Røed, K. (1998),
"Egalitarian Wage Policies and Long-term Unemployment", *Scandinavian Journal of Economics*, 100(3), pp. 611-625.

Røisland, Ø. and R. Torvik (1999),
"Exchange rate versus inflation targeting: A theory of output fluctuations in traded and non-traded sectors", Norges Bank working papers, No. 1999/1, February.

Sandven, T. and K. Smith (1998),
"Understanding R&D intensity indicators, Effects of differences in industrial structure and country size", IDEA paper, No. 14, STEP group, *<http://www.step.no/Projectarea/IDEA/Idea14.pdf>*.

Sollund, S. (1991),
"Norway: 1992 Tax Reform", *European Taxation*, September, pp. 272-278.

Sørensen, P.B. (1998),
"Recent innovations in Nordic tax policy: from the global to the dual income tax", in Sørensen P.B. (ed.), *Tax policy in the Nordic Countries*, pp. 1-27, London.

Statoil (1999),
"Creating value for Statoil and SDFI", report presented to the Minister of Petroleum and Energy by Statoil's board of directors, August, *<http://www.statoil.com/STATOILCOM/SVG00990.nsf/design/frameset?OpenDocument&/statoilcom/inf/svg01429.nsf/design/fs_e_sdoe>*.

Tax Notes International (1998),
"Norwegian Tonnage Tax Regime", Vol. 17, pp. 997-1001 and "Norwegian Taxation of Limited Companies and Shareholders", pp. 1798-1800.

White, W.R. (1998),
"The coming transformation of contintental European banking?", *BIS working paper*, No. 54, <*http://www.bis.org/publ/work54.htm*>.

Youngman, J.M. and J.H. Malme (1994),
An international survey of taxes on land and buildings, Kluwer Law and Taxation Publishers.

Annex 1

Main features of the tax system updated to 1 January 2000

Table A1. **Taxes levied on corporate income: standard regime**

Nature of the tax

- Resident companies are liable to a national income tax (*fellesskatt til staten*) only. The municipal income tax was abolished from 1998.
- In practice, a company is resident if it is incorporated under Norwegian law. The location of the place of effective management in Norway also constitutes residence.
- A company is taxable on world-wide income.
- Income and capital gains are pooled and taxed at the same rate. Capital gains derived from the sale of business assets are normally included in taxable income. However, with respect to sales of assets in depreciation classes *a)* to *d)*, the sale proceeds may instead be deducted from the collective depreciable basis.
- For capital gains and losses realised on other depreciable assets, depreciation classes *e)* to *h)*, and on non-depreciable assets there is also an option. They may be taxed or deducted immediately or they may be booked on a gains and losses account. At the end of each year, at least 20% of any positive balance is included in income and up to 20% of any negative balance deducted from income.

Exemptions, credits and allowances

- Norway has an imputation system. Resident individual and corporate shareholders are taxable on dividends received, but are entitled to a tax credit against their tax liability. The credit is equal to the tax on the dividends.
- The tax credit is computed at the same rate at which corporate income tax is levied.
- Unilaterally, Norway grants residents an ordinary foreign tax credit for income tax paid abroad. The credit is granted on an overall basis.
- If a Norwegian company, either alone or in association with no more than nine other residents of Norway, owns immovable property or business establishments abroad, it may be treated, subject to the approval of the Ministry of Finance, as if it owned such property and business establishments directly. The company may then be able to claim tax credits for the tax paid by the non-resident company. This only applies if double taxation is not relieved by a tax treaty.
- Excess foreign tax credits may be carried forward for 10 years.
- Foreign tax on business income may be deducted as an alternative to taking a tax credit.
- Where a tax treaty applies, the taxpayer may, in general, choose between unilateral and treaty relief.
- Losses may be carried forward to be set off against profits for the next 10 years. When a company liquidates, losses of the year of liquidation may be carried back to be set off against profits of the preceding 2 years.
- A limited and general deduction for doubtful debts may be made.
- Depreciable business assets are divided into eight classes. The maximum rates of annual depreciation are:

Class	Description	Rate (%)
a)	Office machines	30
b)	Goodwill, acquired through purchase or gift or inheritance	30
c)	Trucks, trailers, buses, taxis and vehicles for disabled persons	25
d)	Ordinary motor cars, tractors, other rolling equipment, tools, etc.	20
e)	Ships and other vessels, drilling rigs, etc.	20
f)	Aircraft and helicopters	12
g)	Buildings and plant, hotels, guest houses, restaurants, etc.	4
h)	Office buildings	1

- Business assets in classes *a)* to *d)* are written down on a collective basis, *i.e.* on the aggregate book value of assets in the same class at the beginning of the year, as increased by the cost of assets acquired during the year.
- Business assets in classes *e)* to *h)* are depreciated individually. Gains and losses on disposal are booked on a special gains and losses account, as is a negative balance regarding acquired goodwill, except that gains may be included in taxable income immediately if the taxpayer so elects. If at the end of the year the gains and losses account shows a positive balance, at least 20% of this balance is transferred to taxable income, whereas up to 20% of any negative balance is deductible.

Rates

- For the income year 2000, the rate of the national corporate income tax is 28%. The rate of the municipal corporate income tax is 0% with the effect that no municipal corporate income tax is levied for 2000.

Source: International Bureau of Fiscal Documentation (1999) and Ministry of Finance.

Table A2. **Taxes levied on corporate income: special regimes**

Shipping

Nature of the tax

- A Norwegian-incorporated company may elect to have shipping income exempt from corporate income taxation and, instead, pay a tax on dividends and on net tonnage under a shipping taxation regime. To qualify for the regime a company may hold no assets other than ships in operation, movable equipment to be used for the exploration of petroleum and certain financial assets.
- If a company has opted for the regime its activities are restricted to the leasing or operation of owned or leased ships, and the company's day-to-day business must be carried out by a company not subject to the shipping taxation regime. This is to ensure that only the return on capital investment in a ship is exempt from the normal corporate income taxation.

Rates

- For 2000 the tax on net tonnage is levied as follows:

Net tonnages (tonnes)	Tax per days per 1 000 tonnes (NOK)
Up to 1 000	0
1 001-10 000	72
10 001-25 000	48
Over 25 000	24

The rates may be reduced by up to 25% if the ship meets certain environmental criteria.

- Profits from shipping activity are not taxed on an annual basis. However dividends are taxed at a rate of 28% levied on the *company,* unless the distribution is made out of taxed (net financial) income. This tax is in addition to the net tonnage tax. The taxable base is an amount equal to 139% of the distribution. This is to ensure that the imputation tax credit granted to resident shareholders is given only for tax actually paid. Taxed income is considered distributed before untaxed income.
- Net financial income is taxed at the normal corporate income tax rate of 28%.

Petroleum exploration and exportation activities

Nature of the tax

- Resident and non-resident companies engaged in the exploration for an exploitation of submarine petroleum resources and related activities, including pipeline transport of petroleum produced, are subject to the following taxes:
  a) income tax levied at a rate of 28%; and
  b) the petroleum tax on these activities.
- Generally speaking, the Petroleum Tax Act extends only to the Norwegian offshore area. Non-resident companies are fully taxable but tax treaty rules may result in an exemption from the petroleum tax depending on whether their activities constitute a permanent establishment which subjects them to Norwegian taxation.
- The petroleum tax is payable on that part of the net income which exceeds 5% of the cost of depreciable assets which have been put into ordinary use for these activities and which have been acquired before the beginning of the relevant year (but not more than 6 years ago).
- Taxable income (and capital) is computed according to special rules which include:
  a) gross income and the value of stocks of petroleum are assessed according to norm prices;
  b) deduction for ordinary depreciation may be claimed up to a maximum of 16.67% per annum; and
  b) operating losses may be carried forward for 15 years subject to certain conditions.
- Petroleum produced is valued at "norm" prices determined quarterly by the competent authorities. Norm prices apply only to produced oil (not gas) and are "the equivalent of the price at which petroleum would have been sold between independent parties in a free market".
- No deduction is allowed for dividends declared out of the annual profits and for losses arising from other activities.

Rates

- The petroleum tax rate is currently 50%. The basis of assessment is the income included in the computation of national income tax.
- In respect of oil fields whose development plans were accepted before 1 January 1986, a royalty (*produksjonsavgift*) must also be paid. The royalty is levied at fixed rates between 8% and 16% on the value of the quantity of oil produced. The royalty will be phased out over a period of 6 years. A CO_2–tax is charged on petroleum that is burnt and natural gas that is discharged to air. The rate is 70 øre per Sm^3 o.e. or per litre of oil.

Source: International Bureau of Fiscal Documentation (1999) and Ministry of Finance.

Table A3. **Taxes levied on household and other business income: national and municipal taxes on net income**

Nature of the tax

- The national income tax (*fellesskatt til staten*) and the municipal income tax (*inntektsskatt til kommunen*) are levied on world-wide net income of resident individuals. Net income is also referred to as *ordinary income* and comprises, *inter alia,* income from present and past employment, income from business or profession, annuities, fringe benefits and income from immovable and movable property.
- Capital gains realised on the sale of movable or immovable property are generally taxable (and capital losses are deductible).
- Income from immovable property includes the rental value of owner-occupied dwellings (2.5% of the assessed value, but 5% of the part of the value that exceeds NOK 451 000. The taxable value of the taxpayer's sole or main residence is reduced by NOK 51 250.
- The incomes of a married couple are generally aggregated and taxed together. However, if both spouses have income, they will normally be taxed separately (class 1).

Exemptions, credits and allowances

- An imputation system applies to dividends distributed by resident companies to resident individuals and shareholders. Dividends received are included in the tax base of the shareholders and taxed at the general rate of 28%, but the shareholder is entitled to a tax credit equal to the tax on dividends.
- Capital gains on personal immovable property (subject to certain conditions) are exempt. Capital gains on shares in resident companies are exempt in so far as these reflect the company's retained profits and to the extent they are attributable to the shares owned by the taxpayer.
- Unilaterally, Norway grants residents an ordinary tax credit for income tax paid abroad. Where a tax treaty applies, the taxpayer may, in general, choose between unilateral and treaty relief.
- Premiums for private pension schemes and contributions to certain saving schemes are deductible up to a limit of NOK 40 000.
- The personal allowance for jointly assessed married couples and for single persons with dependants is NOK 55 400. The allowance for other persons is NOK 27 700. A taxpayer is entitled to child deductions, amounting to NOK 1 820 for each child below 18 years of age.
- For expenses incurred in carrying out an employment, a minimum deduction equal to 22% of employment and pension income, with a floor of NOK 4 000 and a ceiling of NOK 36 600, is granted. If actual expenses exceed these limits, they may be deducted instead.
- As from 2000 a new allowance of NOK 30 600 is introduced. The allowance is deductible against ordinary income. The taxpayer will have to choose between the existing minimum deduction and the new allowance.
- Childcare expenses for children under the age of 12 may be deducted up to a ceiling of NOK 25 000 for one child and NOK 30 000 for two or more children.

Rates

- The national and municipal income taxes on net income are levied at a combined maximum rate of 28%. This rate comprises national income tax of 17.1% (a reduced rate of 7.4% applies to certain remote regions) and municipal and county income tax at maximum rate of 10.9%.
- The municipal rate may vary between 16.35% and 17.1%, but in practice the maximum rate applies.

Source: International Bureau of Fiscal Documentation (1999) and Ministry of Finance.

Table A4. **Taxes levied on household and other business income: national income tax on gross income**

Nature of the tax

- The additional national income tax is levied on gross income from certain sources only (*toppskatt til staten*). Gross income is also referred to as *personal income,* and includes:
  - Wages and salaries and other remuneration for work performed in or outside an employment situation, including fees, commissions, tips, etc.
  - Remuneration as a member of a board of directors, etc.; benefits in kind received with the above sources; pension and life annuities in relation to employment; various social security benefits (sick pay, unemployment benefits); personal income of owners of a one-man company, of active shareholders of a company, or derived from active participation in partnerships.
- The personal income of persons engaged in a business and of active participants and shareholders in companies and other entities is computed in the so called split model by eliminating any capital income or losses from the business entity's total income and by subtracting that part of the income which is deemed to be a remuneration of the capital used in the business, and by further subtracting 20% of the employment income paid to any employees of the business.

Exemptions, credits and allowances

No special allowances or deductions are granted in computing the tax base

Rates

Class 0 and 1	Class 2	%	Class	Taxpayers
Up to NOK 277 799	Up to NOK 328 999	0	0	Most non-residents
NOK 277 800-762 699	NOK 277 800-762 699	13.5	1	Single persons without dependants and married persons assessed separately
Over NOK 762 700	Over NOK 762 700	19.5	2	Married couples assessed jointly and single persons with dependants

Source: International Bureau of Fiscal Documentation (1999) and Ministry of Finance.

Table A5. **Social security contributions**

National insurance premiums	Employers' social security contributions
Nature of the tax	
• The taxable income basis is identical to that of the national income tax on gross income (see Table A4).	• Contributions are levied on a company's payroll.
Exemptions, credits and allowances	
No special allowances or deductions are granted in computing the tax base.	
Rates	
Category %	• Employers' social security contributions are payable at a rate of 14.1%. Lower rates of 0%, 5.1%, 6.4% and 10.6% apply with respect to most of the employees who live in northern Norway and in developing areas. There is an additional 12.5% employers social security contribution if an employee's salary from the same company or group of companies exceeds 16 × G[1] (the employee's contribution is not increased). Social security contributions are deductible for corporate income tax purposes.
All employment income. 7.8	
Pension and life annuities, employment income of persons under 17 or over 69 years of age. 3.0	
Labour income of self-employed individuals up to an income of 12 x G.[1,2] 10.7	

1. G is the weighted average basic amount for social security purposes. G = NOK 46 423 from 1 May 1999.
2. The rate is 7.8% on income from self-employment over 12 × G.

Source: International Bureau of Fiscal Documentation (1999) and Ministry of Finance.

Table A6. **Taxes levied on consumption and investment**

Value-added tax	Investment tax	Excise duties
Nature of the tax		
• Registered entrepreneurs, selling taxable goods and services in Norway, and importers are liable to value added tax *(merverdiavgift).* • The taxable base is the consideration received for sales of taxable goods and services, and value at importation including import duties. In computing an entrepreneur's final turnover tax liability, the tax paid on his purchases and imports may be deducted so that, in effect, only the value added is taxed.	• The taxable base is the value of fixed business assets purchased by entrepreneurs for lasting use in their business, on which the value added tax is deductible. In practice, the investment tax operates as a limitation on the deductibility of input VAT. As the rate of the investment tax is 7% this means that only 16% of the 23% input VAT may be deducted if the investment tax applies.	• Taxes on alcoholic and non-alcoholic beverages, beverage packaging, tobacco, vehicles, petrol and diesel, electricity, mineral products, waste, lubricating oil, pesticides, sweets, sugar, real property transactions, air travel, radio frequencies.
Exemptions, credits and allowances		
• Exemptions exist for charity, cultural services, education, financial services, insurance, lottery and gambling, medical and dental care, postal services, property letting, supply of land and buildings, broadcasting, legal services, public passenger transport, public cemetery, burial and funeral services and sporting events.	• The law provides for exemptions with respect to, *inter alia,* assets used in manufacturing goods; lorries, trailers, quarrying, construction of sewerage systems by municipal authorities, and construction of buildings and installations in northern Norway.	
Rates		
• The rate is 23%. • To some goods a zero rate is applied (mainly books, periodicals, newspapers and electricity in the northern parts of the country).	• The rate is 7%.	

Category[1]	NOK per	Lowest	Highest
Vehicles[2]	Kilo	25.16	117.05
Car fuel[3]	Litre	3.74	4.34
CO_2 car fuel[4]	Litre	0.47	0.94
Wine-spirits	Litre and % of alcohol	3.65	7.04
Beer[5]	Litre	1.58	16.37
Beverage packing[6]	Unit	1.00	4.00

1. Selected products; list is not exhaustive.
2. Progressive rate structure, maximum rate is the marginal rate for cars weighing over 1 500 kg (from 2000).
3. Diesel – unleaded petrol.
4. Mineral oil – petrol.
5. Alcohol-free – up to 4.75% alcohol. Beer with more than 4.75% alcohol has a tax rate of NOK 3.65 per litre and % of alcohol (from 2000).
6. Carton – metal and glass (from 2000).

Source: International Bureau of Fiscal Documentation (1999) and Ministry of Finance.

Table A7. **Taxes levied on property**

Real estate	Net wealth	Inheritance and gifts
Nature of the tax		
• Immovable property situated in Norway is subject to municipal real estate tax (*eiendomsskatt*) provided the local municipality has made use of the authority to levy the tax (this is the case in most urban areas in Norway), but in principle companies can be subject to a real estate tax also outside urban areas. The taxable base is the assessed value of the immovable property, which usually is between 20% and 50% of the fair market value. • Both corporations and individuals are liable to the tax.	• Resident individuals are subject to national net wealth tax (*formuesskatt til staten*) and municipal net wealth tax (*formuesskatt til kommunen*) with respect to their world-wide net wealth. The net wealth taxes are levied on the net value of the taxpayer's assets at the end of the tax year. The net wealth of married couples living together is aggregated for tax purposes. • Non-residents are subject to Norwegian net wealth taxes with respect to the property of a Norwegian business carried on by them or in which they participate, and with respect to immovable and tangible movable property located in Norway. The tax liability of non-residents may be further restricted by tax treaty provisions. • The taxable value of financial assets is a fixed share of the assessed market value, according to the following rates: Non-listed shares and listed shares in small and medium sized companies 65% Other 100%	• The person receiving the inheritance or gift is liable for the tax thereon. • The inheritance tax is due with respect to all assets, wherever located, if the deceased at the time of his death was a resident of Norway or a Norwegian national, wherever resident. However, if the deceased was resident in another country, no tax liability will arise on the basis of his Norwegian nationality provided his estate has been subject to an inheritance or similar tax in his country of residence. • When the deceased is a non-resident, Norwegian inheritance tax is due with respect to immovable property located in Norway and business property belonging to a permanent establishment in Norway. • The above rules also apply, *mutatis mutandis*, to taxable gifts, depending on residence and nationality of the donor and location of the assets donated.
Exemptions, credits and allowances		
	• Immovable property situated abroad is exempt. The same exemption may apply if the taxpayer, alone or together with not more than nine other residents of Norway, owns at least 95% of the share capital of a non-resident company owning such immovable property abroad. • An ordinary tax credit is granted for net wealth tax paid abroad, but only against Norwegian net wealth tax.	• An exemption applies with respect to immovable property located abroad and business property belonging to a permanent establishment abroad, provided an inheritance or similar tax has been paid in the country where such property is located.

Table A7. **Taxes levied on property** *(cont.)*

Real estate	Net wealth	Inheritance and gifts
Rates		
• Tax is levied at fixed rates ranging from 0.2% to 0.7% depending on the municipality. Real estate tax is deductible for the purposes of the corporate income tax.	• The national net wealth tax is levied at progressive rates and the threshold amounts depend on the classification of the taxpayer. In this case the classification is different from the classification for the purposes of the national income tax on gross income in that married persons assessed separately belong to class 2 rather than 1. For 2000 the rates are: Taxable net wealth (NOK) Class 0 and 1 / Class 2 / Rate (%) Up to 120 000 / Up to 150 000 / 0 120 000-540 000 / 150 000-580 000 / 0.2 Over 540 000 / Over 580 000 / 0.4 • The municipal net wealth tax is levied only on net wealth in excess of NOK 120 000 (non-residents are taxed on the entire amount of their net wealth, however). The rate is 0.7% in virtually all municipalities, although it may vary, in principle, between 0.4% and 0.7%. If the total of income tax and the net wealth tax exceed 80% of the taxpayer's ordinary income, the national net wealth tax and, if necessary, the municipal net wealth tax are reduced accordingly. The marginal tax rate on net wealth exceeding NOK 1 000 000, can never fall below 0.6%.	• On inheritance and gifts to children, foster children and parents the rates are: Taxable amount (NOK) / Rate (%) Up to 200 000 / 0 200 000-500 000 / 8 Over 500 000 / 20 • On inheritance and gifts to other persons the rates are: Taxable amount (NOK) / Rate (%) Up to 200 000 / 0 200 000-500 000 / 10 Over 500 000 / 30

Source: International Bureau of Fiscal Documentation (1999) and Ministry of Finance.

Annex II

Calendar of main economic events

1999

January

After increasing its key interest rates by 4.5 percentage points between March and August 1998, Norges Bank reduces its rates by 0.5 percentage point. The overnight deposit and lending rates stand after the cut at 7.5 and 9.5 per cent, respectively.

The Tax Reform Commission 1999 (Flat Tax Commission) appointed by the Ministry of Finance presents its analysis concerning a new, simpler and flatter tax system. The tax reform commission prefers a reduction in the income surtax on high income brackets from 13.5 to 4.5 per cent and an increase of the flat-rate tax on net income from 28 to 29.5 per cent. The new tax system would reduce taxes on high incomes substantially and raise taxes on capital income and on low to medium wage income. The preferred reform leaves tax revenues unchanged. The commission warns against a too sharp increase in the tax rate on net income given the possible negative effects on investments.

February

Storebrand and the Swedish insurance company Skandia agree to merge their non-life insurance divisions. The Finnish insurance company Pohjola joins the merger in June leading to the largest general insurance company of the Nordic countries with 20 per cent market share. The head office of the new company will be located in Stockholm, though the private market and maritime and energy insurance divisions will be administered from Norway.

A report released by the Ministry of Labour and Government Administration indicates that central and municipal governments could realise a 10 per cent cost reduction by establishing electronic procurement routines. Implementation of electronic procurement systems would enable governments to simplify their procedures and coordinate their procurements.

The Directorate of Taxes releases a report on Norway's grey economy. It estimates the volume of Norway's unregistered economy at around 6 per cent of GDP.

March

Norges Bank reduces its key interest rates by 0.5 percentage point.

The government decides to double the cut in oil production from 100 000 barrels per day to 200 000 barrels from 1 April onwards.

Den norske Bank (DnB) and the Postal Bank agree to merge. This merger will result in a financial institution with approximately NOK 315 billion in total assets, making it Norway's largest financial institution and number eight in Scandinavia.

The Arntsen Committee, with high level representatives of the social partners and the government, recommends to limit the wage increase in 1999 to 4½ per cent and to bring down wage increases in 2000 to the rate of the trading partners.

April

Norges Bank lowers its interest rates for the third time in 1999 by 50 basis points. This brings the central bank's overnight lending rate to 8.5 per cent and the deposit rate to 6.5 per cent.

After six months of operation, disorder in Gardermoen airport has cost airlines around half a billion kroner. Since the beginning of the year, 8 500 flights have been delayed and 3 300 have been cancelled.

The Norwegian Confederation of Trade Unions (LO) and the Norwegian Confederation of Business and Industry (NHO) reach a pay deal for the manufacturing sector. Wages are increased by 0.4 per cent from 1 April 1999. Including the carry-over from 1998, the wage increase is 3¼ per cent, and if the estimated wage drift is taken into account, close to 5 per cent.

May

Since the approval of the 1999 Budget, the government has reduced the intended allocation to the Government Petroleum Fund from nearly NOK 50 billion to less than NOK 27 billion due to low oil prices and production, lower non-oil revenues and higher government expenditures than anticipated. The revised 1999 Budget estimates that the central government surplus will decline by NOK 25½ billion to NOK 30 billion compared to the approved 1999 Budget, while the non-oil deficit will increase from NOK 6½ billion to NOK 10.9 billion.

The total amount of investments and bank deposits tops one trillion kroner. Since 1996, private debt has risen by around NOK 100 billion, while savings have gone up by NOK 150 billion.

Unemployment increases for the first time in six years. The manufacturing and construction sectors are losing jobs, while job offers are increasing in the service and the public sector, especially in health and education.

June

Norges Bank reduces its interest rates by another 0.5 percentage point, the fourth cut since the beginning of the year.

A joint offer by Norsk Hydro and Statoil for Saga Petroleum, of NOK 20 billion, tops bidding by French oil company Elf.

The government launches a national joint effort for "A simpler Norway". The aim of the two-year project is to simplify relations between the public and the national administration.

Seventeen thousand persons went on disability pensions in the first six months of 1999, and sick leave has increased by 6 per cent.

The government publishes a white paper on the distribution of income and living conditions in Norway. With the paper, the government wishes to reintroduce redistribution

as a key political objective. During the 1990s, most people have become more prosperous, but at the same time the gap between the richest and the poorest has widened. The government aims at reversing this trend.

July

Every day one ton of meat is smuggled from Sweden to Norway, according to the Customs and Excise Administration.

Wilhelm Wilhelmsen Co. merges with the Swedish Wallenius Lines to form the world's largest shipping line for motor vehicles and other types of rolling cargo. The new shipping line has an anticipated turnover of NOK 11 billion and will operate a fleet of 80 ships.

August

Norwegian health and social services lack around 6 000 employees. The shortage of qualified personnel has escalated since the beginning of the year. Four out of ten institutions are unable to recruit the personnel they need to maintain the required level of activity, and it is feared that the lack of qualified personnel will force more than 10 per cent to reduce their activity.

September

Norges Bank lowers its interest rates for the fifth time in 1999 by 50 basis points. This brings the central bank's overnight lending rate to 7.5 per cent and the deposit rate to 5.5 per cent.

A takeover bid by MeritaNordbanken for partly state-owned Kreditkassen/Christiania Bank "raises difficult questions" concerning the guidelines for state ownership of financial institutions.

The government announces a revision of the guidelines for the Government Petroleum Fund whereby NOK 1 billion of the capital of the Petroleum Fund will be transferred into an independent fund which is to be managed according to a separate set of environmental guidelines.

The order books are nearly empty for industries supplying equipment to the oil industry, particularly the shipbuilding industry. Around 2 000 workers at yards which traditionally built oil platforms and other types of offshore equipment have been laid off temporarily or permanently in the first nine months of 1999. According to the Federation of Norwegian Engineering Industries, another 15 000 to 25 000 jobs may be lost in the coming years. The government proposes a special package to prevent the loss of thousands of jobs in offshore related industries along the coast.

The Ministry of Petroleum and Energy announces the 16th licensing round on the Norwegian continental shelf. The round aims at contributing to responsible resource management and predictability of oil and gas production. The goal is also to promote exploration activity, maintain the activity level on the continental shelf and to secure employment in the industry.

Conservatives gain seats in municipal elections and the Labour Party loses but remains the biggest party. The election results are as follows: Labour 28.7 per cent (down by 6.3 percentage points from previous municipal elections), Conservatives 21.4 (up by 7.2 points), Progress 12.0 per cent, Socialist Left 7.8 per cent, Centre Party 8.3 per cent, Christian Democrats 9.4 per cent and Liberals 4.3 per cent. At less than 60 per cent, voter turnout is at a record low.

October

The government aims to maintain its ownership in Kreditkassen/Christiania Bank and Den norske Bank. The government proposes, however, an exception from the general 10 per cent ownership limitation for holdings up to 25 per cent in combination with strategic co-operation and alliances. The key elements of the government's ownership policy are: maintain a long-term Norwegian ownership in main Norwegian financial institutions, ensure sufficient Norwegian ownership in the two largest commercial banks by keeping a stake of one third of the shares.

A merger agreement between Telenor and Telia is signed. The Swedish and Norwegian employees' representatives will have two seats each on the board of directors. The signing ceremony was postponed by 24 hours due to need for further clarification.

The government presents the draft 2000 Budget to Parliament. The draft projects a doubling in the central government surplus in 2000 due mainly to a surge in petroleum revenues. The non-oil deficit is estimated to narrow only by NOK 2 billion to around NOK 13 billion.

November

The Oslo Stock Exchange signs a letter of intention to participate in the NOREX stock exchange alliance. The NOREX alliance forms a common market place between independent stock exchanges for financial instruments. The purpose of the alliance is to make use of synergy benefits through the increased distribution and to improve cost efficiency.

December

The government, keeping in step with OPEC, decides to extend the current reduction of 200 000 barrels per day in oil production until the end of the first quarter of 2000. Norway produces about 3 million barrels of oil per day and is the second largest oil exporter in the world after Saudi Arabia. Norway first introduced a restriction of 100 000 barrels per day in May 1998 and doubled the cut to 200 000 barrels from the beginning of April 1999.

State-owned Telia-Telenor launches a hostile takeover bid for Esat Telecom, Ireland's second biggest telecoms operator.

A majority in the Parliament agrees to pool state interests in DnB and Kreditkassen/Christiania Bank into one entity, and to keep the ownership of at least one third of this entity. Based on the discussion in Parliament, the Ministry of Finance asks the Government Bank Investment Fund to consider the options of structural solutions for DnB and Kreditkassen with the objectives to establish a strong financial solution with stable national ownership and to keep main strategic functions (headquarters) in Norway.

The planned merger between Swedish Telia and Norwegian Telenor collapses. A partial privatisation of the public telecommunications operator Telenor was planned for early 2000. The Norwegian authorities are still intending a partial privatisation of Telenor. A proposal will be submitted to Parliament early in 2000.

STATISTIQUES DE BASE :

COMPARAISONS INTERNATIONALES

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
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7. Data refer to the Belgo-Luxembourg Economic Union.
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